

AKADEMIKA

**ACL-AM: AMPLITUDE MODULATION
TRANSMITTER KIT**

&

**ACL-AD :AMPLITUDE DEMODULATION
RECEIVER KIT**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



Today's system designers are faced with tomorrow's problems. ANALOG COMMUNICATION is one of the leading subjects of this century & expanding its horizon in coming century too. It is our vision to provide you with the product you need, ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership. As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that their integrity is checked with respect to the packing list. Also ensure that no visible damage as such appears on any accessories.

Before connecting the power supply to the kit, be sure that the jumpers and connecting Cords are connected correctly as per experiment.

In **AMPLITUDE MODULATION TRANSMITTER TRAINER KIT (ACL-AM) & AMPLITUDE DEMODULATION RECEIVER TRAINER KIT (ACL-AD)** modules signals are to be monitored with an oscilloscope as explained in the manual. The scope input channel should be ac coupled, unless otherwise indicated. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180Ω resistance in series with normal oscilloscope probe.

A frequency counter should be used for all frequency measurements. Use the trimming tool, supplied with the kits, for trimming inductors. Never use a screwdriver, as this may damage the inductors core. Also take care not to turn any inductors core past its end stop, as they may also result in damage.

This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer kit, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kit is liable to malfunction/under-perform if it is not operated under following conditions;

- **Ambient temperature** : from 0 to 45 ° C.
- **Relative humidity** : from 20 to 80 %.

Avoid any immediate/significant change of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting Cords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions a bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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INTRODUCTION

ACL-AM & ACL-AD covers the principal of Amplitude Modulation with Single Side Band and Double Side Band Modulation & Demodulation techniques. This system offers choice of various Modulators and Demodulators, While working on these kits the students learn to apply the concept of signal generation, Audio preamplifier, RF amplifier, voltage controlled oscillator, Balanced Modulators, IF filters, Ceramic filters, Envelope Detector, Mixers, Output amplifier, AGC, Selective amplifier and Audio amplifier, which allows student to compare between different methods of Amplitude Modulation and Demodulation.

The range of available modules enables the gradual development of the program starting from the study of single circuits (e.g. the AM modulator) up to the analysis of complete systems. These Modulated signals may be relayed to **ACL-AD** kit either by means of screened cable or from the on-board Antenna.

Utmost care has been laid in the design and quality control of all circuits, to ensure the repeatability of the results of the experiments. The trainer kits are provided with extensive test points allowing students to investigate the various aspects of system operation. Each kit is equipped with correlated courseware to guide students through the application and demonstration of communication techniques and concepts. In addition circuit description manual guides the students through working of each blocks and circuits wise explanation of each kit.

The system is highly innovative from a technological as well as an educational point of view. The modules are used as “basic blocks” to build up in a flexible way the different communication systems, and to examine all the peculiar operating characteristics.

The only external instruments required are a power supply unit and an oscilloscope. The remaining circuits and instruments (Function generator, RF, Filters, Microphone, Loudspeaker, etc.) are already included in the modules.

TECHNICAL SPECIFICATION

ACL-AM: AMPLITUDE MODULATION TRANSMITTER KIT

SINE WAVE GENERATOR

Frequency : 100 Hz to 10 KHz
Amplitude : 0 to 2 volt

AUDIO PREAMPLIFIER : Audio Preamplifier with Microphone and with adjustable output level

VOLTAGE CONTROLLED OSCILLATOR (VCO)

Output Signal	: Sine wave
Frequency Range	: 1) 400 KHz to 500 KHz 2) 400 KHz to 1500 KHz
Amplitude	: Adjustable from 0 to 2 Vp-p
Output Impedance	: 50 ohm

AM/DSB/SSB MODULATOR

BALANCED MODULATOR 1

Modulation	: Amplitude Modulation Double Side Band Single Side Band (USB & LSB)
Carrier Input	: 1 – 1000 KHz
Modulating Input	: 0.1 – 100 KHz
Carrier Null	: Adjustable
Output Amplitude	: Adjustable

BALANCED MODULATOR 2

Modulation

	Double Side Band
	Single Side Band (USB & LSB)
Carrier Input	: 1 MHz
Modulating Input	: 400 – 500 KHz
Carrier Null	: Adjustable
Output Amplitude	: Adjustable

CERAMIC FILTER

Center Frequency : 460 KHz
Bandwidth : 10 +/- 3 KHz

COLPITT'S OSCILLATOR

Frequency : 1 MHz
Amplitude : 0 – 2 Vp-p

BAND PASS FILTER

Center Frequency : 1.455 MHz
Bandwidth : 10 +/- 3 KHz

OUTPUT AMPLIFIER

Range : 600 KHz to 1650 KHz
Gain : Adjustable

ANTENNA : MW Coil

INTERCONNECTION

With the help of 2mm connecting cords provided.

TEST POINTS

Various test points are provided onboard to observe signals on oscilloscope.

SWITCH FAULTS

Switch fault section provides 4 switches. These switches can be used to simulate fault condition in various parts of the circuit.

POWER SUPPLY : GND, +5V, + 12V, - 12V

ACL-AD: AMPLITUDE DEMODULATION RECEIVER KIT**SUPER HETERODYNE RECEIVER
WITH GANG TUNING****RF AMPLIFIER**

Range : 600 KHz to 1650 KHz
Gain : Adjustable

LOCAL OSCILLATOR

Output Signal : Sine wave
Frequency : 900 KHz to 2.1 MHz
Amplitude : Adjustable from 0 to 2Vp-p

FREQUENCY CONVERTER (MIXER)

Dual gate MOSFET
Inputs : Local Oscillator and RF Signal
Output Frequency : 455 KHz adjustable
FI Filter : Dual Tune LC

BEAT FREQUENCY OSCILLATOR

Frequency : Adjustable from 400 to 500 KHz with Tuned frequency of 457 KHz
Level : 0 to 2 Vp-p

IF AMPLIFIER

Number of stages : 2
Central Frequency : 455 KHz
Load impedance : Variable R-L-C
Gain : 32 dB

AUTOMATIC GAIN CONTROL

Output can be controlled in case of maximum input signal.

ENVELOPE DETECTOR : Detection of the positive and negative Envelope with Variable RC Filter for studying the effect of noise

PRODUCT DETECTOR
Operating Frequency : Adjustable from 400 KHz to 500 KHz SSB
Input amplitude : 1 V p-p

AUDIO AMPLIFIER
Audio Output : Amplifier with headphones
Gain : Adjustable

RECEIVING MEDIA : MW Coil

INTERCONNECTION
 With the help of 2mm connecting cords provided

TEST POINTS
 Various test points are provided onboard to observe signals on oscilloscope.

SWITCH FAULTS
 Switch fault section provides 4 switches. These switches can be used to simulate fault condition in various parts of the circuit.

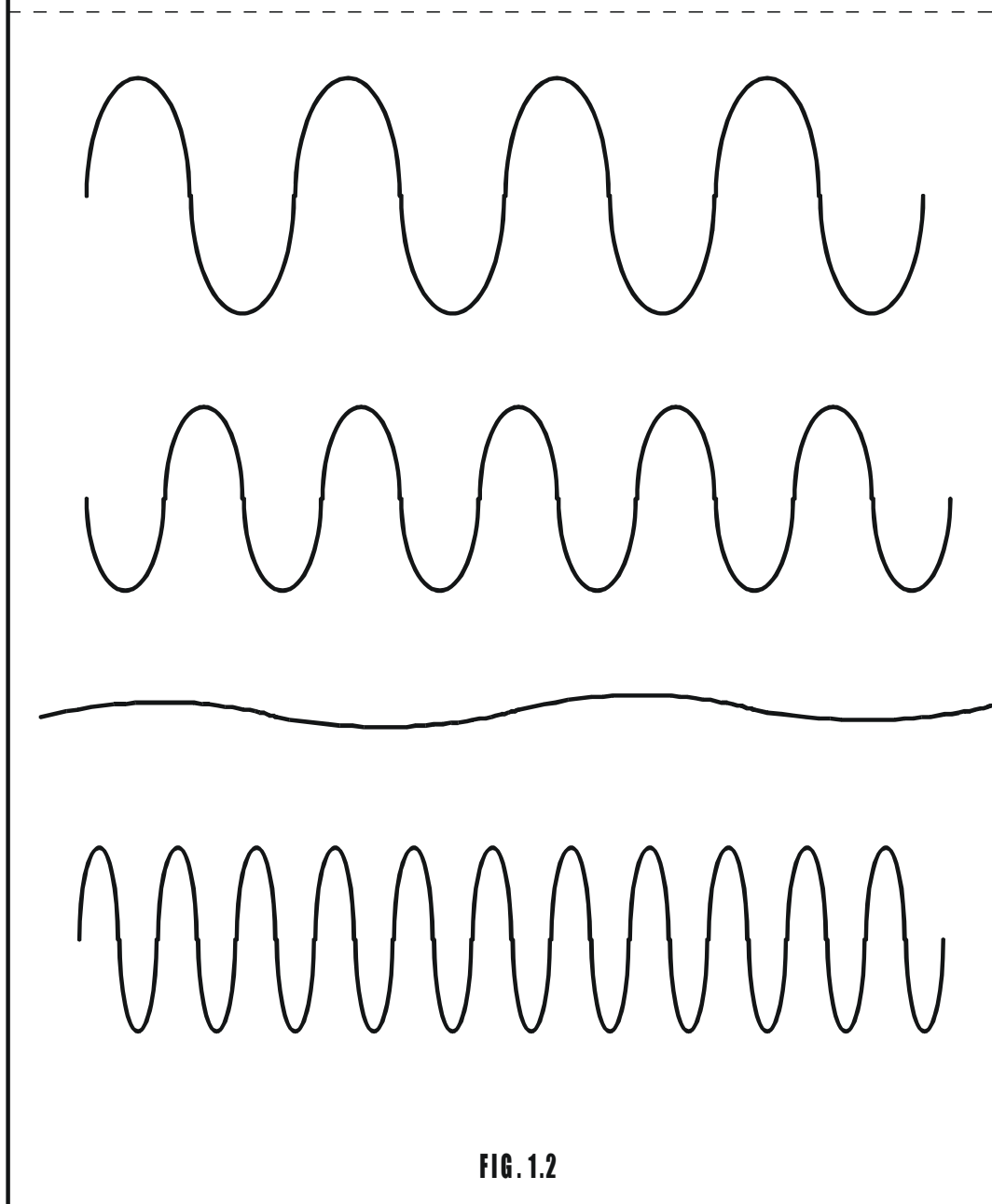
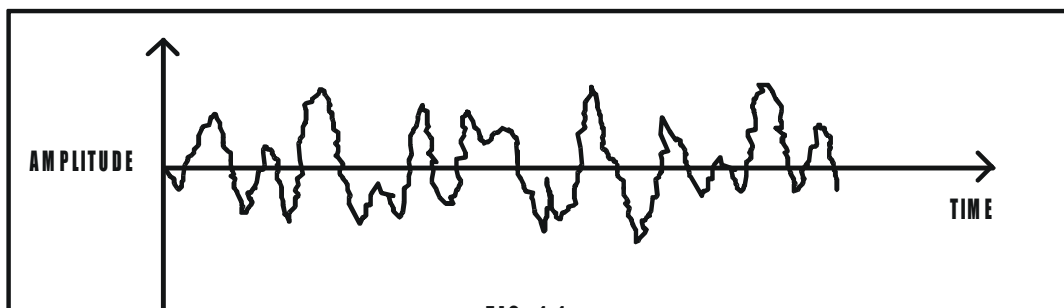
POWER SUPPLY : GND, +5V, + 12V, -12V

ACCESSORIES

ACL-AM			Quantity
1	Mini Links PCS2 Patch Cord	Male To Male	7
2	Short Links PCS2 Patch Cord	Male To Male	3
3	Long Links PCS2 Patch Cord	Male TO Male	2
4	Headphone		1
5	ACL-AM & AD Experimental Manual		1
6	Power supply		1
7	Power Cord		1

ACL-AD			
1	Mini Links PCS2 Patch Cord	Male To Male	10

EXPERIMENT NO.1



EXPERIMENT NO: 1

NAME:-

DOUBLE SIDE BAND AM GENERATION

OBJECTIVE:-

- A. To study the operation of a DSB AM Modulator
- B. To calculate the modulation index of an AM modulated wave

THEORY:-

1.1 THE FREQUENCY COMPONENTS OF THE HUMAN VOICE:

When we speak, we generate sound that is very complex and changes continuously so at a particular instant of time the waveform may appear as shown in FIG.1.1 below.

However complicated the waveform looks, we can show that it is made of many different sinusoidal signals added together.

To record this information we have a choice of three methods. The first is to show the original waveform as we did in FIG.1.1.

The second method is to make a list of all the separate sinusoidal waveforms that were contained within the complex waveform (these are called 'components' or 'frequency components'). This can be seen in FIG.1.2. Only four of the components of the audio signal in FIG.1.1 are shown. The actual number of components depends on the shape of the signal being considered and could be a hundred or more if the waveform was very complex.

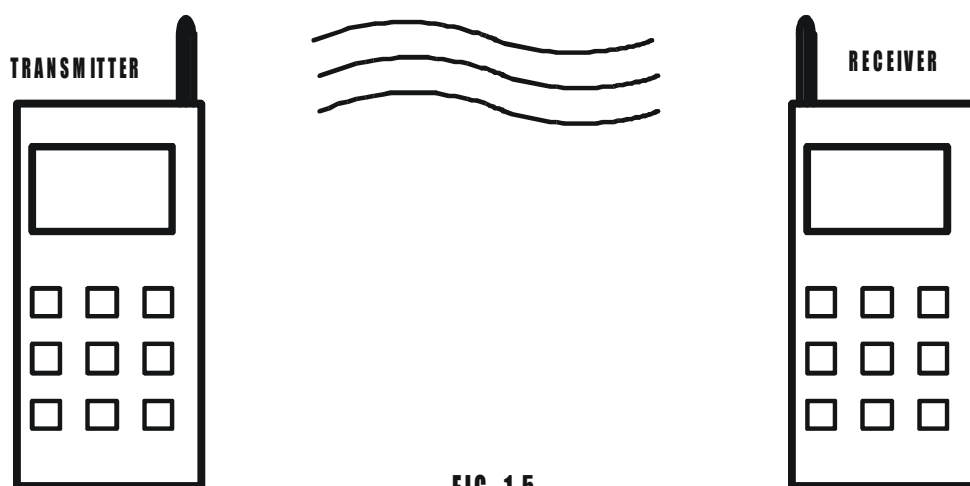
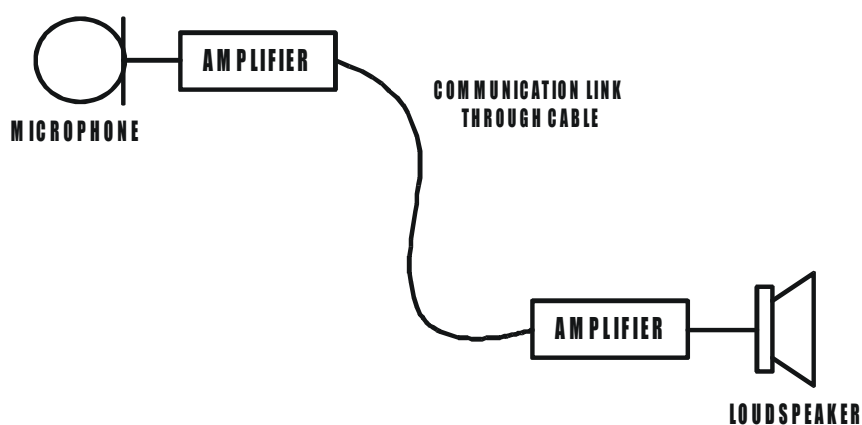
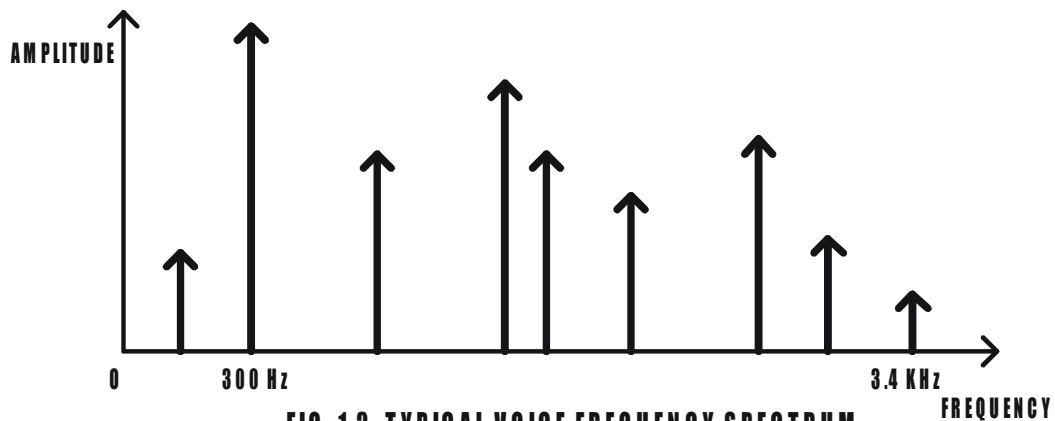
The third way is to display all the information on a diagram. Such a diagram shows the frequency spectrum. It is a graph with amplitude plotted against frequency. Each separate frequency is represented by a single vertical line, the length of which represents the amplitude of the sine wave. Such a diagram is shown in FIG.1.3. Note that nearly all speech information is contained within the frequency range of 300Hz to 3.4kHz.

1.2 SIMPLE COMMUNICATION SYSTEM:

Once we are out of shouting range of another person, we must rely on some communication system to enable us to pass information.

The only essential parts of any communication system are a transmitter, a communication link and a receiver. In the case of speech, this can be achieved by a length of cable with a microphone and an amplifier at one end and a loudspeaker and an amplifier at the other as shown in FIG.1.4.

For long distances or for when it is required to send signals to many destinations at the same time, it is convenient to use a radio communication system.



1.3 THE FREQUENCY PROBLEM:

To communicate by radio over long distances we have to send a signal between two antennas, one at the sending or transmitting end and the other at the receiver as shown in FIG.1.5.

The frequencies used by radio systems for AM transmissions are between 200KHz and 25MHz.

A typical radio frequency of, say, 1MHz is much higher than the frequencies present in the human voice.

We appear to have two incompatible requirements. The radio system uses frequencies like 1MHz to transmit over long distances, but we wish to send voice frequencies between 300Hz and 3.4KHz, which are quite impossible to transmit by radio signals.

1.4 AMPLITUDE MODULATION:

In Amplitude Modulation the amplitude of high frequency sine wave (carrier) is varied in accordance with the instantaneous value of the modulating signal. Refer FIG.1.6.

Consider a sine signal $v_m(t)$ with frequency f (FIG.1.7).

$$V_m(t) = B \cdot \sin(2\pi f \cdot t)$$

And another sine signal $v_c(t)$ is called modulating signal, the signal $v_c(t)$ is called carrier signal.

$$V_c(t) = A \cdot \sin(2\pi F \cdot t)$$

The signal $v_m(t)$ is called modulating signal; the signal $v_c(t)$ is called carrier signal.

Vary the amplitude of the carrier $v_c(t)$ adding the modulating signal $v_m(t)$ to A . You obtain a signal $v_M(t)$ amplitude modulated, which can be expressed by:

$$v_M(t) = [A + k \cdot B \cdot \sin(2\pi f \cdot t)] \cdot \sin(2\pi F \cdot t) = A \cdot [1 + m \cdot \sin(2\pi f \cdot t)] \cdot \sin(2\pi F \cdot t)$$

With k = constant of proportionality.

Percentage modulation signal is defined as the value:

$$m = \frac{K \cdot B}{A} \cdot 100$$

With reference to FIG.1.7, the modulation index m can be calculated in this way

$$m = \frac{H-h}{H+h} \cdot 100\%$$

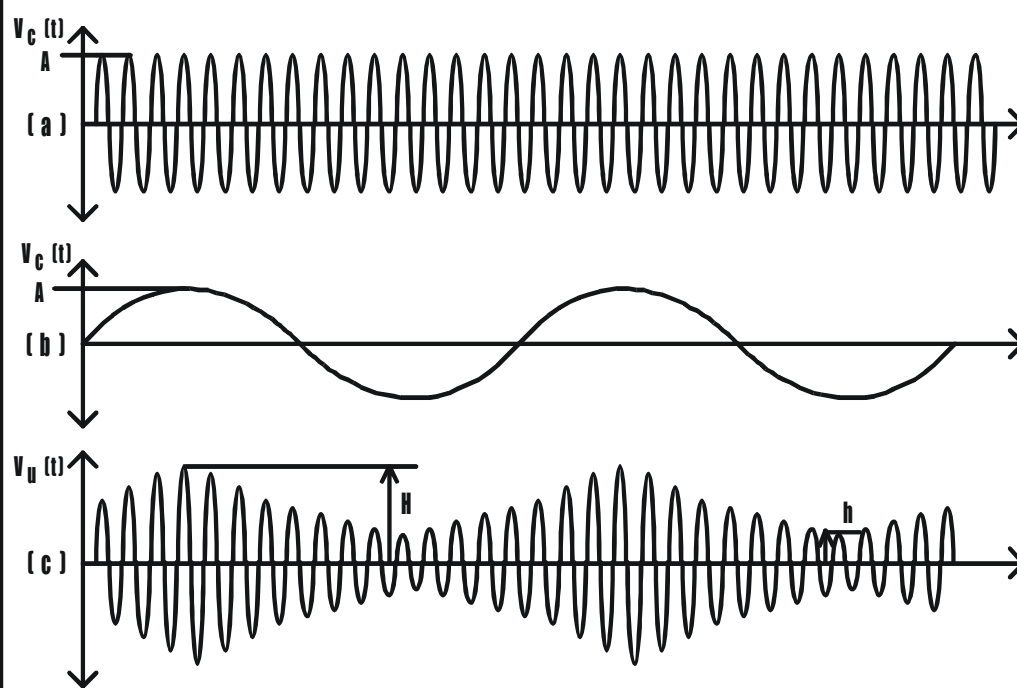
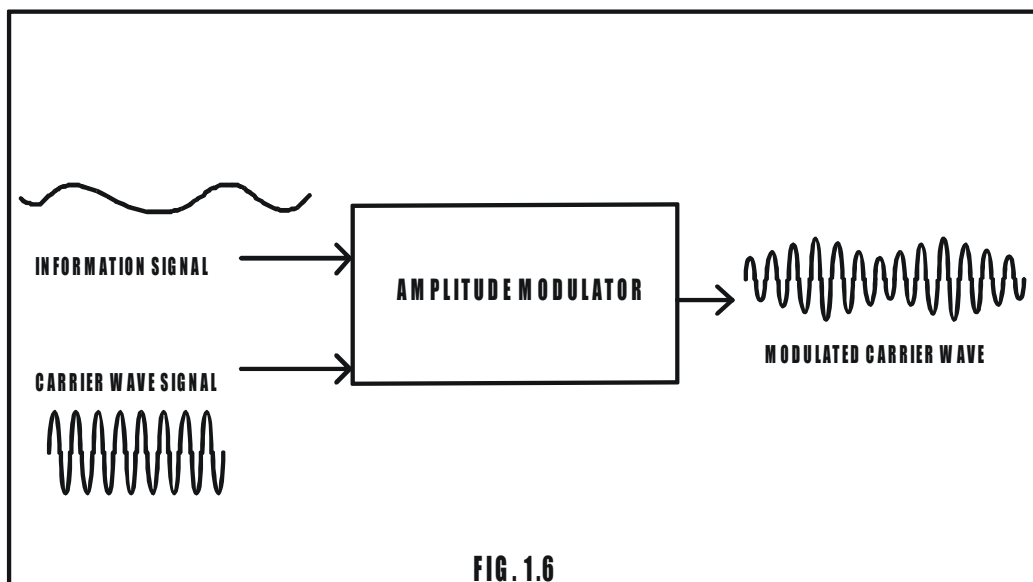


FIG. 1.7 A) CARRIER SIGNAL B) MODULATING SIGNAL C) MODULATED SIGNAL

1.4.1 SPECTRUM OF THE MODULATED SIGNAL:

With simple trigonometric passages, the relation expressing the modulated signal v_M becomes:

$$v_M(t) = A \cdot \sin(2\pi F \cdot t) + m \cdot \frac{A}{2} \cdot \cos[(2\pi(F-f) \cdot t)] - m \cdot \frac{A}{2} \cdot \cos[(2\pi(F+f) \cdot t)]$$

From which we can deduce that the signal modulated in amplitude by a sine modulator consists of three sine components:

$A \cdot \sin(2\pi F \cdot t)$	Carrier
$m \cdot \frac{A}{2} \cdot \cos[(2\pi(F-f) \cdot t)]$	Lower side band
$m \cdot \frac{A}{2} \cdot \cos[(2\pi(F+f) \cdot t)]$	Upper side band

Particularly effective is the representation of the modulated signal into an Amplitude/frequency diagram. FIG.1.8 reports the different components of the AM signal, in the amplitude/frequency diagram as well as the amplitude/time diagram.

1.4.2 POWER OF THE MODULATED SIGNAL:

The total power of an AM signal is the sum of the contributes related to the carrier and to the upper and lower side bands.

Considering a sine modulating signal and a load resistance R , the different components supply the following powers:

$P_C = A^2 / 2 \cdot R$	Power associated to the carrier
$P_L = (m \cdot A)^2 / 8 \cdot R$	Power associated to the lower side band
$P_U = (m \cdot A)^2 / 8 \cdot R$	Power associated to the upper side band

It is important to note that:

The power associated to the carrier is fixed and does not depend on the modulation. The power associated to each side band depends on the index of modulation and reaches at maximum 25% of the power of the carrier (50% of the two side bands together).

1.4.3 NON-SINUSOIDAL MODULATING SIGNAL SPECTRUM

Consider a modulating signal not constituted by a single sine wave, but a generic signal with frequency spectrum ranging between f_1 and f_2 . With the amplitude modulation this spectrum is moved over and under the carrier (FIG.1.9).

It is evident that the larger is the spectrum of the modulating signal, the larger is the band BW occupied by the modulating signal. BW results equal to the double of the modulating signal:

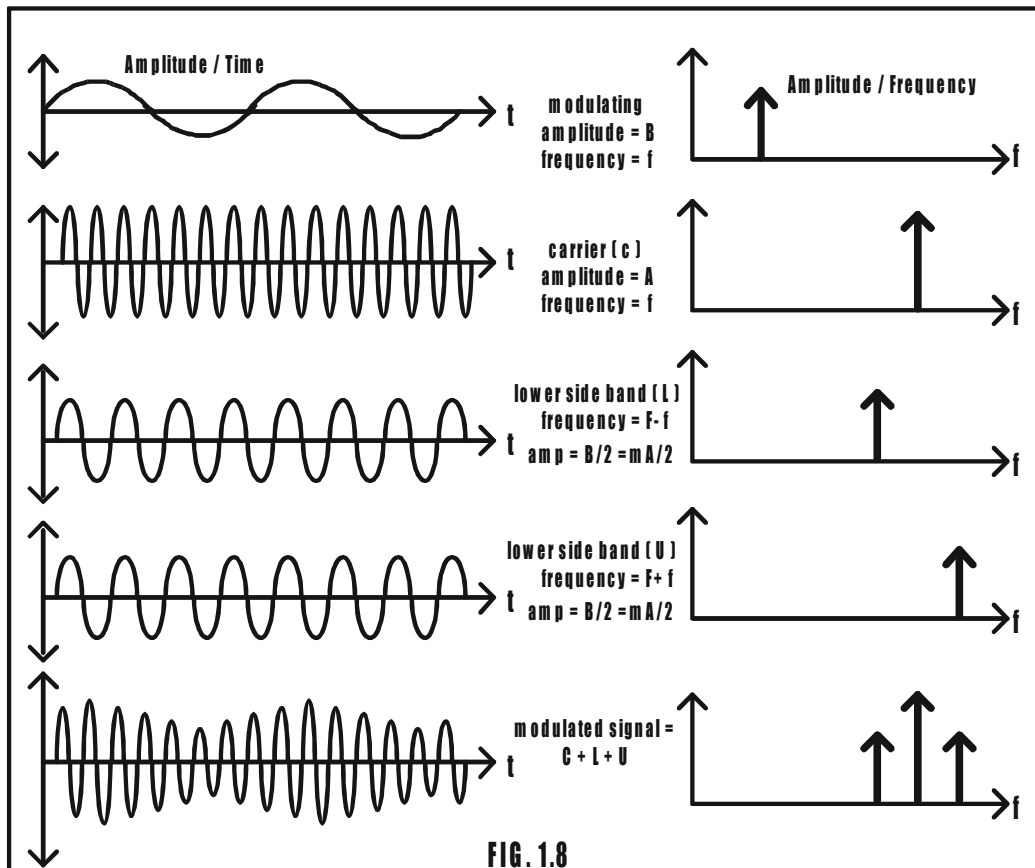
$$BW = 2 \cdot f_2.$$


FIG. 1.8

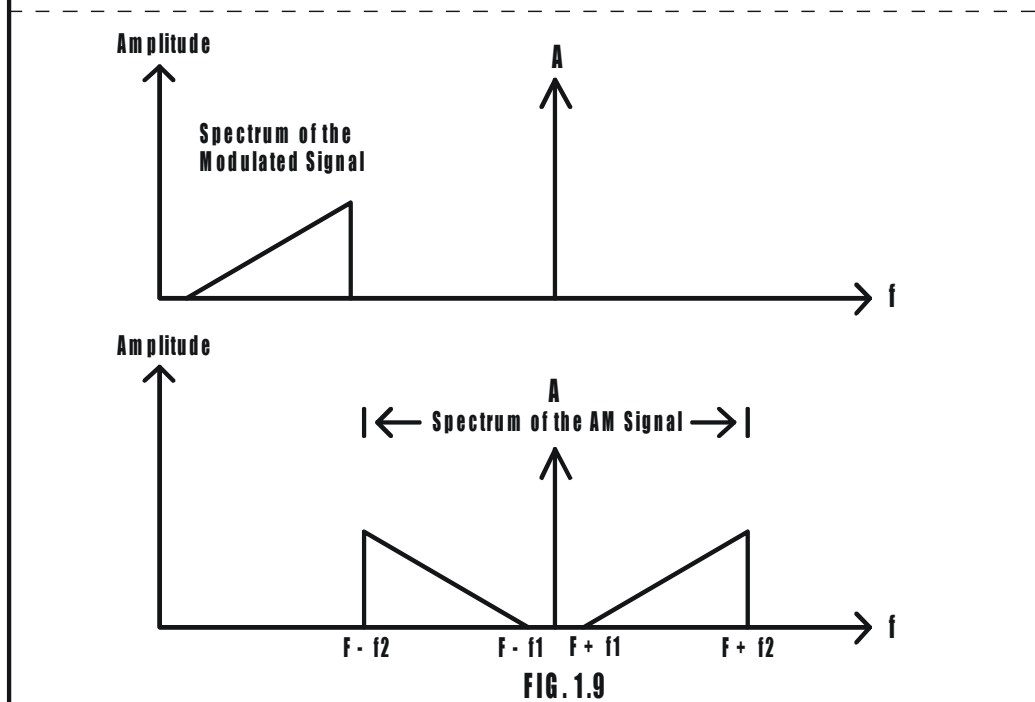


FIG. 1.9

1.4.4 AMPLITUDE MODULATION GENERATION:

The circuits used to generate an amplitude modulation must vary the amplitude of a high frequency signal (carrier) as function of the amplitude of a low frequency signal (modulating signal).

In an AM transmitter we speak of:

High-level modulation, if the modulation is carried out directly in the last power stage, which is generally an amplifier in class C.

Low frequency modulation, when the modulation is carried out by stages, which are before the final power amplifier.

Semiconductor devices can be used, in case of low power, when the required power is high.

In the circuit used for the exercises, the amplitude modulation is generated by a differential amplifier, whose gain is varied by the modulating signal. This circuit, contained in the integrated circuit LM1496, can be used also to generate the amplitude modulation with suppressed carrier.

1.5 SIDEBANDS:

If the information signal consists of a range of frequencies, each separate frequency will create its own upper side frequency and lower side frequency.

As an example, let us imagine that a carrier frequency of 1MHz is amplitude modulated by an information signal consisting of frequencies 500Hz, 1.5KHz and 3KHz.

As each modulating frequency produces its own upper and lower side frequency there is a range of frequencies present above and below the carrier frequency. All the upper side frequencies are grouped together and referred to as the upper sideband (USB) and all the lower side frequencies form the lower side and (LSB).

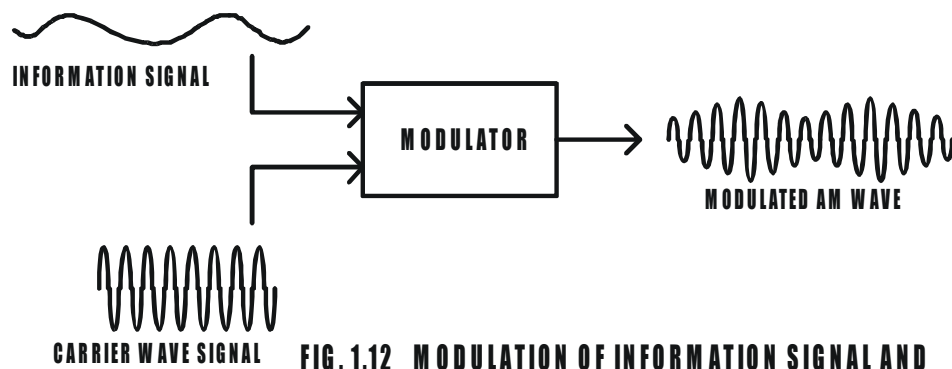
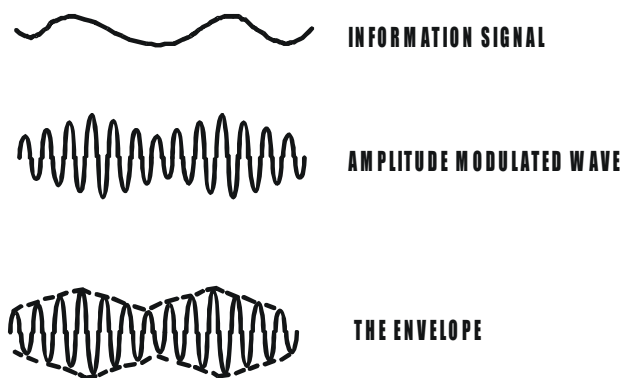
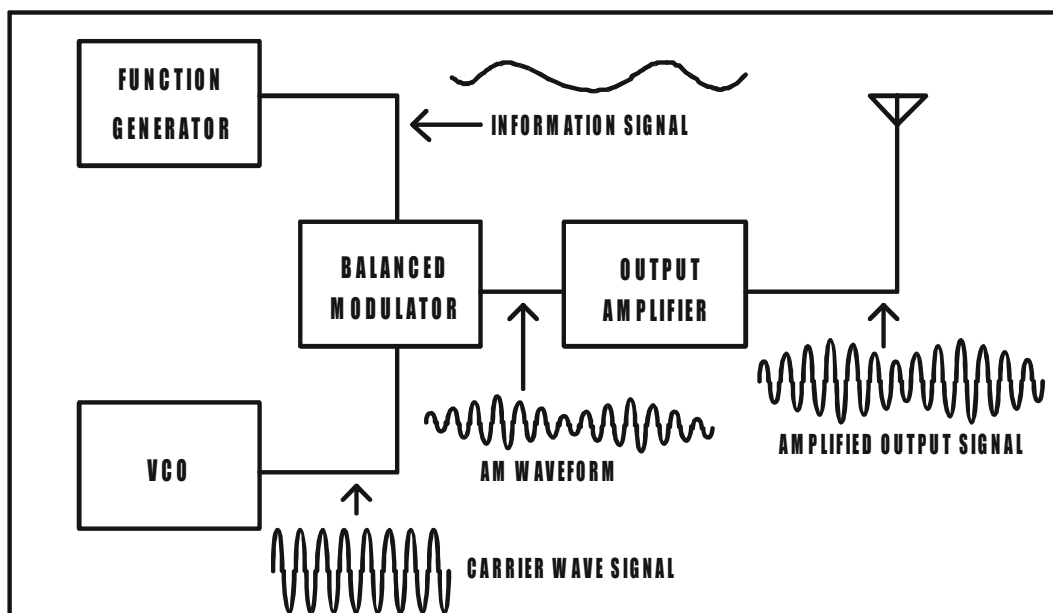
This amplitude modulated wave would have a frequency spectrum as shown in FIG.1.8.

Because the frequency spectrum of the AM waveform contains two sidebands, this type of amplitude modulation is often called a double-sideband transmission or DSB.

1.5.1 POWER IN THE SIDEBANDS:

1000 is for the carrier power and 0.6 for the modulation depth. We could have used the figure 60% instead of 0.6 but this way makes the calculations slightly easier.

$$\begin{aligned}
 \text{Total power} &= (1000) \left[1 + \frac{0.6^2}{2} \right] \\
 \text{Total power} &= (1000) \left[1 + \frac{0.36}{2} \right] \\
 &= 1000 \times (1 + 0.18) \\
 &= 1000 \times 1.18 \\
 &= 1180W
 \end{aligned}$$



The carrier power was 1000W and the total power of the modulated wave is 1180W so the two sidebands must, between them, contain the other 180W. The power contained in the upper and lower sidebands is always equal and so each must contain $\frac{180}{2} = 90W$

The greater the depth of modulation, the greater is the power contained within the sidebands. The highest usable depth of modulation is 100% (above this the distortion becomes excessive).

Since at least twice as much power is wasted as is used, this form of modulation is not very efficient when considered on a power basis. The good news is that the necessary circuits at the transmitter and at the receiver are simple and in expensive to design and construct.

1.6 DSB TRANSMITTER THE DOUBLE SIDEBAND TRANSMITTER:

The transmitter circuits produce the amplitude modulated signals, which are used to carry information over the transmission path to the receiver. The main parts of the transmitter are shown in FIG.1.10.

In FIG.1.10 & FIG.1.11, we can see that the peak-to-peak voltages in the AM waveform increase and decrease in sympathy with the audio signal.

To emphasize the connection between the information and the final waveform, a line is sometimes drawn to follow the peaks of the carrier wave as shown in FIG.1.11. This shape, enclosed by a dashed line in our diagram, is referred to as an 'envelope' or a 'modulation envelope'. It is important to appreciate that it is only a guide to emphasize the shape of the AM waveform.

We will now consider the action of each circuit as we follow the route taken by the information that we have chosen to transmit.

The first-task is to get hold of the information to be transmitted.

1.6.1 THE INFORMATION SIGNAL:

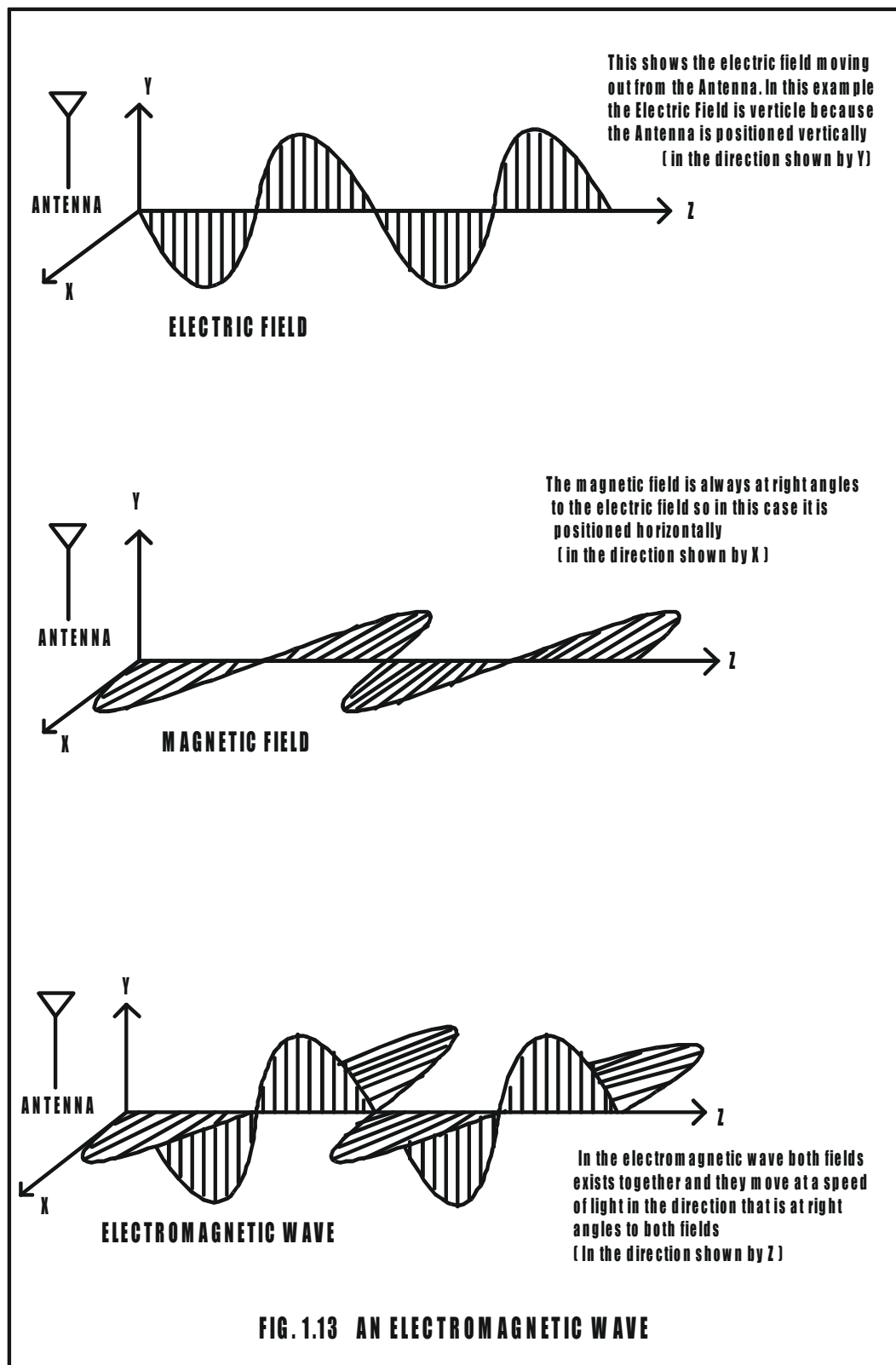
In test situations it is more satisfactory to use a simple sinusoidal information signal since its attributes are known and of constant value. We can then measure various characteristics of the resultant AM waveform, such as the modulation depth. Such measurements would be very difficult if we were using a varying signal from an external source such as a broadcast station.

The next step is to generate the carrier wave.

1.6.2 THE CARRIER WAVE:

The carrier wave must meet two main criteria.

First it should be of a convenient frequency to transmit over the communication path in use. Radio link transmissions are difficult to achieve at frequencies less than 15KHz and few radio links employ frequencies above 10GHz. Outside of this range the cost of the equipment increases rapidly with very few advantages.



Remember that although 15KHz is within the audio range, we cannot hear the radio signal because it is an electromagnetic wave and our ears can only detect waves, which are due to changes of pressure.

The second criteria are that the carrier wave should also be a sinusoidal waveform.

Can you see why a sinusoidal signal contains only a single frequency and when modulated by a single frequency, will give rise to just two side frequencies, the upper and the lower side frequencies. However, if the sine wave were to be a complex wave containing many different frequencies, each separate frequency component would generate its own side frequencies. The result is that the overall bandwidth occupied by the transmission would be very wide and on the radio, would cause interference with the adjacent stations.

1.6.3 THE MODULATOR:

There are many different designs of amplitude modulator. They all achieve the same result. The amplitude of the carrier is increased and decreased in sympathy with the incoming information signal as we saw in FIG.1.12.

The signal is now nearly ready for transmission.

If the modulation process has given rise to any unwanted frequency components then a band pass filter can be employed to remove them.

1.6.4 OUTPUT AMPLIFIER (OR POWER AMPLIFIER)

This amplifier is used to increase the strength of the signal before being passed to the antenna for transmission. The output power contained in the signal and the frequency of transmission are the two main factors that determine the range of the transmission.

1.6.5 THE ANTENNA

An electromagnetic wave, such as a light ray, consists of two fields, an electric field and a magnetic field. These two fields are always at right angles to each other and move in a direction, which is at right angles to both the magnetic, and the electric fields, this is shown in FIG.1.13.

The antenna converts the power output of the Output Amplifier into an electromagnetic wave.

How does it do this?

The output amplifier causes a voltage to be generated along the antenna thus generating a voltage difference and the resultant electric field between the top and bottom. This causes an alternating movement of electrons on the transmitting antenna, which is really an AC current. Since an electric current always has a magnetic field associated with it, an alternating magnetic field is produced.

The overall effect is that the output amplifier has produced alternating electric and magnetic fields around the antenna. The electric and magnetic fields spread out as an electromagnetic wave at the speed of light (3×10^8 meters per second).

For maximum efficiency the antenna should be of a precise length. The optimum

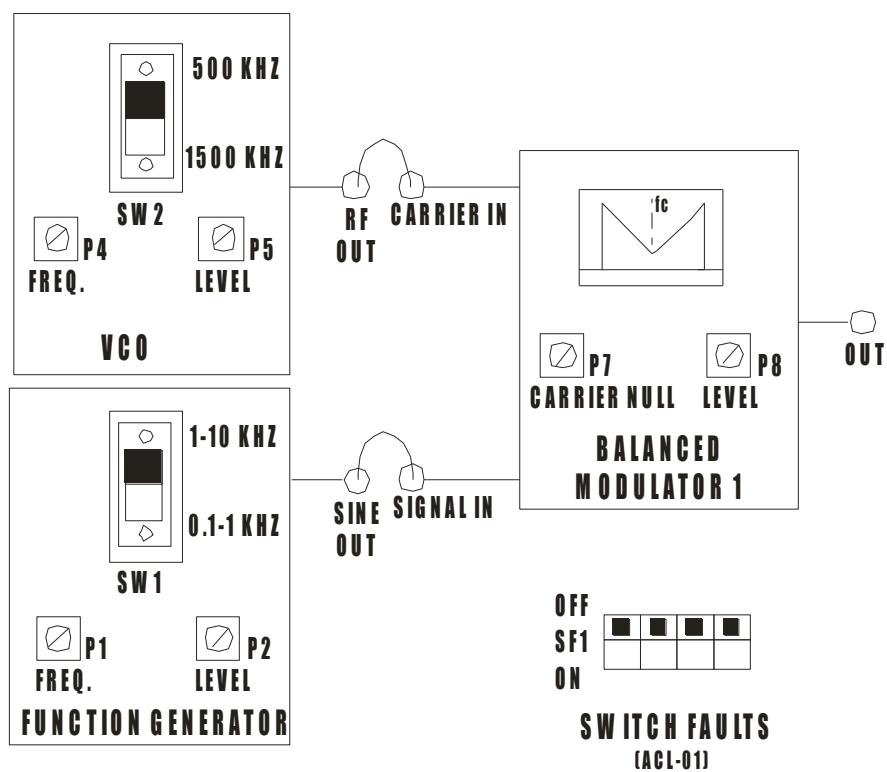


FIG.1.14 BLOCK DIAGRAM FOR STUDY OF DSB AM GENERATION

size of antenna for most purposes is one having an overall length of one quarter of the wavelength of the transmitted signal.

This can be found by:

$$\frac{\lambda}{f} = \frac{v}{f} \quad \text{where } v = \text{speed of light, } \lambda = \text{wavelength and } f = \text{frequency in Hertz}$$

In the case of the ACL-AM, the transmitted carrier is upto 1.5MHz and so if we set it to 1MHz the ideal length of antenna is:

$$\lambda = \frac{3 \times 10^8}{1 \times 10^6}$$

$$\lambda = 300\text{m}$$

One quarter of this wavelength would be 75 meters (about 245 feet).

We can now see that the antenna provided on the ACL-AM is necessarily less than the ideal size!

1.6.6 POLARIZATION:

If the transmitting antenna is placed vertically, the electrical field is vertical and the magnetic field is horizontal (as seen in FIG.1.13). If the transmitting antenna is now moved by 90° to make it horizontal, the electrical field is horizontal and the magnetic field becomes vertical. By convention, we use the plane of the electric field to describe the orientation, or polarization, of the em (electromagnetic) wave. A vertical transmitting antenna results in a vertically polarized wave, and a horizontal one would result in a horizontally polarized em wave.

A. TO STUDY THE OPERATION OF A DSB AM MODULATOR.

EQUIPMENT:-

- Modules ACL-AM
- Power supply
- 20MHz Oscilloscope.
- Connecting Links.
- Frequency counter.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG. 1.14 & Carry out the following connections.
Connect **SINE OUT** post of **FUNCTION GENERATOR SECTION** (ACL-AM) to the i/p of Balance Modulator1 (ACL-AM) **SIGNAL IN** Post.
2. Connect o/p of **VCO** (ACL-AM) **RF OUT** post to the input of Balance modulator1 **CARRIER IN** post (ACL-AM).
3. Connect the power supply with proper polarity to the kit ACL-AM & ACL-AD, while connecting this; ensure that the power supply is OFF.
4. Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR:** LEVEL about 0.5Vpp; FREQ. about 1 KHz.

- **VCO:** LEVEL about 1 Vpp; FREQ. about 450 KHz, Switch on 500KHz.
 - **BALANCED MODULATOR1:** CARRIER NULL completely rotated
 - Clockwise or counter clockwise, so as to “unbalance” the modulator and to obtain an AM signal with not suppressed carrier across the output; OUT LEVEL in fully clockwise.
5. Connect the oscilloscope to the inputs of the modulator post (SIG and CAR) and detect the modulating signal and the carrier signal (FIG. 1.16 A/B).
 6. Move the probe from post SIG to post OUT (output of the modulator), where signal modulated in amplitude is detected (FIG. 1.16 C). Note that the modulated signal envelope corresponds to the wave form of the DSB AM modulating signal.
 7. Vary the amplitude of the modulating signal and check the 3 following conditions: Modulation percentage lower than the 100% (FIG. 1.16 C), equal to the 100% (FIG. 1.16 D), superior to 100% (over modulation, FIG. 1.16 E).
 8. Vary the frequency and amplitude of the modulating signal, and check the corresponding variations of the modulated signal.
 9. Vary the amplitude of the modulating signal and note that the modulated signal can result saturation or over modulation.

SWITCH FAULTS

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R57) in parallel with the bias resistor. The balance modulator 1 output gets distorted.

B. TO CALCULATE THE MODULATION INDEX OF AN AM MODULATED WAVE.

EQUIPMENTS:-

- Modules ACL-AM
- Power supply
- 20MHz Oscilloscope
- Connecting Links

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG. 1.14 & Carry out the following connections.
Connect **OUT** post of **FUNCTION GENERATOR** (ACL-AM) to the i/p of Balance Modulator 1 (ACL-AM) **SIGNAL IN** Post.
2. Connect o/p of **VCO** (ACL-AM) **OUT** post to the input of Balance modulator1 **CARRIER IN** post (ACL-AM).
3. Connect the power supply with proper polarity to the kit ACL-AM, while connecting this; ensure that the power supply is OFF.
4. Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR:OUT** post LEVEL about 0.5Vpp; FREQ. About 1 KHz.
 - **VCO:** LEVEL about 1 V p-p; FREQ. about 450 KHz, Switch on 500KHz.

- **BALANCED MODULATOR 1:** CARRIER NULL completely rotated clockwise or counter clockwise, so as “unbalance” the modulator and to obtain an AM signal with not suppressed carrier across the output; OUT LEVEL in fully clockwise.
5. Obtain the AM modulated wave as shown in FIG. 1.15.
 6. Using the oscilloscope measure from the waveforms of the amplitude B of the modulated signal at post **OUT** of balance modulator (ACL-AM), amplitudes H and h of the modulated signal and the amplitude C of the envelope of the modulated signal at post **OUT** of balance modulator 1(ACL-AM).
 7. Calculate the constant k of the modulator, equal to: $k = C/B$ You find a value a little over 1.
 8. Calculate the amplitude A of the carrier, equal to:

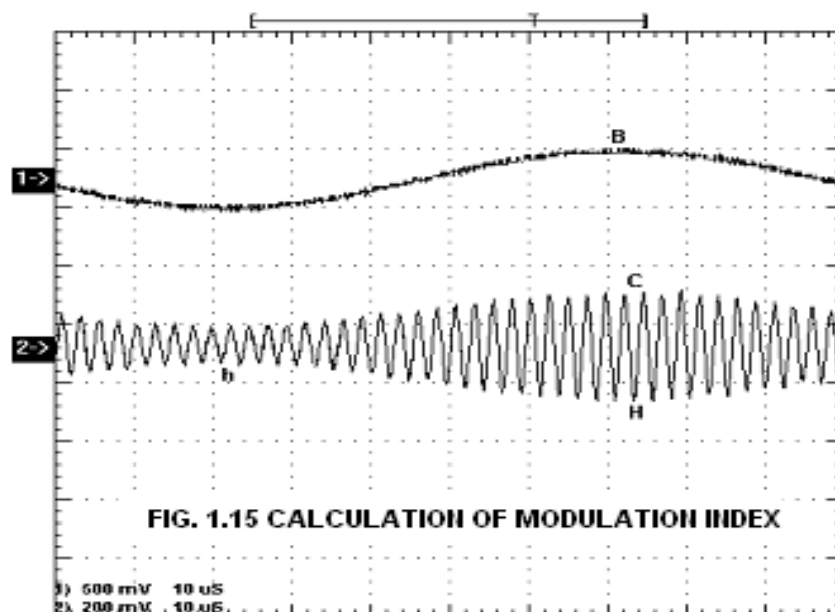
$$A = \frac{H+h}{2}$$
 9. Calculate the percentage index of modulation m, equal to:

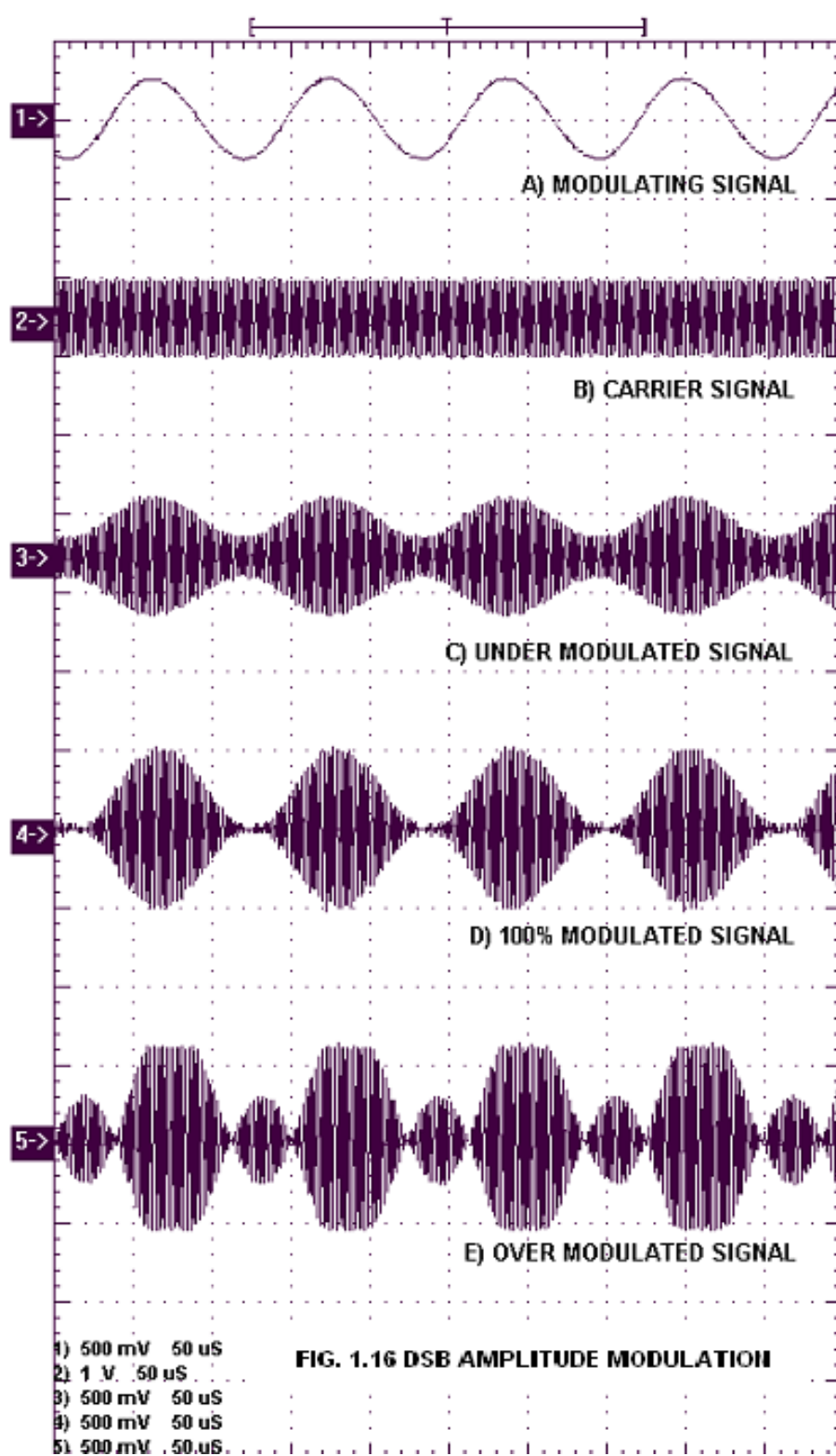
$$m = \frac{H-h}{H+h} \bullet 100\%$$

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R57) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.





EXPERIMENT NO.2

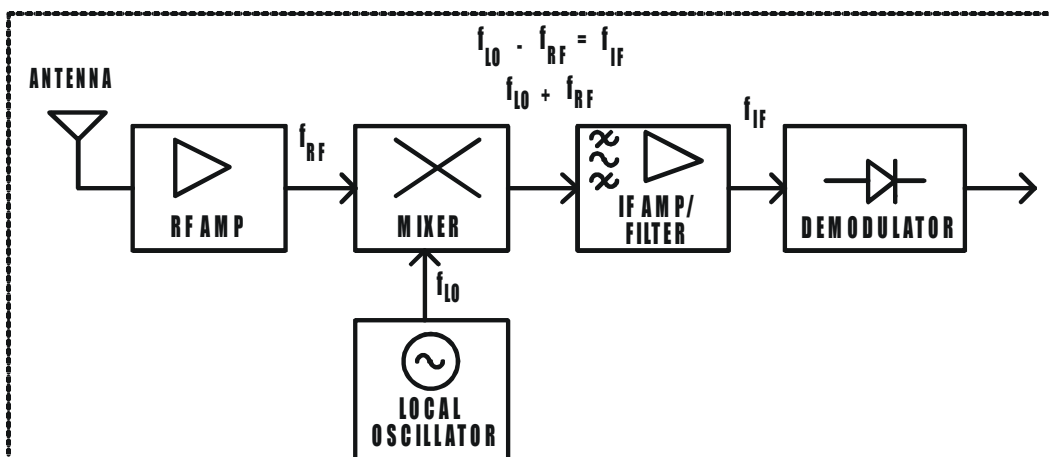


FIG. 2.1 RADIO RECEIVER BLOCK DIAGRAM

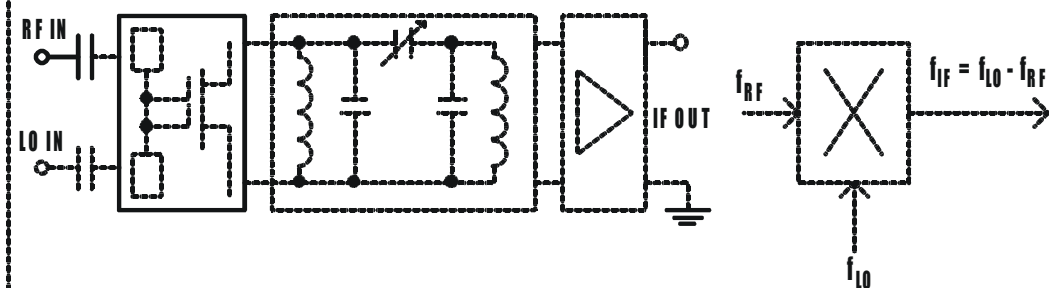


FIG. 2.2 BLOCK DIAGRAM OF A MIXER WITH DUAL GATE MOSFET

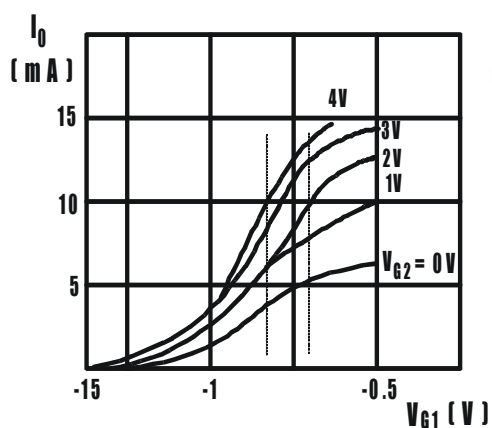
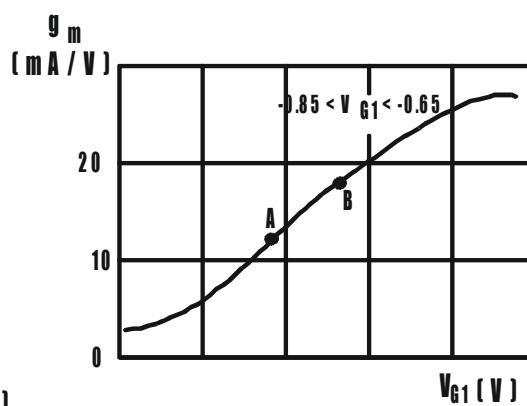


FIG. 2.3 CHARACTERISTIC CURVES

FIG 2.4
TRANSCONDUCTANCE g_m

EXPERIMENT NO: 2

NAME:-

STUDY OF DOUBLE SIDEBAND AM RECEPTION.

OBJECTIVE:-

- A. STUDY OF DOUBLE SIDEBAND AM RECEPTION USING ENVELOPE DETECTOR VIA CABLE.
- B. STUDY OF DOUBLE SIDEBAND AM RECEPTION USING ENVELOPE DETECTOR VIA ANTENNA.

THEORY:-

2.1 THE DSB RECEIVER

The am wave from the transmitting antenna will travel to the receiving antenna, carrying the information with it.

We will continue to follow our information signal as it passes through the receiver.

2.2 THE RECEIVING ANTENNA

The receiving antenna operates in the reverse mode to the transmitter antenna. The electromagnetic wave strikes the antenna and generates a small voltage in it.

Ideally, the receiving antenna must be aligned to the polarization of the incoming signal. So generally, signal from a vertical transmitting antenna will be received best by using a vertical receiving antenna.

The actual voltage generated in the antenna is very small – usually less than 50 milli volts and often only a few micro volts. The voltage supplied to the loudspeaker at the output of the receiver is up to ten volts.

We clearly need a lot of amplification.

2.3 THE RADIO FREQUENCY (RF) AMPLIFIER

The antenna not only provides very low amplitude input signals but it picks up all available transmissions at the same time. This would mean that the receiver output would include all the various stations on top of each other, which would make it impossible to listen to any one transmission.

The receiver circuits generate noise signals, which are added to the wanted signals. We hear this as a background hiss and are particularly noticeable if the receiver is tuned between stations or if a weak station is being received.

The RF amplifier is the first stage of amplification. It has to amplify the incoming signal above the level of the internally generated noise and also to start the process of selecting the wanted station and rejecting the unwanted ones.

2.4 SELECTIVITY

A parallel tuned circuit has its greatest impedance at resonance and decreases at higher and lower frequencies. If the tuned circuit is included in the circuit design of an amplifier, it results in an amplifier, which offers more gain at the frequency of resonance and reduced amplification above and below this frequency. This is called selectivity. We can see the effects of using an amplifier with selectivity.

The radio receiver is tuned to a frequency of 820KHz and, at this frequency; the amplifier provides a gain of five. Assuming the incoming signal has an amplitude of 10mV as shown, its output at this frequency would be $5 \times 10\text{mV} = 50\text{mV}$. The stations being received at 810KHz have a gain of one each. With the same amplitude of 10mV, this would result in outputs of $1 \times 10\text{mV} = 10\text{mV}$. The stations at 800KHz and 840KHz are offered a gain of only 0.1 (approx). This means that the output signal strength would be only $0.1 \times 10\text{mV} = 1\text{mV}$.

The overall effect of the selectivity is that whereas the incoming signals each have the same amplitude, the outputs vary between 1mV and 50mV so we can select, or 'tune', the amplifier to pick out the desired station.

The greatest amplification occurs at the resonance frequency of the tuned circuit. This is sometimes called the center frequency.

Generally in all radio receivers, the capacitor value is adjusted by means of the local oscillator control to select various signals.

2.5 THE LOCAL OSCILLATOR

This is an oscillator producing a sinusoidal output similar to the carrier wave oscillator in the transmitter. In this case, however, the frequency of its output is adjustable.

The same tuning control is used to adjust the frequency of both the local oscillator and the center frequency of the RF amplifier. The local oscillator is always maintained at a frequency, which is higher, by a fixed amount, than the incoming RF signals. The local oscillator frequency therefore follows or tracks the RF amplifier frequency.

2.6 THE MIXER (FREQUENCY CONVERTER):

In many applications it is convenient to operate a frequency translation of a signal. In radio electronics this technique is largely used in superheterodine receiver, whose block diagram is shown in FIG. 2.1.

The radio frequency signal (RF) detected by the antenna is amplified by an RF amplifier and transferred, via a frequency mixer and a local oscillator (LO), to a lower and fixed frequency (Intermediate Frequency IF). The translation to a fixed and lower frequency than the received one enables to:

- Use a fixed band amplifier for the amplification of all the received signals, independently from their frequency.
- Operate on lower frequency signals.

The frequency translation is some times used also on communication systems, where for example the modulation is carried out at an intermediate frequency which value is lower to the effectively transmitted frequency.

As already said, to carry out the frequency translation a locally generated translation

is necessary, called local oscillation, which interacts with the radio frequency signal so to produce the new frequency.

If the oscillation is produced by the same device operating the circuit translation it is normally called Converter; in the reverse case we speak respectively of Local Oscillator and Mixer. The frequency translation is differently indicated such as frequency conversion, mixing, beating, heterodyne process. The techniques used are many, and depend on the device used and on the operating frequencies. During the exercises we use a frequency mixer carried out with Dual Gate MOSFET.

2.6.1 FREQUENCY MIXER WITH DUAL GATE MOSFET:

The Block diagram of a mixer is shown in FIG. 2.2. The RF and LO signals are separately applied to two gates of the MOSFET. The behavior of the MOSFET can be analyzed by presuming that, when the signal LO has a higher amplitude than the signal RF, the transconductance g_m as function of the Gate to Source voltage (V_{G2}), for V_{G1} included into a narrow range of values, can be defined as;

$$g_m = \frac{\Delta I_D}{\Delta V_{G1}}$$

with I_D as Drain current. If for example V_{G1} (signal RF) is limited to a small value around (+/-100mV) of $-0.75V$, and V_{G2} (signal LO) takes values ranging between 1 and 3V, the transconductance practically depends only on V_{G2} . It is possible to detect then for g_m the representation of FIG. 2.4.

The heterodyne action of the MOSFET depends on its biasing: if V_{G2} acts in a zone where g_m can be expressed as:

$$g_m \approx a + b \cdot V_{G2}$$

$$\text{Set: } V_{G1} = A_{RF} \cdot \cos(\omega_{RF} \cdot t)$$

$$V_{G2} = A_{LO} \cdot \cos(\omega_{LO} \cdot t)$$

The output current I_D becomes:

$$I_D = g_m \cdot V_{G1}$$

$$= a \cdot A_{RF} \cdot \cos(\omega_{RF} \cdot t) + b \cdot A_{RF} \cdot A_{LO} \cdot \cos(\omega_{LO} \cdot t) \cdot \cos(\omega_{RF} \cdot t)$$

$$= a \cdot A_{RF} \cdot \cos(\omega_{RF} \cdot t) + \frac{b \cdot A_{RF} \cdot A_{LO}}{2} \cdot [\cos(\omega_{LO} - \omega_{RF}) \cdot t + \cos(\omega_{LO} + \omega_{RF}) \cdot t]$$

Using the circuit tuned on the Drain you extract the intermediate frequency component, with pulse:

$$\omega_{IF} = \omega_{LO} - \omega_{RF}$$

If V_{G2} does not act in a linear segment of the transconductance g_m , then g_m won't be expressed anymore with a linear relation and the result of the conversion operated by the Mosfet will contain also the harmonics of the frequency LO. A frequency IF can be obtained equal to:

$$WIF = 3 \bullet wLO - wRF$$

We define Conversion Transconductance g_c as the relation between the pulse output current wIF and the pulse input voltage wRF :

$$g_c = \frac{iIF}{vRF}$$

The mixer used for the exercises is composed of three sections (FIG. 2.2):

- The MOSFET, which carries out the interaction between the LO and the RF signal.
- A band pass filter centered to 450KHz, consisting of two tuned circuits coupled between them.
- An output amplifier stage.

The mixer performs a similar function to the modulator in the transmitter.

We may remember that the transmitter modulator accepts the information signal and the carrier frequency, and produces the carrier plus the upper and lower sidebands.

The mixer in the receiver combines the signal from the RF amplifier and the frequency input from the local oscillator to produce three frequencies:

- I. A 'difference' frequency of local oscillator frequency – RF signal frequency.
- II. A 'sum' frequency equal to local oscillator frequency + RF signal frequency.
- III. Component at the local oscillator frequency.

When this is carried out at frequencies which are above the audio spectrum, called 'supersonic' frequencies, the type of receiver is called a 'superheterodyne' receiver. This is normally abbreviated to 'superhet'. It is not a modern idea having been invented in the year 1917.

In Section 2.5, we saw how the local oscillator tracks the RF amplifier so that the difference between the two frequencies is maintained at a constant value. In ACL-AD this difference is actually 455KHz.

As an example, if the radio is tuned to receive a broadcast station, which transmits at 800KHz, the local oscillator will be running at 1.255MHz. The difference frequency is

$$1.255\text{MHz} - 800\text{KHz} = 455\text{KHz}.$$

If the radio is now retuned to receive a different station being broadcast on 700KHz, the tuning control re-adjusts the RF amplifier to provide maximum gain at 700KHz and the local oscillator to 1.155MHz. The difference frequency is still maintained at the required 455KHz.

This frequency difference therefore remains constant regardless of the frequency to which the radio is actually tuned and is called the Intermediate Frequency (IF).

2.6 INTERMEDIATE FREQUENCY AMPLIFIERS (IF AMPLIFIERS):

The IF amplifier in this receiver consists of two stages of amplification and provides the main signal amplification and selectivity.

Operating at a fixed IF frequency means that the design of the amplifiers can be simplified. If it were not for the fixed frequency, all the amplifiers would need to be tunable across the whole range of incoming RF frequencies and it would be difficult to arrange for all the amplifiers to keep in step as they are re-tuned.

In addition, the radio must select the wanted transmission and reject all the others. To do this the band pass of all the stages must be carefully controlled. Each IF stage does not necessarily have the same band pass characteristics, it is the overall response that is important. Again, this is something, which is much more easily achieved without the added complication of making them tunable.

At the final output from the IF amplifiers, we have a 455KHz wave which is amplitude modulated by the wanted audio information.

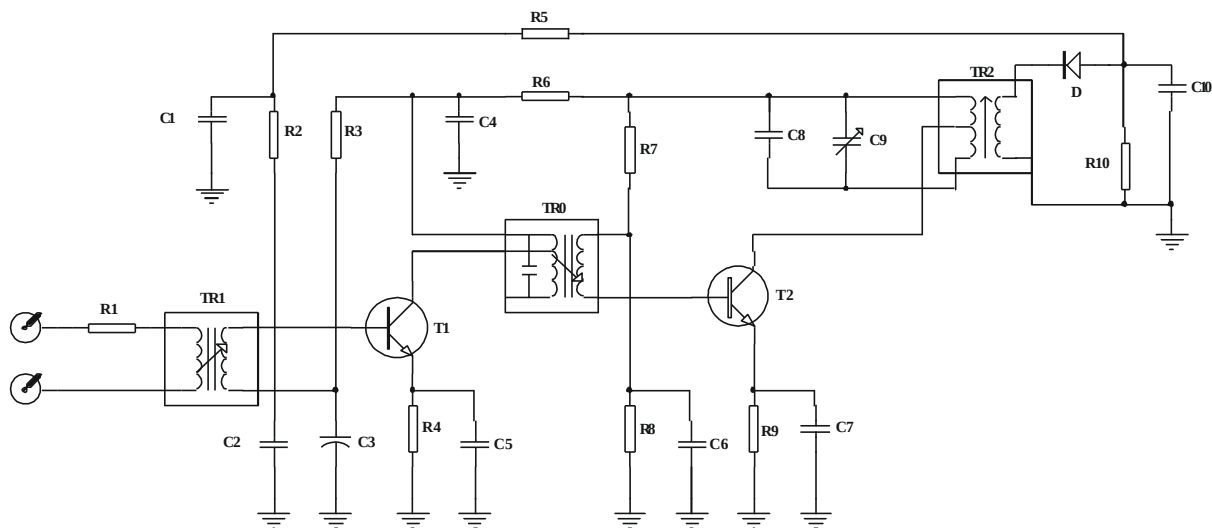


FIG. 2.4 TWO STAGE IF AMPLIFIER WITH DETECTOR AGC

2.6.1 2-STAGE IF AMPLIFIER WITH AM AND THE AUTOMATIC GAIN CONTROL CIRCUIT (AGC):

Refer to FIG. 2.4, which reports the electrical diagram of a 2-stage intermediate Frequency Amplifier, with AM diode detector and AGC (Automatic gain Control). This circuit is typically a used in superheterodyne AM receivers.

The AGC circuit is used to prevent very strong signals from overloading the receiver. It can also reduce the effect of fluctuations in the received signal strength.

The AGC circuit makes use of the mean DC voltage level present at the output of the diode detector.

If the signal strength increases, the mean DC voltage level also increases. If the mean DC voltage level exceeds a predetermined threshold value, a voltage is applied to the RF and IF amplifiers in such a way as to decrease their gain to prevent overload.

The IF signal is applied to the first transformer IF (TR) and by this to the first stage (transistor T). The biasing of T depends on the CAG voltage coming from the detector, and in this way the amplification of the first stage is varied, too. Across the transformer TRO the signal is taken to the second IF stage (transistor T1) and by this, with the transformer TR2, to the detector diode.

The diode is connected so that the negative envelope of the modulated signal is Detected. The detected signal is composed of the low frequency modulating signal and a negative continuous component proportional to the amplitude of the IF signal. The dc component is separated by the signal (via the low pass circuit R-C) and constitutes the CAG voltage used to vary the amplification of the first stage IF.

The selectivity of the IF amplifiers has removed the unwanted components generated by the mixing process.

The result is an output, which contains three components:

- I. The wanted audio information signal.
- II. Some ripple at the IF frequency.
- III. A positive DC voltage level.

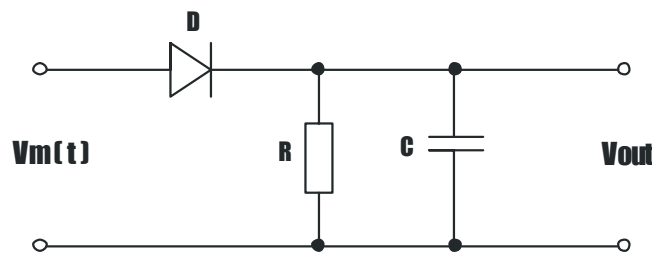


FIG. 2.5 ENVELOPE DETECTOR

2.8 ENVELOPE DETECTOR:

The extraction of the modulating signal from an AM signal can be carried out using an envelope detector. Consider, in fact, the AM signal shown in FIG. 2A, and note that the modulating signal constitutes the envelope of the waveform reported.

The most common envelope detector consists of a diode followed by a RC filter (FIG. 2.5). Its operation is analogous to the one of a half-wave rectifier, as the output voltage follows the maximum values of the carrier. As the amplitude of the carrier is variable, by properly choosing R and C, the output of the detector can faithfully reproduce these variations.

2.8.1 DISTORTIONS OF THE DETECTED SIGNAL:

The demodulated signal can have two kinds of distortions:

- If the time constant $R \cdot C$ is too small in respect to the period of the carrier, the envelope is approximated to a waveform affected by ripple. More the evident, the lower is the value of $R \cdot C$ (FIG. 2B)
- If the time constant $R \cdot C$ is too high in respect to the period of the modulating signal, the detected signal always follows the behavior of the envelope, but sometimes it takes an exponential decreasing law (distortion by diagonal cutting, FIG. 2C)

The distortion presented by the detected signal is known as Distortion by diagonal cutting.

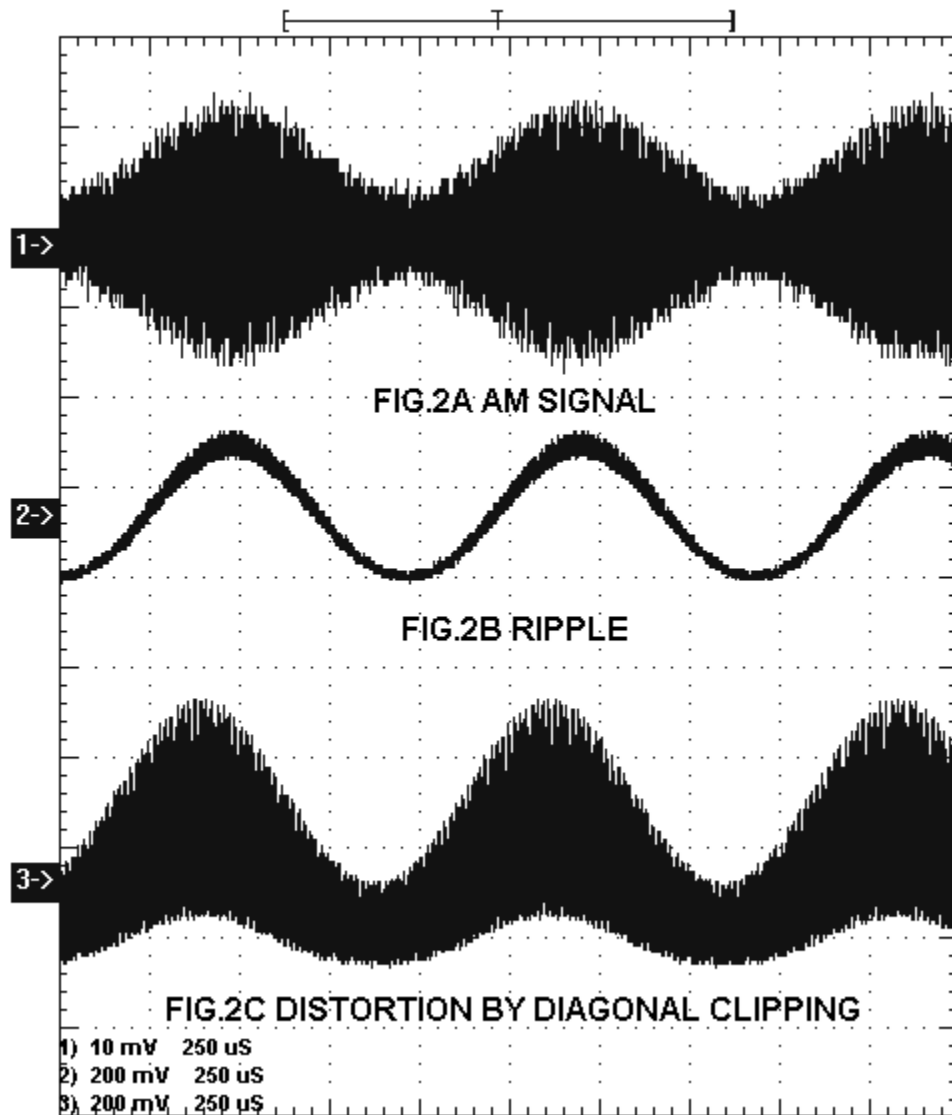
The maximum value of $R \cdot C$ is calculated supposing that the capacitor C discharges on the resistor R to a higher or equal rhythm to the one in which the envelope of the modulated signal drops. The detected condition is the following:

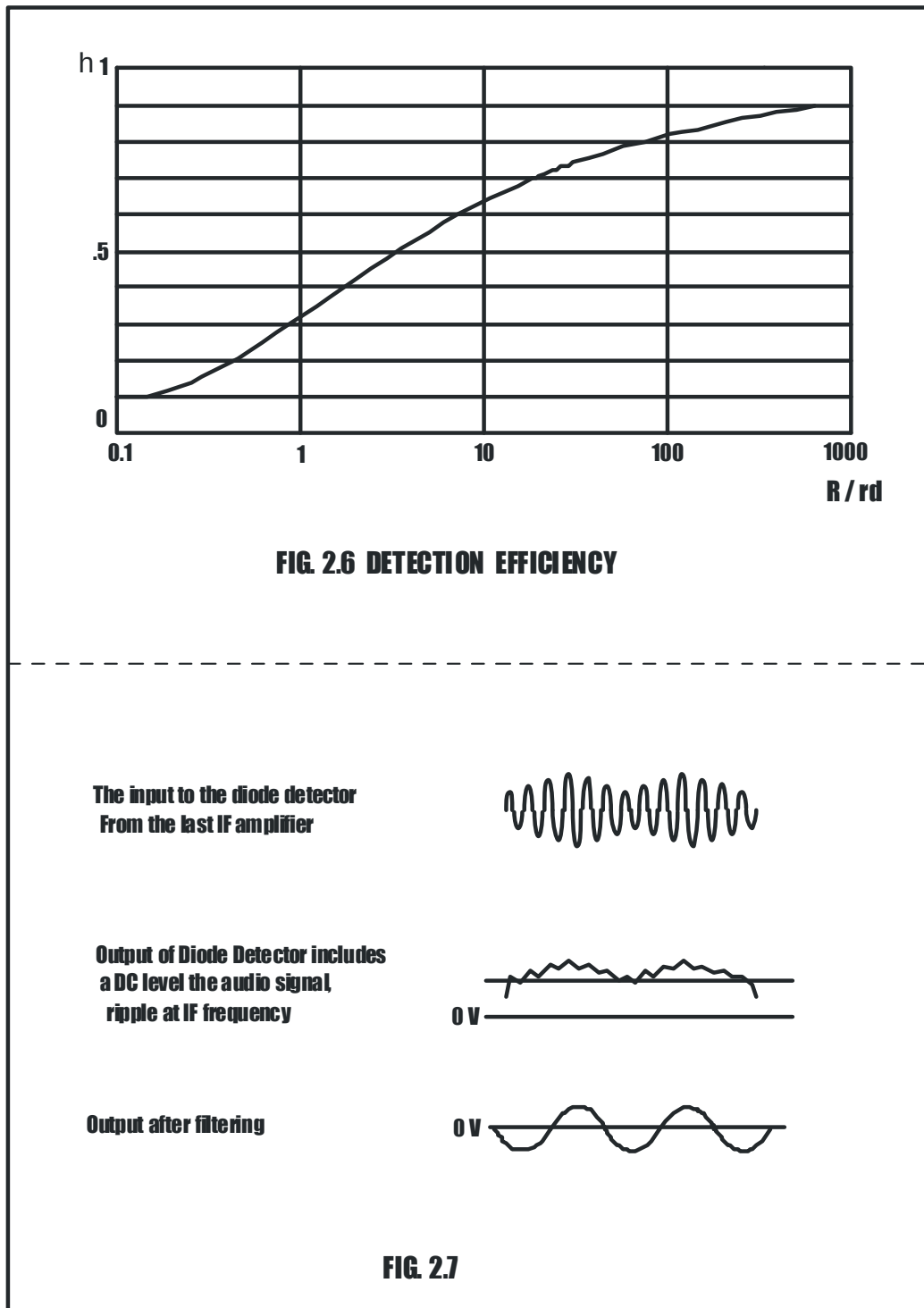
$$R \cdot C \leq \frac{1}{f_{\max}} \cdot \frac{\sqrt{(1-m \cdot 2)}}{m}$$

with m = modulation index

$f_{\max}$ = maximum frequency of the modulating signal.

Note that for $m=1$ the condition cannot be satisfied, and so the detected signal will be certainly distorted.





2.8.2 DETECTION EFFICIENCY

The efficiency η of the diode envelope detector is defined as the ratio between the amplitude of the output signal of the detector and the amplitude of the envelope of the input AM signal. Supposing that at the carrier frequency, the reactance of the capacitor $[1/(2\pi \cdot F \cdot C)]$ is much smaller than the resistance R (hypothesis checked if $RC \gg 1/F$ to reduce the ripple), the efficiency η practically depends on the relation between R and the differential resistance r_d of the diode (FIG. 2.6)

2.8.3 LAST CONSIDERATIONS ON THE CHOICE OF $R \cdot C$ IN THE ENVELOPE DETECTOR:

- To obtain high detection efficiency the resistance R must be much higher than the differential resistance r_d of the diode ($R \gg r_d$).
 - To minimize the ripple in the detected signal the product $R \cdot C$ must be much higher than the period T of the carrier ($R \cdot C \gg T$).
- To avoid distortion by diagonal cutting the product $R \cdot C$ must be smaller than or equal to a value, which depends on the modulating frequency and on the modulation index. In the case of the modulating signal with variable amplitude and frequency, the maximum values of the frequency as well as modulation index must be considered.

2.9 THE AUDIO AMPLIFIER:

At the input to the audio amplifier, a low pass filter is used to remove the If ripple and a capacitor blocks the DC voltage level. FIG. 2.7 shows the result of the information signal passing through the Diode Detector and Audio Amplifier.

The remaining audio signals are then amplified to provide the final output to the loudspeaker.

A. STUDY OF DOUBLE SIDEBAND AM RECEPTION USING ENVELOPE DETECTOR VIA CABLE.

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD
- Power supply
- 20MHz Oscilloscope.
- Connecting Links.
- Frequency counter.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG. 2.8 & Carry out the following connections.
2. Connect o/p of **FUNCTION GENERATOR** section (ACL-AM) **OUT** post to the i/p of Balance Modulator1 (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of VCO (ACL-AM) **OUT** post to the input of Balance modulator 1 (ACL-AM) **CARRIER IN** post.
4. Connect the power supply with proper polarity to the kit ACL-AM & ACL -AD, While connecting this, ensure that the power supply is OFF.
5. Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR:** Sine LEVEL about 0.5 Vpp; FREQ. about 1 KHz.
 - **VCO:** LEVEL about 2Vpp; FREQ. about 850 KHz, Switch on 1500KHz.
 - **BALANCED MODULATOR1:** CARRIER NULL completely rotates Clockwise or counter clockwise, so that the modulator is “unbalanced” and an AM signal with not suppressed carrier is obtained across the output: adjust OUTLEVEL to obtain an AM signal across the output whose amplitude is about 100mVpp
 - **LOCAL OSCILLATOR (ACL-AD):** 1300KHz, 2V.

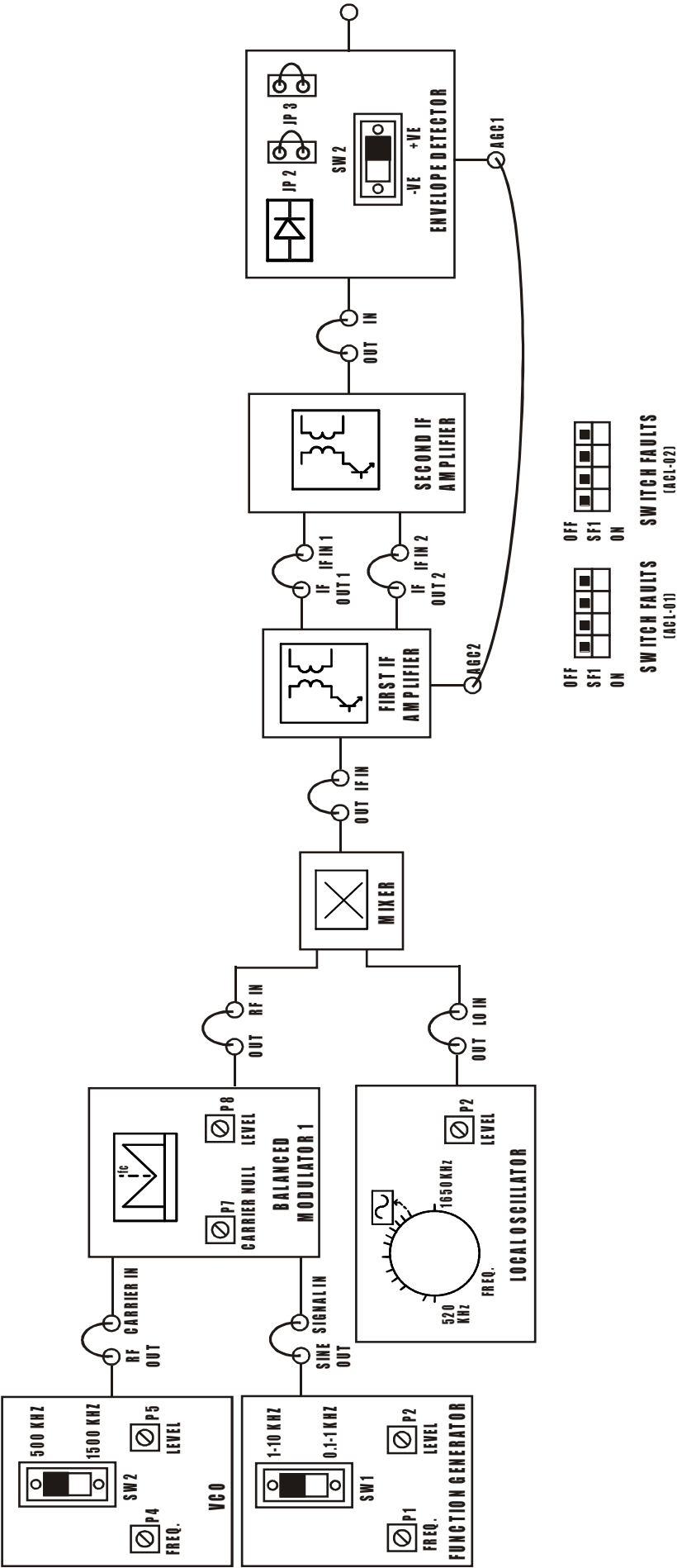


FIG. 2.8 BLOCK DIAGRAM FOR STUDY OF DSB AM RECEPTION USING ENVELOPE DETECTOR VIA CABLE

6. Connect local oscillator **OUT** post to **LO IN** of the mixer section.
7. Connect balance modulator1 out to **RF IN** of mixer section in ACL-AD.
8. Connect mixer **OUT** to **IF IN** of 1st IF AMPLIFIER in ACL-AD.
9. Connect **IF OUT1** of 1st IF to **IF IN 1** and **IF OUT2** of 1st IF to **IFIN 2** of 2ND IF AMPLIFIER.
10. Connect **OUT** post of 2nd IF amplifier to **IN** post of envelope detector.
11. Connect post **AGC₁** to post **AGC₂** and jumper position as per diagram.
12. Observe the modulated signal envelope, which corresponds to the waveform of the modulating signal at **OUT** post of the balanced modulator1 of ACL-AM. Connect the oscilloscope to the **IN** and **OUT** post of envelope detector and detect the AM signal and the detected one (FIG. 2E) If the central frequency of the amplifier and the carrier frequency of the AM signal and local oscillator frequency coincides, you obtain two signals similar to the ones of FIG.2E.
13. Check that the detected signal follows the behavior of the AM signal envelope. Vary the frequency and amplitude of the modulating signal, and check the corresponding variations of the demodulated signal.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time.

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.
3. Put switch 1 of SF1 in ACL-AD switch fault section to ON position. This will open the capacitor (C35) from the emitter of the first IF amplifier. So the gain will reduce which in turn reduces the output amplitude.
4. Put switch 4 of SF1 in ACL-AD switch fault section to ON position. This will open the diode D1 from the envelope detector; O/P will get disturbed in both the envelope. Also there will not be any effect of AGC signal.

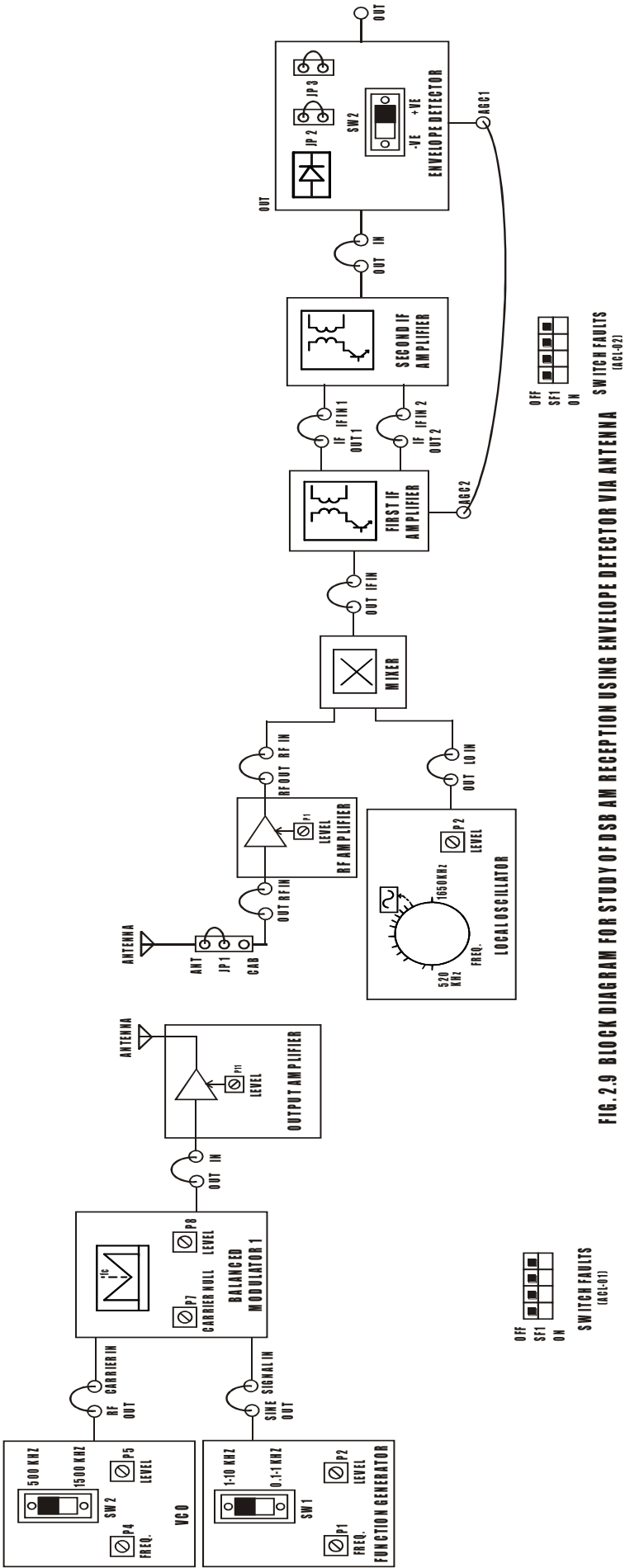


FIG. 2.9 BLOCK DIAGRAM FOR STUDY OF DSB AM RECEPTION USING ENVELOPE DETECTOR VIA ANTENNA

B. STUDY OF DOUBLE SIDEBAND AM RECEPTION USING ENVELOPE DETECTOR VIA ANTENNA.

EQUIPMENTS:

- Modules ACL-AM & ACL-AD.
- Power supply
- Oscilloscope.
- Connecting Links.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

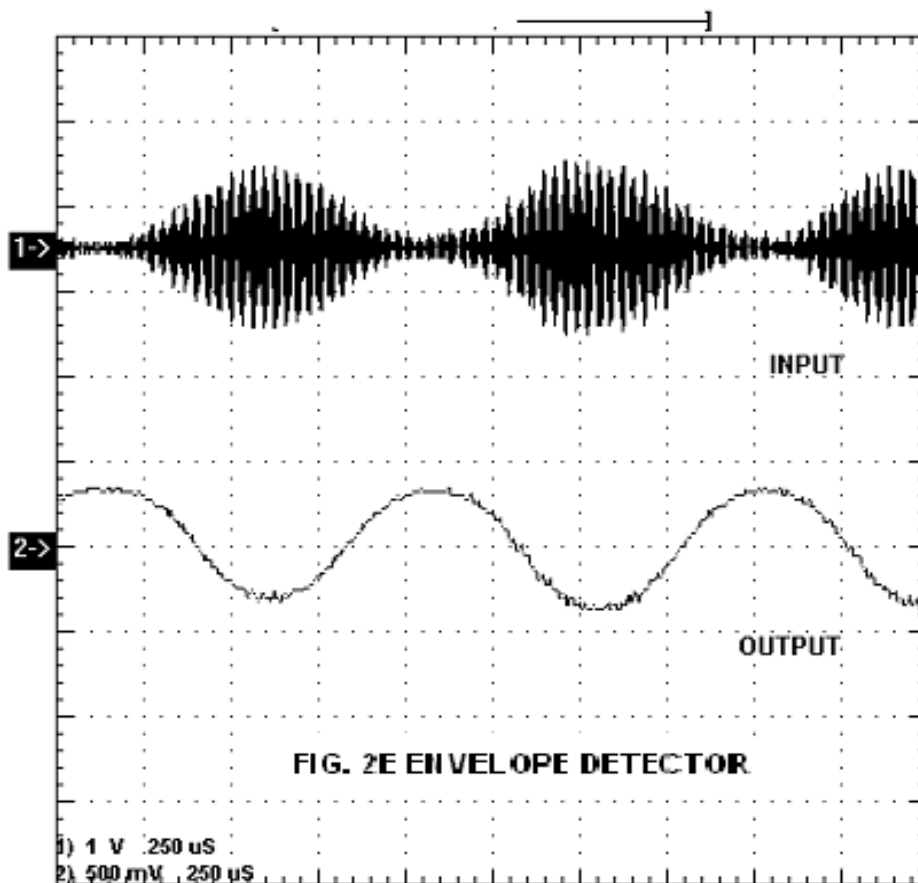
PROCEDURE:-

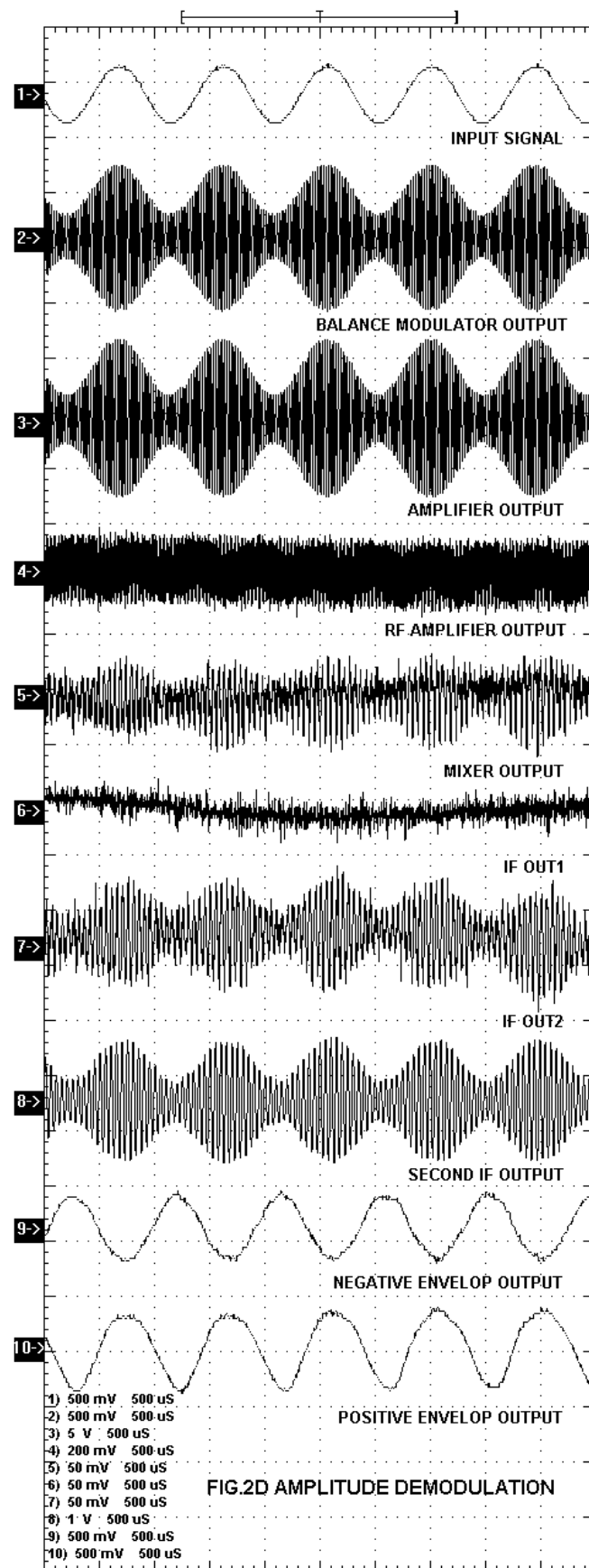
1. Refer to the FIG. 2.9 & Carry out the following connections.
2. Connect o/p of **FUNCTION GENERATOR** section (ACL-AM) **OUT** post to the i/p of Balance Modulator1 (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of VCO (ACL-AM) **OUT** post to the input of Balance modulator (ACL-AM) **CARRIER IN** post.
4. Connect power supply with proper polarity to the kit ACL-AM& ACL -AD. While connecting this, ensure that the power supply is OFF.
5. Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR:** Sine LEVEL about 0.5 Vpp; FREQ. About 1 KHz
 - **VCO:** LEVEL about 2 Vpp; FREQ. About 850 KHz. Switch on 1500KHz.
 - **BALANCED MODULATOR 1:** CARRIER NULL completely rotates clockwise or counter clockwise, so that the modulator is “unbalanced” and an AM signal with not suppressed carrier is obtained across the output: adjust OUTLEVEL to obtain an AM signal across the output whose amplitude is about 500mVpp.
 - **OUTPUT AMPLIFIER (ACL-AM):** LEVEL fully clock wise.
 - **LOCAL OSCILLATOR (ACL-AD):** 1300KHz, 2V.
 - **RF LEVEL (ACL-AD):** on max. position or adjust to achieve as per input signal.
6. Connect the **OUT** post of balance modulator1 to the **IN** post of output amplifier. In which output of amplifier is directly connected to the antenna.
7. Connect **RF OUT** post of RF AMPLIFIER to **RF IN** of MIXER of ACL-AD.
8. Connect local oscillator **OUT** post to **LO IN** of the mixer section.
9. Connect mixer **OUT** post to **IF IN** of 1st IF amplifier.
10. Connect **IF OUT1** of 1st IF to **IF IN 1** and **IF OUT2** of 1st IF to **IFIN 2** OF 2ND IF AMPLIFIER.
11. Connect **OUT** post of 2nd IF amplifier to **IN** post of envelope detector.
12. Connect **AGC₁** to **AGC₂** and jumper position as per the diagram.
13. Connect the oscilloscope to the **IN** and **OUT** post of envelope detector and detect the AM signal and the detected one (FIG. 2E). If the central frequency of the amplifier and the carrier frequency of the AM signal and local oscillator frequency coincides, you will obtain two signals similar to the ones of FIG. 2E.
14. Check that the detected signal follows the behavior of the AM signal envelope. Vary the frequency and amplitude of the modulating signal, and check the corresponding variations of the demodulated signal.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time.

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.
3. Put switch 1 of SF1 in ACL-AD switch fault section to ON position. This will open the capacitor (C35) from the emitter of the first IF amplifier. So the gain will reduce which in turn reduces the output amplitude.
4. Put switch 4 of SF1 in ACL-AD switch fault section to ON position. This will open the diode D1 from the envelope detector; O/P will get disturbed in both the envelope. Also there will not be any effect of AGC signal.





EXPERIMENT NO.3

EXPERIMENT NO: 3

NAME:-

STUDY OF SENSITIVITY, SELECTIVITY, OF A RADIO RECEIVER

OBJECTIVE:-

- A. STUDY OF SENSITIVITY OF A RADIO RECEIVER
- B. STUDY OF SELECTIVITY OF A RADIO RECEIVER

THEORY:-

3.1 SENSITIVITY

The sensitivity of a receiver is its ability to amplify weak signals. Sensitivity of a receiver is the minimum voltage required at the input of the receiver to produce the original signal. It is often defined in terms of the voltage that must be applied to the receiver input terminals to give a standard output power, measured at the output terminal. For AM broadcast receivers, several of the relevant quantities have been standardized.

The most important factors determining the sensitivity of a super heterodyne receiver are the gain of the IF amplifiers and that of the RF amplifier.

3.2 SELECTIVITY

A parallel tuned circuit has its greatest impedance at resonance and decreases at higher and lower frequencies. If the tuned circuit is included in the circuit design of an amplifier, it results in an amplifier, which offers more gain at the frequency of resonance and reduced amplification above and below this frequency. This is called selectivity.

We can see the effects of using an amplifier with selectivity.

The radio receiver is tuned to a frequency of 820KHz and, at this frequency; the amplifier provides a gain of five. Assuming the incoming signal has an amplitude of 10mV, its output at this frequency would be $5 \times 10\text{mV} = 50\text{mV}$. The stations being received at 810KHz each have a gain of one. With the same amplitude of 10mV, this would result in outputs of $1 \times 10\text{mV} = 10\text{mV}$. The stations at 800KHz and 840KHz are offered a gain of only 0.1 (approx). This means that the output signal strength would be only $0.1 \times 10\text{mV} = 1\text{mV}$.

The overall effect of the selectivity is that whereas the incoming signals each have the same amplitude, the outputs vary between 1mV and 50mV so we can select, or 'tune', the amplifier to pick out the desired station.

The greatest amplification occurs at the resonance frequency of the tuned circuit. This is sometimes called the center frequency.

In common with nearly all radio receivers, adjusts the capacitor value by means of the local oscillator control to select various signals.

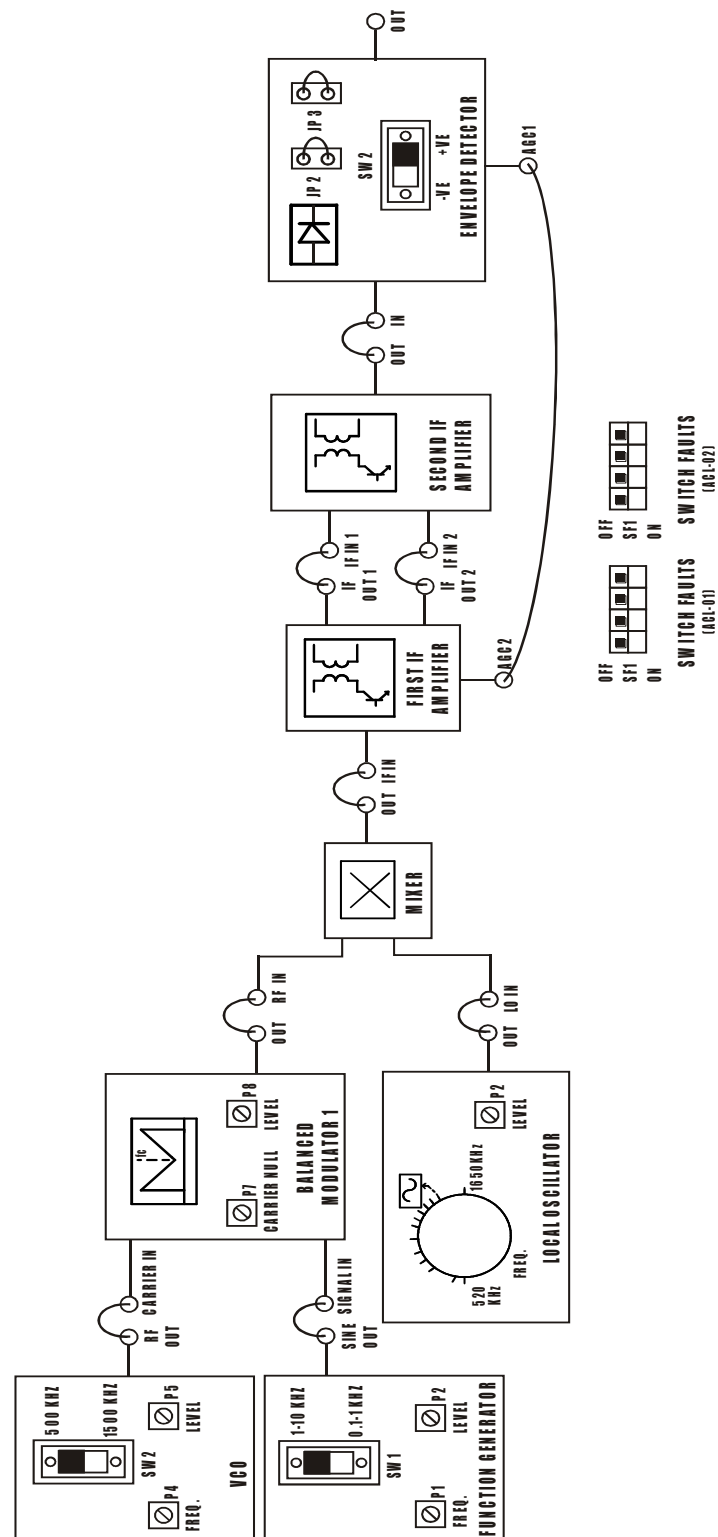


FIG. 3.1 BLOCK DIAGRAM FOR MEASUREMENT OF SENSITIVITY AND SELECTIVITY OF DSBBM RECEPTION VIA CABLE

A1. STUDY OF SENSITIVITY OF A RADIO RECEIVER VIA CABLE

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD.
- Power supply GND, +5V, +/-12 V.
- 20MHz Oscilloscope.
- Connecting Links.
- Frequency counter.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION

PROCEDURE:-

1. Refer to the FIG.3.1 & Carry out the following connections.
2. Connect o/p of **FUNCTION GENERATOR** section (ACL-AM) **OUT** post to the i/p of Balance Modulator (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of VCO (ACL-AM) **OUT** post to the input of Balance modulator (ACL-AM) **CARRIER IN** post.
4. Connect the power supply with proper polarity to the kit ACL-AM&ACL-AD, While connecting this, ensure that the power supply is OFF.
5. Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR**: sine LEVEL about 0.5V Vpp; FREQ. about 1 kHz
 - **VCO**: LEVEL about 2Vpp; FREQ. about 600 kHz, Switch on 1500KHz
 - **BALANCED MODULATOR 1**: CARRIER NULL completely rotates Clockwise or counter clockwise, so that the modulator is “unbalanced” and an AM signal with not suppressed carrier is obtained across the output: adjust OUTLEVEL to obtain an AM signal across the output which amplitude is about 200mVpp.
 - **LOCAL OSCILLATOR** (ACL-AD): 1050KHz, 2V.
6. Connect local oscillator **OUT** post to **LO IN** of the mixer section.
7. Connect balance modulator1 out to **RF IN** of mixer section in ACL-AD.
8. Connect mixer **OUT** to **IF IN** of 1st IF AMPLIFIER in ACL-AD.
9. Connect **IF OUT1** of 1st IF to **IF IN 1** and **IF OUT2** of 1st IF to **IFIN 2** of 2ND IF AMPLIFIER.
10. Connect **OUT** post of 2nd IF amplifier to **IN** post of envelope detector.
11. Connect post **AGC1** to post **AGC 2** and jumper position as per diagram.
12. Observe the modulated signal envelope, which corresponds to the waveform of the modulating signal at **OUT** post of the balanced modulator1 of ACL-AM. Connect the oscilloscope to the **IN** and **OUT** post of ENVELOPE DETECTOR and detect the AM signal.
13. Check that the detected signal follows the behavior of the AM signal envelope vary the amplitude of the Balanced modulator output and check the corresponding variations at the demodulated signal.
14. Adjust the input to **RF IN** post by varying the output of BM1 in such a way that you should get minimum detected output of about 0.3Volt at the output of Envelope detector.
15. You can take the readings as per the table mentioned below for various carrier frequencies and corresponding Local Oscillator frequency settings.

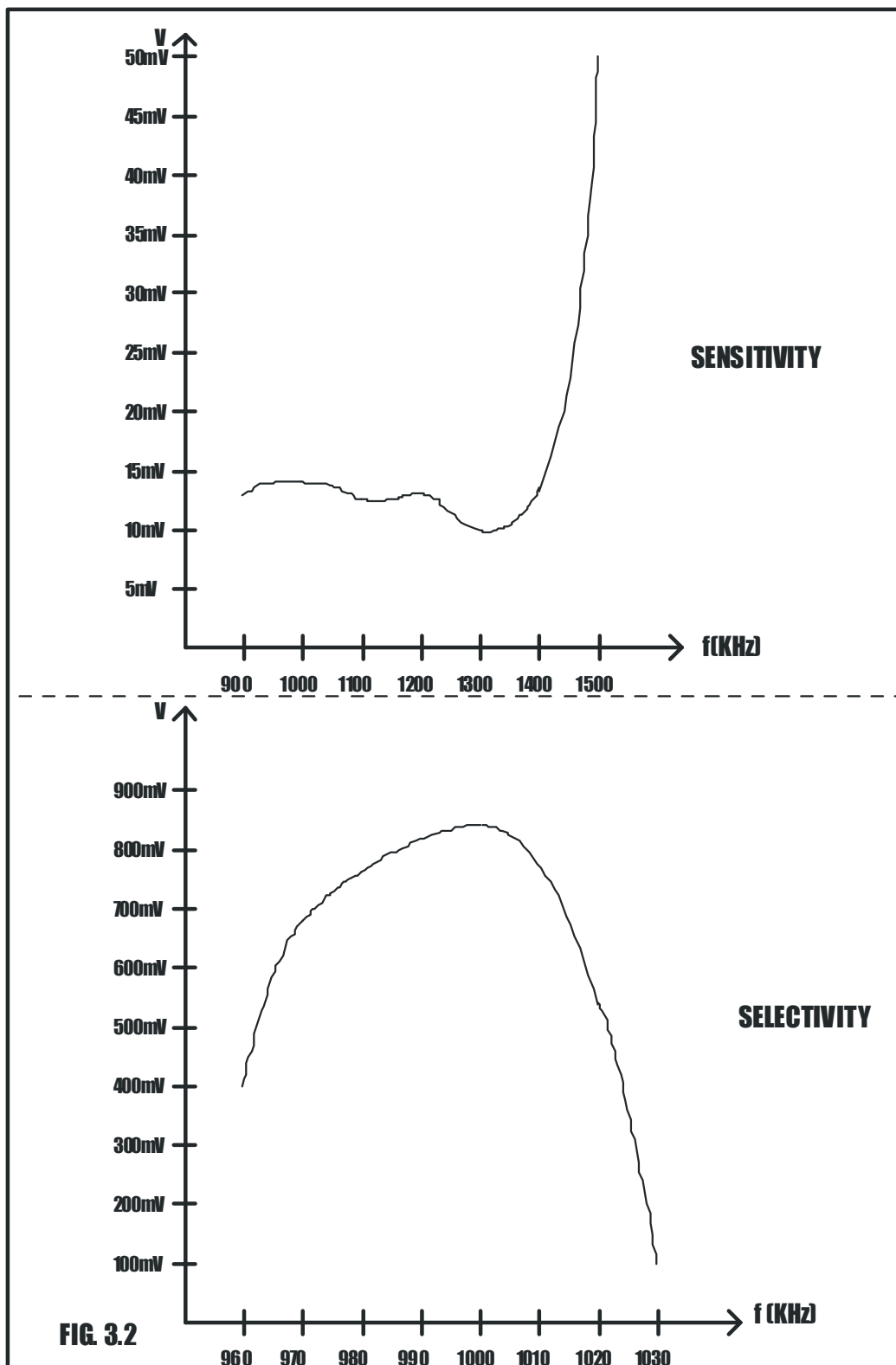
CARRIER FREQ. (KHz)	600	700	800	900	1000	1100	1200	1300
LOCAL OSC. FREQ. (KHz)	1050	1150	1250	1350	1450	1550	1650	1750
OUTPUT OF MIXER								

Plot the graph of OUTPUT OF BM1 ON Y-AXIS AND LOCAL OSCILLATOR FREQUENCY ON X-AXIS (FIG. 3.2).

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output . The faults are normally used one at a time

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.
3. Put switch 1 of SF1 in ACL-AD switch fault section to ON position. This will open the capacitor (C35) from the emitter of the first IF amplifier. So the gain will reduce which in turn reduces the output amplitude.
4. Put switch 4 of SF1 in ACL-AD switch fault section to ON position. This will open the diode D1 from the envelope detector; O/P will get disturbed in both the envelope. Also there will not be any effect of AGC signal.



B1. STUDY OF SELECTIVITY OF A RADIO RECEIVER VIA CABLE

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD
- Power supply
- Oscilloscope.
- Connecting Links.
- Frequency counter.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG.3.1 & Carry out the following connections.
2. Connect o/p of **FUNCTION GENERATOR** section (ACL-AM) **OUT** post to the i/p of Balance Modulator (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of VCO (ACL-AM) **OUT** post to the input of Balance modulator (ACL-AM) **CARRIER IN** post.
4. Connect the power supply with proper polarity to the kit ACL-AM&ACL-AD, while connecting this, ensure that the power supply is OFF.
5. Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR**: sine LEVEL about 0.5Vpp; FREQ. About 1 KHz
 - **VCO**: LEVEL about 2Vpp; FREQ. about 850 kHz, Switch on 1500KHz
 - **BALANCED MODULATOR1**: CARRIER NULL completely rotates Clockwise or counter clockwise, so that the modulator is “unbalanced” and an AM signal with not suppressed carrier is obtained across the output: adjust OUTLEVEL to obtain an AM signal across the output which amplitude is about 50mVpp.
 - **LOCAL OSCILLATOR** (ACL-AD): 1300KHz, 2V.
6. Connect local oscillator **OUT** post to **LO IN** of the mixer section.
7. Connect balance modulator1 out to **RF IN** of mixer section in ACL-AD.
8. Connect mixer **OUT** to **IF IN** of 1st IF AMPLIFIER in ACL-AD.
9. Connect **IF OUT1** of 1st IF to **IF IN 1** and **IF OUT2** of 1st IF to **IFIN 2** of 2ND IF AMPLIFIER.
10. Connect **OUT** post of 2nd IF amplifier to **IN** post of envelope detector.
11. Connect post **AGC1** to post **AGC 2** and jumper position as per diagram.
12. Observe the modulated signal envelope, which corresponds to the wave form of the modulating signal at **OUT** post of the balanced modulator of ACL-AM Connect the oscilloscope to the **IN** and **OUT** post of envelope DETECTOR and detect the AM signal.
13. Check that the detected signal follows the behavior of the AM signal envelope measure the detected signal amplitude.
14. Vary the Carrier frequency of (ACL-AM) in terms of 10KHz on both sides from 860KHz to 940 KHz.
15. Take readings of Amplitude levels at Envelope Detector output for various Carrier frequencies.

Table Similar to the One given below can be observed at the users end.

CARRIER FREQ. (KHz)	ENVELOPE DETECTOR OUTPUT (mV)
860	6
870	40
880	120
890	350
900	500
910	380
920	150
930	80
940	20

Plot the graph accordingly as per the readings taken. For the carrier frequency and o/p Amplitude (FIG. 3.2).

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.
3. Put switch 1 of SF1 in ACL-AD switch fault section to ON position. This will open the capacitor (C35) from the emitter of the first IF amplifier. So the gain will reduce which in turn reduces the output amplitude.
4. Put switch 4 of SF1 in ACL-AD switch fault section to ON position. This will open the diode D1 from the envelope detector; O/P will get disturbed in both the envelope. Also there will not be any effect of AGC signal.

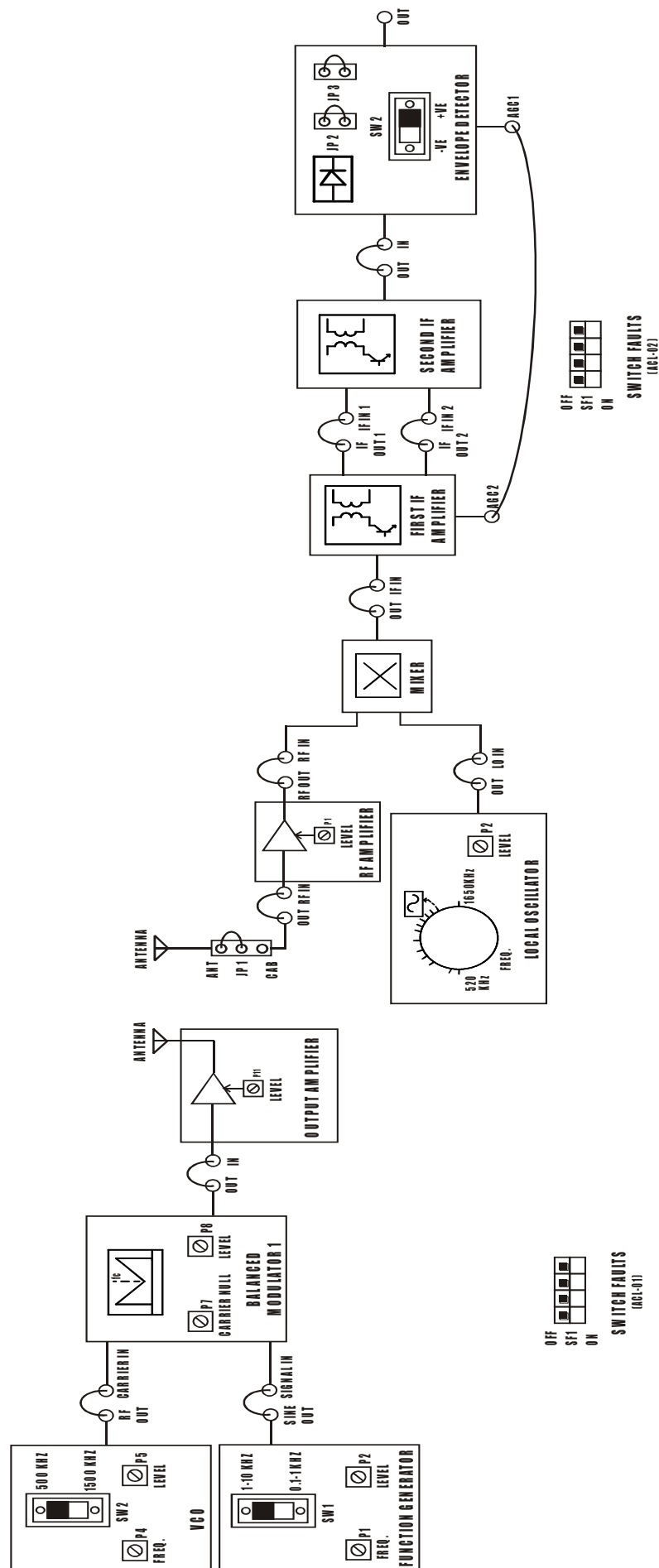


FIG. 3.3 BLOCK DIAGRAM FOR MEASUREMENT OF SENSITIVITY AND SELECTIVITY OF DSB AM RECEPTION VIA ANTENNA

A2. STUDY OF SENSITIVITY OF A RADIO RECEIVER VIA ANTENNA

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD
- Power supply
- 20MHz Oscilloscope.
- Connecting Links.
- Frequency counter.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG.3.3 & Carry out the following connections.
2. Connect o/p of **FUNCTION GENERATOR** section (ACL-AM) **OUT** post to the i/p of Balance Modulator1 (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of VCO (ACL-AM) **OUT** post to the input of Balance modulator 1 (ACL-AM) **CARRIER IN** post.
4. Connect the power supply with proper polarity to the kit ACL-AM&ACL-AD, while connecting this, ensure that the power supply is OFF.
5. Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR:** sine LEVEL about 0.5V V p-p; FREQ. About 1 kHz
 - **VCO:** Level about 2Vpp; FREQ. about 850 kHz, Switch on 1500KHz
 - **BALANCED MODULATOR1:** CARRIER NULL completely rotates Clockwise or counter clockwise, so that the modulator is “unbalanced” and an AM signal with not suppressed carrier is obtained across the output: adjust OUTLEVEL to obtain an AM signal across the output which amplitude is about 500mVpp.
 - **OUTPUT AMPLIFIER (ACL-AM):** Max. (Fully Clockwise Position.)
Switch **SW3** on **DSB** Position
 - **LOCAL OSCILLATOR(ACL-AD):** About 1300KHz, 2V.
 - **RF LEVEL (ACL-AD):** on max. position or adjust to achieve as per input signal
6. Connect the **OUT** post of balance modulator1 to the **IN** post of output amplifier, in which **OUT** post of amplifier is directly connected to the antenna.
7. Connect local oscillator **OUT** post to **LO IN** of the mixer section.
8. Connect **RF OUT** post to **RF IN** of MIXER of ACL-AD.
9. Connect mixer **OUT** to **IF IN** of 1st IF AMPLIFIER in ACL-AD.
10. Connect **IF OUT1** of 1st IF to **IF IN 1** and **IF OUT2** of 1st IF to **IFIN 2** of 2ND IF AMPLIFIER.
11. Connect **OUT** post of 2nd IF amplifier to **IN** post of envelope detector.
12. Connect post **AGC1** to post **AGC 2** and jumper position as per diagram.
13. Observe the modulated signal envelope, which corresponds to the waveform of the modulating signal at **OUT** post of the balanced modulator of ACL-AM. Connect the oscilloscope to the **IN** and **OUT** post of envelope DETECTOR and detect the AM signal.
14. Check that the detected signal follows the behavior of the AM signal envelope. Vary the amplitude of the Balanced modulator output and check the corresponding variations at the demodulated signal.
15. Adjust the output **LEVEL OF BM1** in such a way that that you should get minimum detected output of about 0.3Volt at the output of Envelope detector.

You can take the readings as per the table mentioned below for various carrier frequency and corresponding Local Oscillator frequency settings.

CARRIER FREQ. (KHz)	700	800	900	1000	1100	1200	1300	1400
LOCAL OSC. FREQ. (KHz)	1155	1255	1355	1455	1555	1655	1755	1855
OUTPUT OF BM1 (mV)								

Plot the graph of OUTPUT OF BM1 ON Y-AXIS AND LOCAL OSCILLATOR FREQUENCY ON X-AXIS (FIG. 3.4)

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.
3. Put switch 1 of SF1 in ACL-AD switch fault section to ON position. This will open the capacitor (C35) from the emitter of the first IF amplifier. So the gain will reduce which in turn reduces the output amplitude.
4. Put switch 4 of SF1 in ACL-AD switch fault section to ON position. This will open the diode D1 from the envelope detector; O/P will get disturbed in both the envelope. Also there will not be any effect of AGC signal.

B2. STUDY OF SELECTIVITY OF A RADIO RECEIVER VIA ANTENNA

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD.
- Power supply
- Oscilloscope.
- Connecting Links.
- Frequency counter.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG.3.3 & Carry out the following connections.
2. Connect o/p of **FUNCTION GENERATOR** section (ACL-AM) **OUT** post to the i/p of Balance Modulator1 (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of VCO (ACL-AM) **OUT** post to the input of Balance modulator1 (ACL-AM) **CARRIERIN** post.
4. Connect the power supply with proper polarity to the kit ACL-AM&ACL-AD, while connecting this, ensure that the power supply is OFF.

5. Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR:** sine LEVEL about 0.5Vpp; FREQ. About 1 kHz
 - **VCO:** LEVEL about 1Vpp; FREQ. about 850 kHz, Switch on 1500KHz
 - **BALANCED MODULATOR1:** CARRIER NULL completely rotates Clockwise or counter clockwise, so that the modulator is “unbalanced” and an AM signal with not suppressed carrier is obtained across the output: adjust OUTLEVEL to obtain an AM signal across the output which amplitude is about 400mVpp.
 - **OUTPUT AMPLIFIER (ACL-AM):** Max. (Fully Clockwise Position.)
Switch **SW3** on **DSB** Position
 - **LOCAL OSCILLATOR(ACL-AD):**1300KHz, 1V.
 - **RF LEVEL (ACL-AD):** on max. position or adjust to achieve as per input signal.
6. Connect the **OUT** post of balance modulator1 to the **IN** post of output amplifier. In which output of amplifier is directly connected to the antenna.
7. Connect local oscillator **OUT** post to **LO IN** of the mixer section.
8. Connect **RF OUT** post to **RF IN** of MIXER of ACL-AD.
9. Connect mixer **OUT** to **IF IN** of 1st IF AMPLIFIER in ACL-AD.
10. Connect **IF OUT1** of 1st IF to **IF IN 1** and **IF OUT2** of 1st IF to **IFIN 2** of 2ND IF AMPLIFIER.
11. Connect **OUT** post of 2nd IF amplifier to **IN** post of envelope detector.
12. Connect post **AGC1** to post **AGC 2** and jumper position as per diagram.
13. Observe the modulated signal envelope, which corresponds to the wave form of the modulating signal at **OUT** post of the balanced modulator of ACL-AM Connect the oscilloscope to the **IN** and **OUT** post of envelope DETECTOR and detect the AM signal.
14. Check that the detected signal follows the behavior of the AM signal envelope measure the detected signal amplitude.
15. Vary the Carrier frequency of (ACL-AM) in terms of 10KHz on both sides from 860KHz to 940 KHz.
16. Take readings of Amplitude levels at Envelope Detector output for various Carrier frequencies.

Table Similar to the One given below can be observed at the users end.

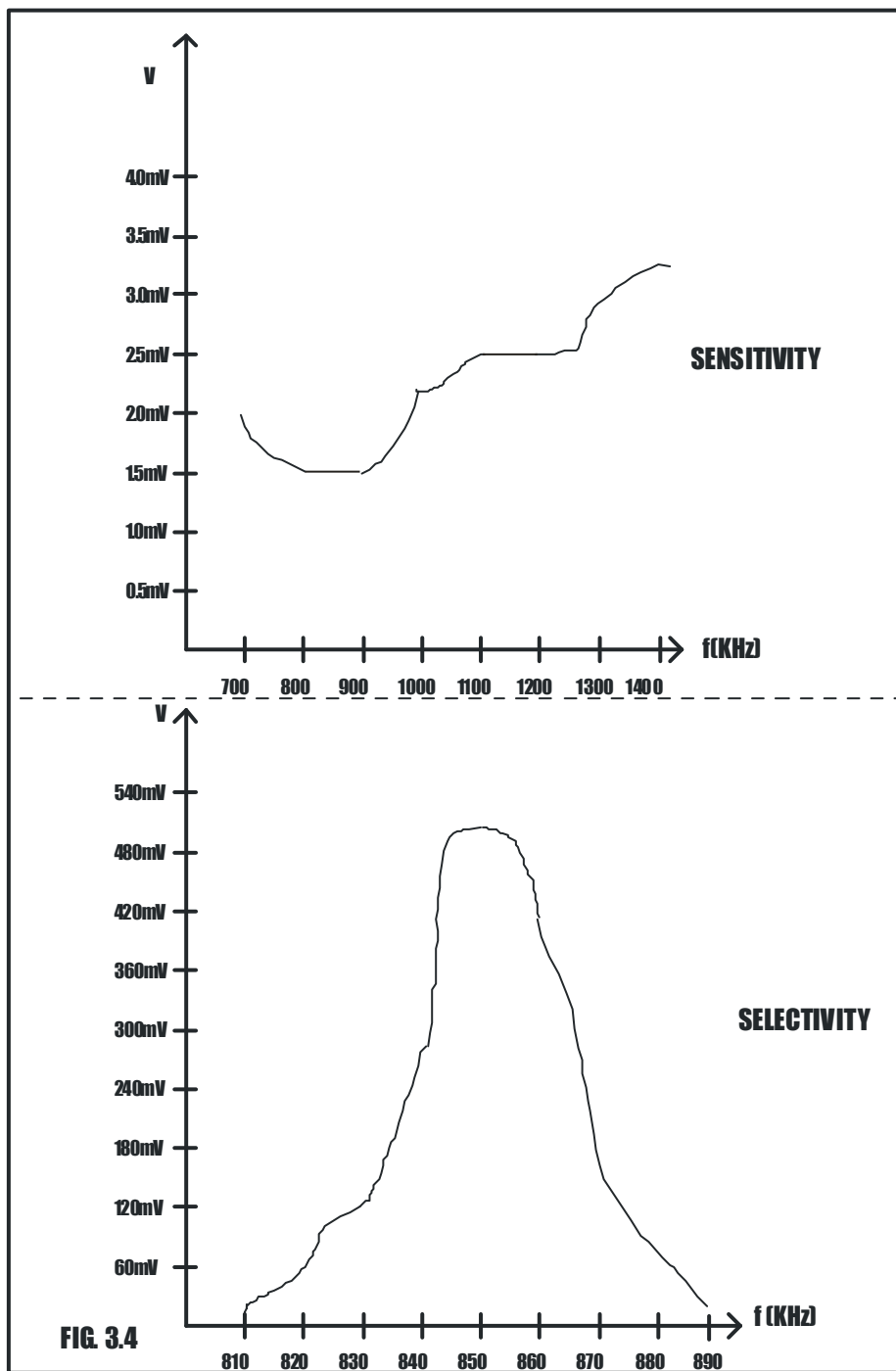
CARRIER FREQ. (KHz)	ENVELOPE DETECTOR OUTPUT (mV)
810	6
820	30
830	100
840	400
850	500
860	300
870	150
880	80
890	20

Plot the graph accordingly as per the readings taken. For the carrier frequency and o/p Amplitude (FIG. 3.4).

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.
3. Put switch 1 of SF1 in ACL-AD switch fault section to ON position. This will open the capacitor (C35) from the emitter of the first IF amplifier. So the gain will reduce which in turn reduces the output amplitude.
4. Put switch 4 of SF1 in ACL-AD switch fault section to ON position. This will open the diode D1 from the envelope detector; O/P will get disturbed in both the envelope. Also there will not be any effect of AGC signal.



EXPERIMENT NO.4

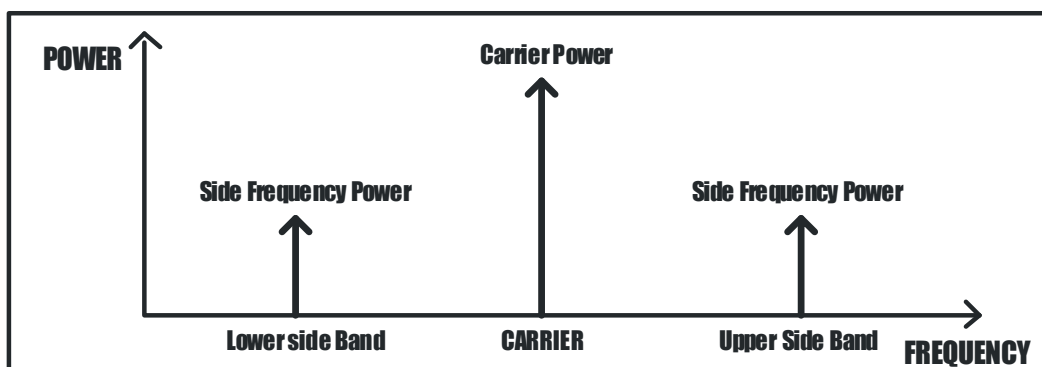


FIG. 4.1 DSB SPECTRUM

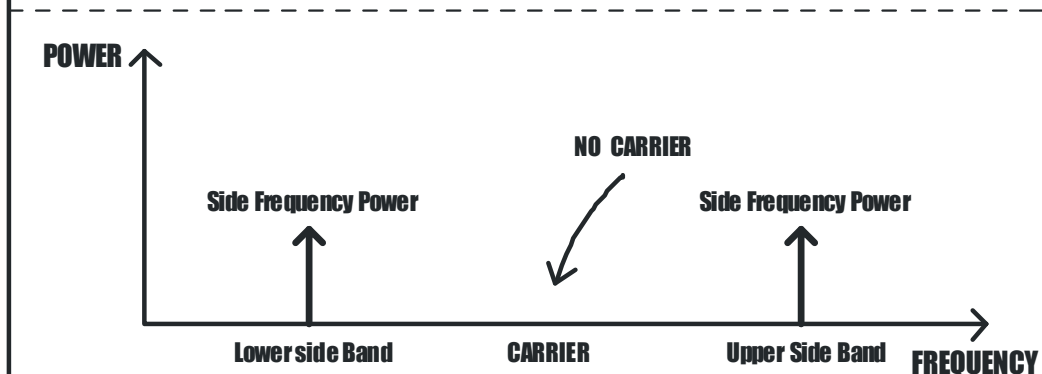


FIG. 4.2 DSBSC SPECTRUM

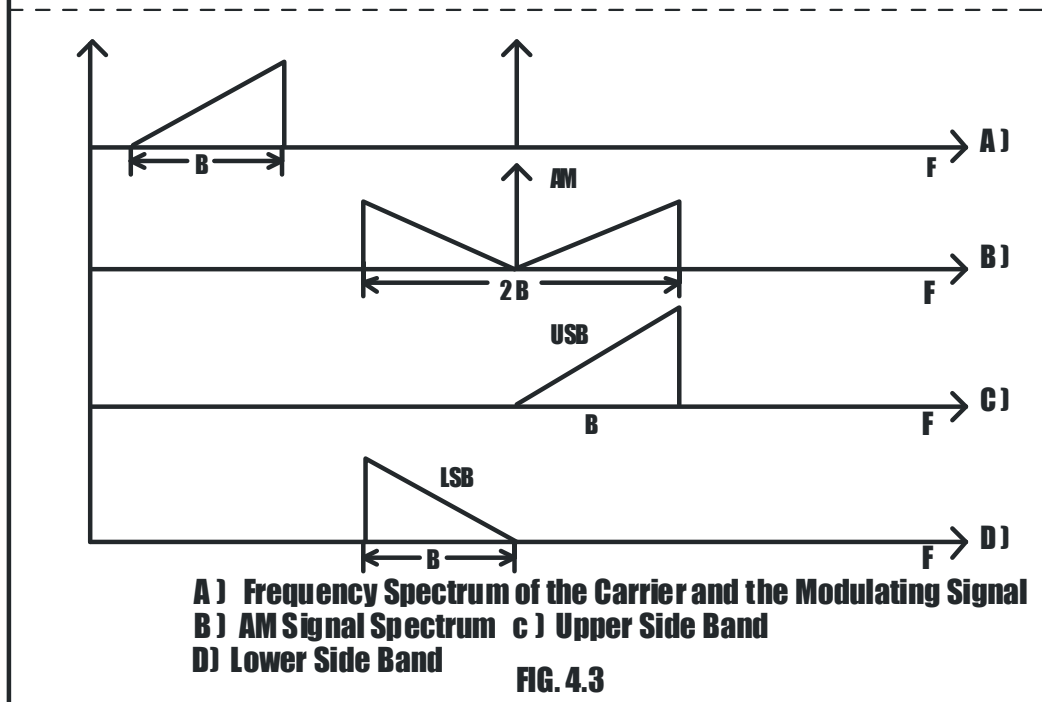


FIG. 4.3

EXPERIMENT NO: 4

NAME:-

SINGLE SIDE BAND MODULATION (SSB)

OBJECTIVE:-

- A. To study the operation of Amplitude modulator with suppressed carrier.
- B. To study Single Side Band Generation.

THEORY:-

4.1 A LOOK AT DSB TRANSMISSION:

Double sideband transmissions were the first method of modulation developed and for broadcast stations, is still the most popular. Indeed, for medium and long range broadcast stations it is the only system in use.

The reason for such widespread use is that the receiver design can be very simple and reliable. None of the characteristics are particularly critical, so reception is still possible even in adverse conditions.

In this context, a broadcast is information transmitted for entertainment or information available for use by anyone with a receiver. It never requires a response or acknowledgment for the receiving station. So in many ways it is similar to a newspaper or magazine which is published and distributed to anyone who is interested in reading a copy.

Radio is also used for communications in which the signal is addressed to receiving station or a group of stations. Using the written word, this would correspond to a private letter or perhaps business or military information being exchanged. For this type of communication other systems are used.

There are two serious drawbacks to the DSB AM system.

4.1.1 DSB IS WASTEFUL OF POWER:

The first problem is to do with the power distribution in DSB amplitude modulated wave.

In FIG. 4.1, we are transmitting a total power of 1.64KW. Out of this power, the carrier contains 1KW and does not contain any of the information being transmitted. The side frequencies each have a power of 320W and each carries a copy of the same information signal. So, in this example, 1.64KW is being used in order to transmit only 320W.

4.1.2 DSB HAS A WIDE BANDWIDTH:

When we amplitude modulated a carrier wave with a range of frequencies we generated an upper sideband consisting of the carrier frequency plus each of the components in the information wave together with the carrier wave minus each of the components.

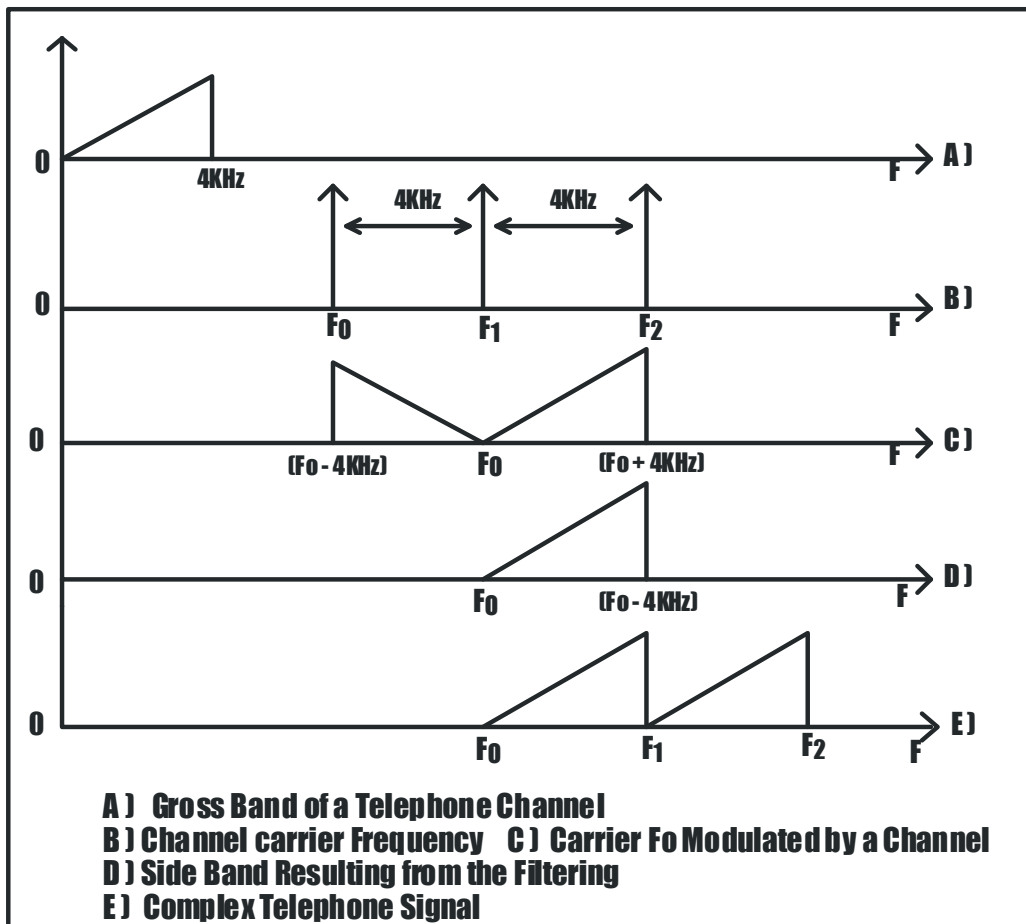
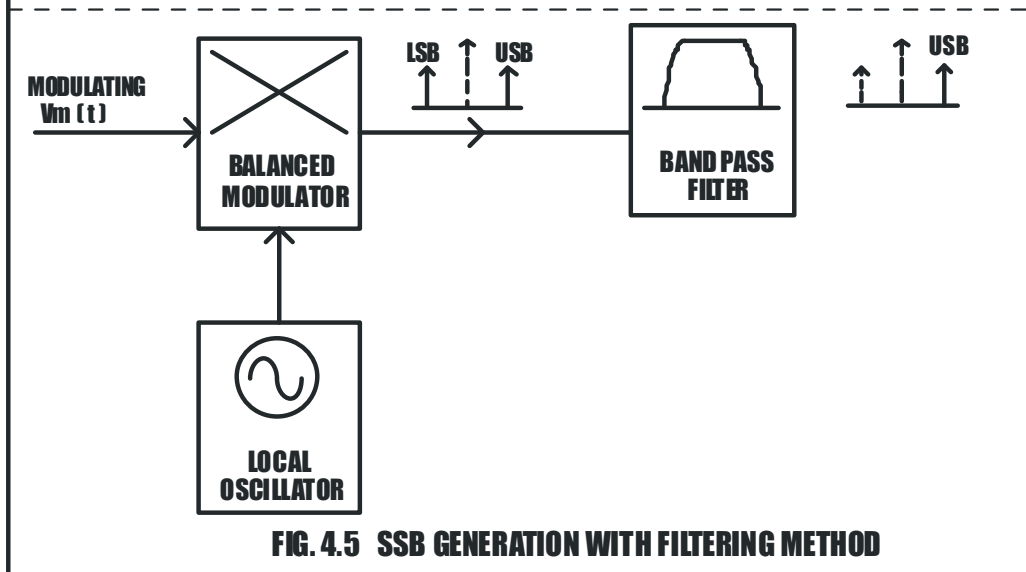


FIG. 4.4



4.2 How much of the DSB AM wave is really needed?

The whole purpose of the modulation system is to transfer information from one place to another. **How efficiently does it achieve this?**

We are transmitting two sidebands and a carrier.

The carrier contains no useful information at all and yet contains over half the total power. This is clearly a waste.

Even the sidebands can be improved. We can remember that combining the information signal and the carrier wave rise to an upper and a lower sideband, each of which contains a copy of the information being transmitted.

There is no necessity to send two copies of the same information. So this is a waste of power and bandwidth.

4.2.1 A waste of bandwidth?

There are more stations seeking permission to transmit than there are frequencies available. Within a band of say, 100KHz, we can transmit only 5 signals that occupy 20KHz, but 10 stations if they agree to limit their transmitted bandwidth to 10KHz. It is for this reason that we limit the highest frequency component within the information wave. High quality music transmissions on the medium waveband are therefore not allowed.

If so much of the transmitted wave is not required, then why transmit it?

4.3 DOUBLE SIDEBAND SUPPRESSED CARRIER TRANSMISSION (DSBSC):

If we avoided using the carrier frequency shown in FIG. 4.2, we would save ourselves 1KW of the transmitted power.

An example spectrum of the transmitted wave is shown in FIG. 4.2.

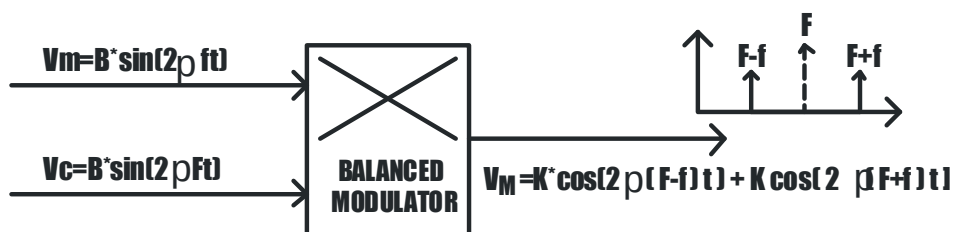
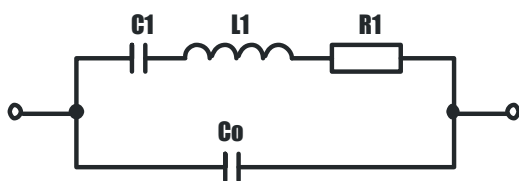
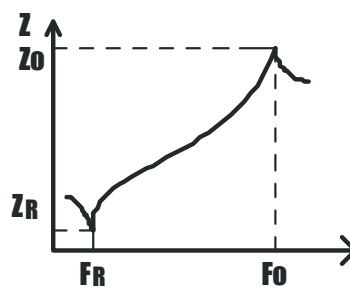
You may be thinking 'that's a bit strange – how come the carrier can suddenly be removed when it was so important before!

Well, the carrier has done its job – in the modulator. That is where we needed it to move or translate the audio signals up to radio frequency values, which can be radiated by the antenna. This shifting, or translating of frequencies is the main function of a modulator.

At the transmitter, the carrier can easily be removed by a band stop filter designed to eliminate the carrier frequency whilst allowing the two sidebands to be transmitted.

At the receiver, the carrier must be re-inserted to produce the modulation envelope to enable the detector to extract the information signal.

The carrier has to be re-inserted at exactly the correct frequency to reproduce the original AM waveform (within a few Hertz). If it is not, there are serious problems with the reception.

**FIG 4.6 BALANCED MODULATOR****A) TWO TERMINAL FILTER****B) THREE TERMINAL FILTER****C) EQUIVALENT ELECTRICAL CIRCUIT**

F_R — Resonance frequency
 F_O — Anti Resonance Frequency
 Z — Impedence

D) IMPEDANCE**FIG 4.7**

Take a situation in which the upper and lower side frequencies are spaced 3KHz either side of the carrier at:

$$600 - 3 = 597\text{KHz and } 600 + 3 = 603\text{KHz}$$

Now, let's assume that the receive carrier were to be re-inserted at an incorrect value of 601KHz. This would result in a spacing of only 3KHz between the carrier and the upper side frequency and 5KHz between the carrier and the lower side frequency.

4.3.1 What effect would this have?

Remembering our previous exercise in which we created an AM envelope by plotting a graph, we can see that these incorrect side frequency spacing will give rise to a badly deformed modulation envelope and hence a distorted output sound.

With this type of transmission, the receiver would be carefully tuned in to the correct frequency and the station would be received. A few moments later, the re-inserted carrier frequency would drift slowly off tune. We would have to reach over and retune the radio and settle back to enjoy the next few seconds of broadcast until the drift starts again.

The frequency control necessary to ensure that the re-inserted carrier stays at exactly the correct value regardless of changes of temperature, vibration etc. would make the receiver too complex and expensive for domestic use.

For this reason, DSBSC is very seldom used. Overall, the waste of transmitted power to send the carrier is less expensive than the additional cost of perhaps several million high quality receivers.

Such receivers are used for professional (and amateur) communications but are expensive, between ten and a hundred times the cost of a standard radio receiver.

4.4 SINGLE SIDE BAND MODULATION (SSB):

The AM modulation process produces a carrier and two side bands.
See $v_m(t)$ is a modulation signal:

$$v_m(t) = B \sin(2\pi f \cdot t)$$

and $v_c(t)$ the carrier:

$$v_c(t) = A \sin(2\pi F \cdot t)$$

The modulated signal $v_M(t)$ consists of three sine components:

$$A \sin(2\pi F \cdot t)$$

Carrier

$$m \frac{A}{2} \cos[(2\pi(F-f) \cdot t]$$

Lower side band

$$m \frac{A}{2} \cos[(2\pi(F+f) \cdot t]$$

Upper side band

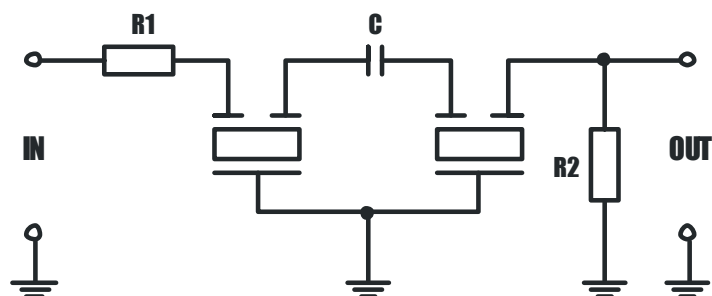


FIG. 4.8 CERAMIC FILTER WITH COUPLED CELLS

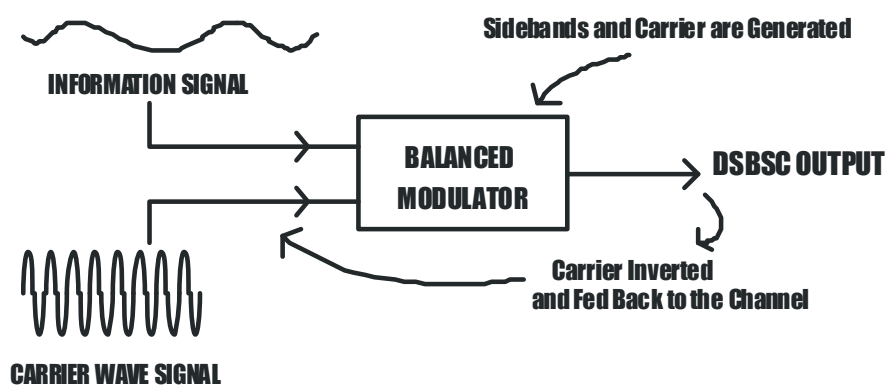


FIG. 4.10 BALANCED MODULATOR

If the modulating signal does not consist on a single sine wave, but of a generic signal with frequency spectrum ranging between f_1 and f_2 , with the amplitude modulation this spectrum is moved over or under the carrier (FIG. 4.3).

It is evident that the carrier takes no information, as it keeps constant in amplitude as well as in frequency independently from the modulating signal.

It is also evident that the two side bands are an exact image of the each other. The amplitudes of both vary in the same way according to $m \cdot A/2$ and so also the frequencies are different from the frequency of the carrier of the same quantity f .

You can see that all the information can be transmitted using one single side band: The carrier is superfluous and the other side band redundant.

4.5 SSB AGAINST AM:

In respect to the amplitude modulation the single side band presents the following advantages:

- The band of the modulated signal is reduced to half. This means that, e.g., in the same range of frequencies, there can be a double number of communication channels.
- All power emitted by the transmitter is associated with the information to be transmitted, differently from the AM in which most of the power is associated with the carrier.

Among the most evident advantages of the SSB in respect to the AM we can mention:

- Higher circuit complexity in the modulator as well as in the demodulator.
- Need to locally generate the carrier in the receiver to detect the modulated signal correctly.

USE OF THE SSB:

- Radio transmissions.
- Multichannel telephone transmission (FIG. 4.4), with the technique of Frequency Division Multiplexing (FDM).
- High-speed data transmission (modem V35/V36/V37).

4.6 SSB GENERATION BY FILTERING:

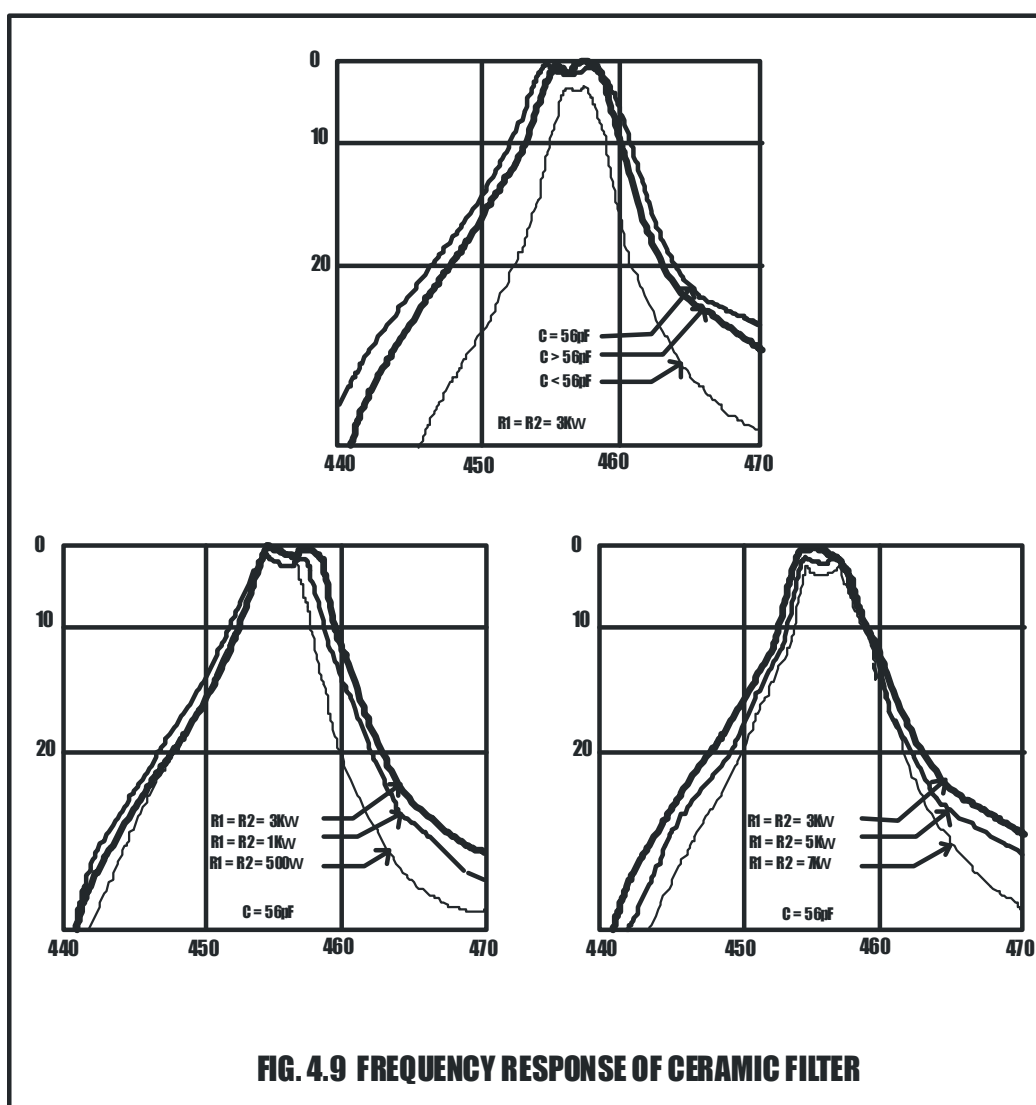
The purpose of the methods used for the generation of an SSB signal to suppress the carrier and one side band.

The most used method is the filtering, which is actuated in two consequent phases (FIG. 4.5).

- An amplitude modulation with suppressed carrier is generated, known also as Double Side Band (DSB).
- A next band pass filter extracts one of the two side bands.

The most common modulators are the balanced modulators and the ring modulators, described hereafter. The filters used for the separation of one side band must present a high steepness of the attenuation curve between the pass band and the attenuated band. Quartz, ceramic, active or passive L-C filters are used as function of the operating frequencies and the applications.

To obtain the lower or the upper side band, the frequency of the carrier generator is varied, keeping the filter as it is.



4.7 BALANCED MODULATOR:

The “balanced modulator” is a circuit, which can generate an amplitude modulation with suppressed carrier, consisting of single side bands. To obtain this result, it is sufficient to multiply the carrier signal and the modulating one between them. If $v_m(t)$ is the modulating signal:

$$V_m(t) = B \cdot \sin(2\pi f \cdot t)$$

And $v_c(t)$ is the carrier:

$$V_c(t) = A \cdot \sin(2\pi F \cdot t)$$

The modulated signal $v_M(t)$ obtained by their multiplication is constituted by two sine components (FIG. 4.6).

$$K \cdot \cos[(2\pi (F-f) \cdot t)]$$

Lower side band

$$K \cdot \cos[(2\pi (F+f) \cdot t)]$$

Upper side band

An integrated circuit, commonly used as signal multiplier and then as balanced modulator, is the 1496. This consists in a quadruple differential amplifier, driven by a further differential amplifier. The output signal consists of a constant (gain), which multiplies the product of two input signals.

4.8 CERAMIC FILTER:

A ceramic filter is a band pass filter using a piezoelectric ceramic material as Electro-mechanic transducer and as mechanic resonator. The symbols of the cells with 2 and 3 terminals are reported in FIG. 4.7. The same figure reports also the equivalent circuit of the filter with 2 terminals and its impedance at frequency variations. In many cases the filter consists of 2 cells coupled using a small capacity.

Important parameters of the ceramic filter are the input impedance, the output impedance and the coupling capacity between two sections. FIG. 4.9. Shows the frequency response of the ceramic filter with 455 KHz (the same used for the exercises) at variation of the coupling capacity and input and output impedance matching.

4.9 THE SSB TRANSMITTER:

The design of the SSB transmitter is accomplished in two stages. First we generate a DSBSC signal and then remove the lower sideband to achieve the final SSB result.

4.10 GENERATING THE DSBSC SIGNAL:

To do this, we use a Balanced Modulator. The principle of this circuit is shown in FIG. 4.10.

Internally, the balanced modulator generates the AM waveform, which includes the carrier and both sidebands. It then offers the facility to feed a variable amount of the carrier back into the modulator in anti-phase to cancel the carrier output. In this way we can balance out the carrier to suppress it completely leaving just the required DSBSC waveform.

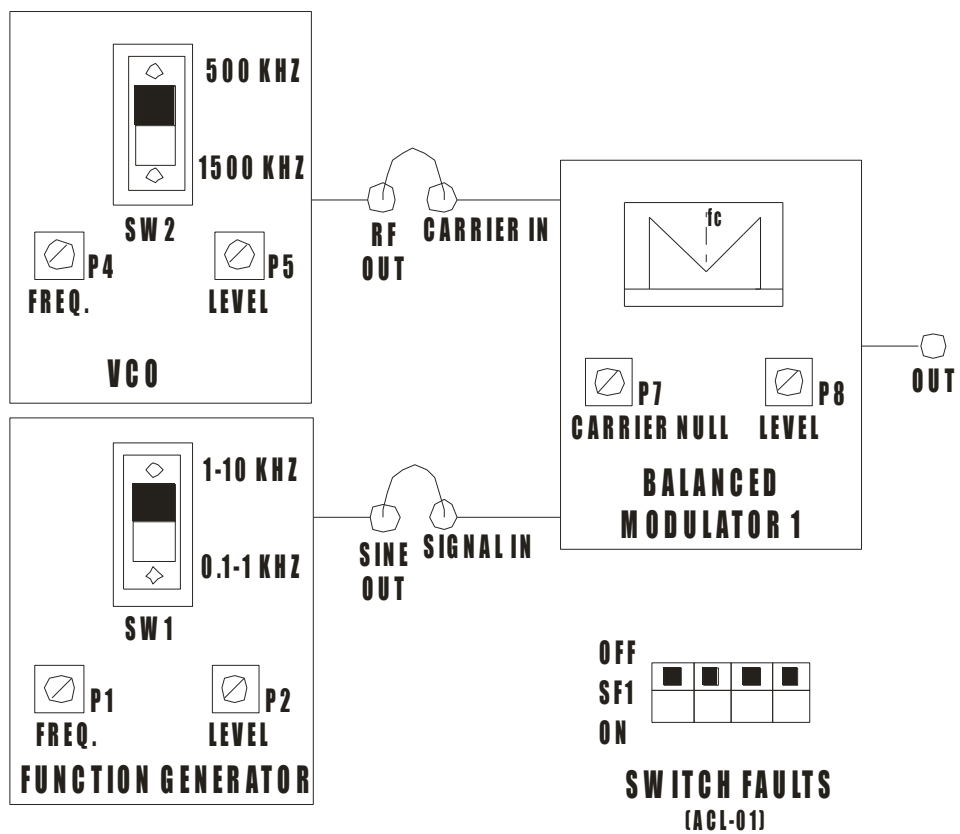


FIG.4.11 BLOCK DIAGRAM FOR STUDY OF AM GENERATION WITH SUPPRESSED CARRIER

4.11 FROM DSBSC TO SSB:

This is basically the same pattern of events as we met in section 4.8. The DSBSC signal consists of the two sidebands, one of which can be removed by passing them through a band pass filter.

On the ACL-AM this is achieved as shown in FIG. 4.5.

The inputs to the balanced modulator comprise the audio inputs from the audio oscillator, which extend from 100Hz to 10KHz, and the carrier input. On the ACL-AM kit this carrier oscillator, although marked as VCO, actually needs to operate at a frequency, which is a little less than this, around 453KHz.

It is to ensure that the upper sideband can pass through the ceramic band pass filter but the lower sideband cannot pass through.

In FIG. 4.3 the upper sideband can be seen to be within the pass band of the ceramic filter but the lower sideband is outside and will therefore be rejected. The sideband frequencies are quite close to each other and a good quality ceramic filter is required. A ceramic filter passes only a narrow range of frequencies with a sharp cut-off outside of its pass band.

4.12 TRANSMITTING THE SSB SIGNAL:

So far, we have got an SSB signal but it is at a frequency around 455KHz. This transmission is fed to receiver for demodulation.

A. TO STUDY THE OPERATION OF AMPLITUDE MODULATOR WITH SUPPRESSED CARRIER

EQUIPMENTS:-

- Modules ACL-AM.
- Power supply
- 20MHz Oscilloscope.
- Connecting Links.
- Frequency meter.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

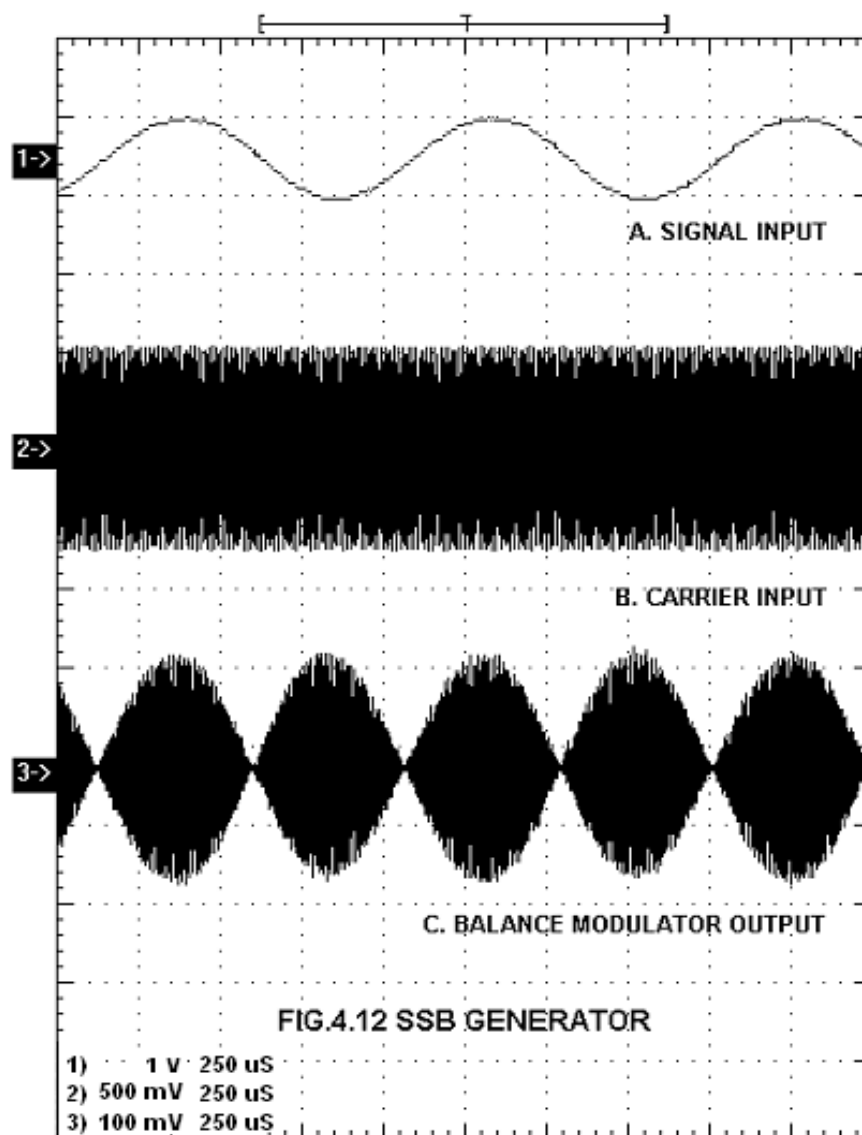
1. Refer to the FIG. 4.11 & Carry out the following connections.
2. Connect o/p of SINEWAVE section (ACL-AM) **SINE OUT** post to the i/p of Balance Modulator (ACL-AM) **SIGNALIN** post.
3. Connect o/p of VCO (ACL-AM) **RF OUT** post to the input of Balance Modulator **CARRIER IN** (ACL-AM) post.
4. Connect power supply with proper polarity to the kit ACL-AM, while connecting this; ensure that the power supply is OFF.
5. Switch on the power supply and Carry out the following presetting:
 - **SINEWAVE SECTION:SINE OUT POST:** LEVEL about 1 Vpp; FREQ. About 3 KHz.

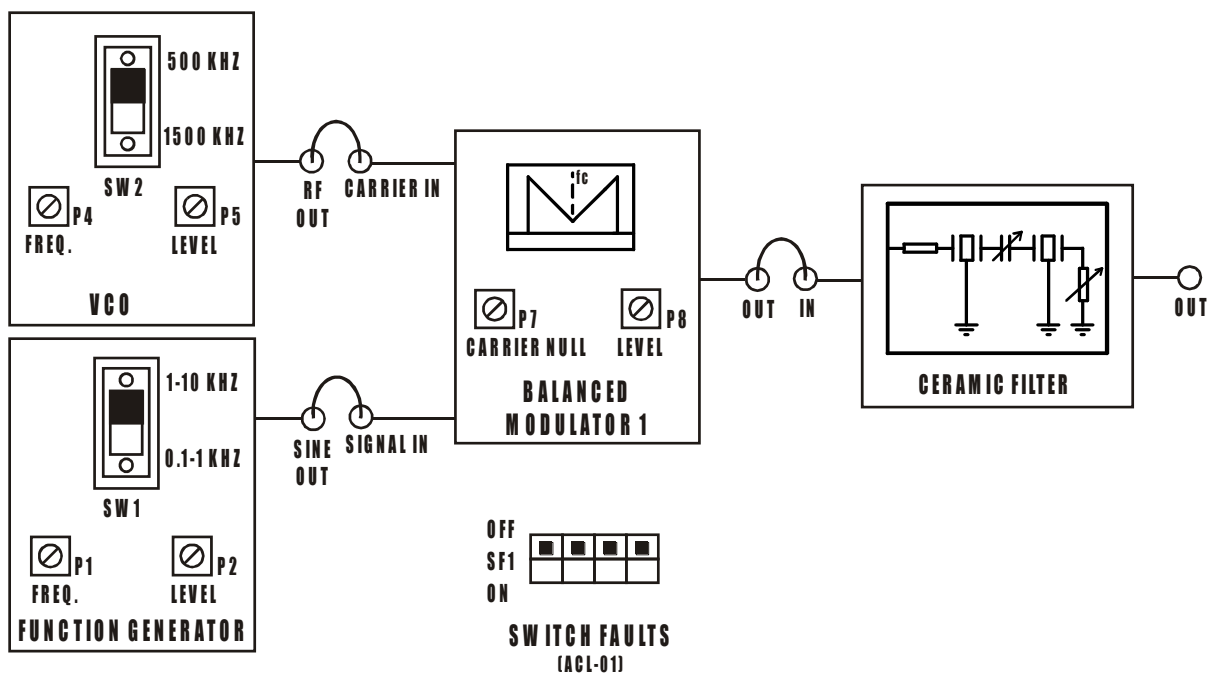
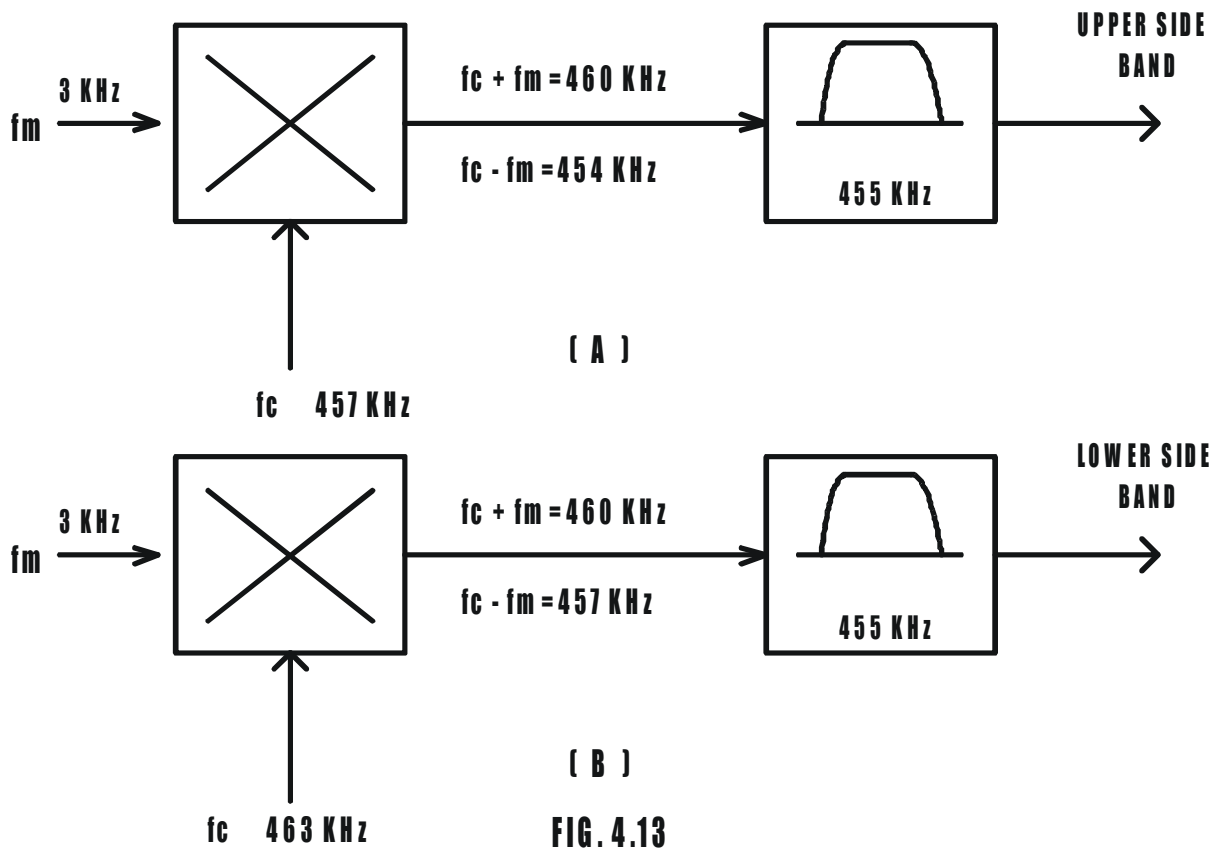
- **VCO:** LEVEL about 2 V p-p; FREQ. about 457KHz. Switch on 500KHz
 - **BALANCED MODULATOR:** CARRIER NULL in central position or such that you should get waveform as shown in FIG. 4.12(c), so that the modulator is “balanced” and obtain an AM signal across the output with suppressed carrier. OUT LEVEL in fully clockwise position.
6. Connect the oscilloscope to the inputs of the modulator (posts **SIGNALIN** and **CARRIER IN**) and detect the modulating signal and the carrier signal (FIG. 4.12 A/B).
 7. Move the probe from post **SIGNAL IN** to post **RF OUT** (Modulator output), where the modulated signal is detected (FIG. 4.12 C). Reset the level of the modulating signal, and adjust the CARRIER NULL to obtain the minimum of the output carrier of the modulator. Take to about 0.5Vpp the amplitude of the modulating signal. Note that the waveform of the modulating signal does not correspond to the envelope of the modulated signal, as it occurs, instead, in case of signal AM.
 8. Vary the amplitude of the modulating signal and check the corresponding variation of the modulated signal amplitude. Note that, differently from the AM modulation where the modulated signal is never null, the modulated signal annuls when the modulating signal is null.
 9. Vary frequency and waveform of the modulating signal, and check the corresponding variations of the modulated signal.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time.

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.





B. TO STUDY SINGLE SIDE BAND GENERATION.

EQUIPMENTS:-

- Modules ACL-AM.
- Power supply
- Oscilloscope.
- Connecting Links.
- Frequency meter.

NOTE : KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG. 4.14 & Carry out the following connections.
2. Connect o/p of **FUNCTION GENERATOR** section (ACL-AM) **SINE OUT** post to the i/p of Balance Modulator1 (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of **VCO OUT** post (ACL-AM) to the input of Balance modulator 1 **CARRIER IN** post (ACL-AM).
4. Connect power supply with proper polarity to the kit ACL-AM&ACL-AD, while connecting this; ensure that the power supply is OFF.
5. Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR:SINEOUT** POST LEVEL about 1Vpp; FREQ. About 3 KHz.
 - **VCO:** LEVEL about 2Vpp; FREQ. about 457 KHz.
 - **BALANCED MODULATOR1:** CARRIER NULL in central position or such that you should get signal as shown in FIG. 4.12(c), so that the modulator is “balanced” and obtain an AM signal across the **output with suppressed carrier, OUT LEVEL in clockwise position**
6. Connect **OUT** post of Balanced modulator1 to **IN** post of ceramic filter.
7. Observe the SSB signal at the **OUT** post of ceramic filter. You can observe that the filter extracts only one of the two components (sidebands) generated by balance modulator.
8. Measure the frequency f_c of the carrier (post **CARRIER IN**), f_m of the modulating signal (post **SIGNAL IN**) and f_{ssb} of the SSB signal across the output of the filter (post **OUT**).
9. Check that:

$$f_{ssb} = f_c + f_m$$

This means that the band extracted by the filter corresponds to the Upper Side Band (FIG. 4.13 A).
10. Repeat the last measurements setting the frequency of the carrier to 463 KHz. You obtain:

$$f_{ssb} = f_c - f_m$$

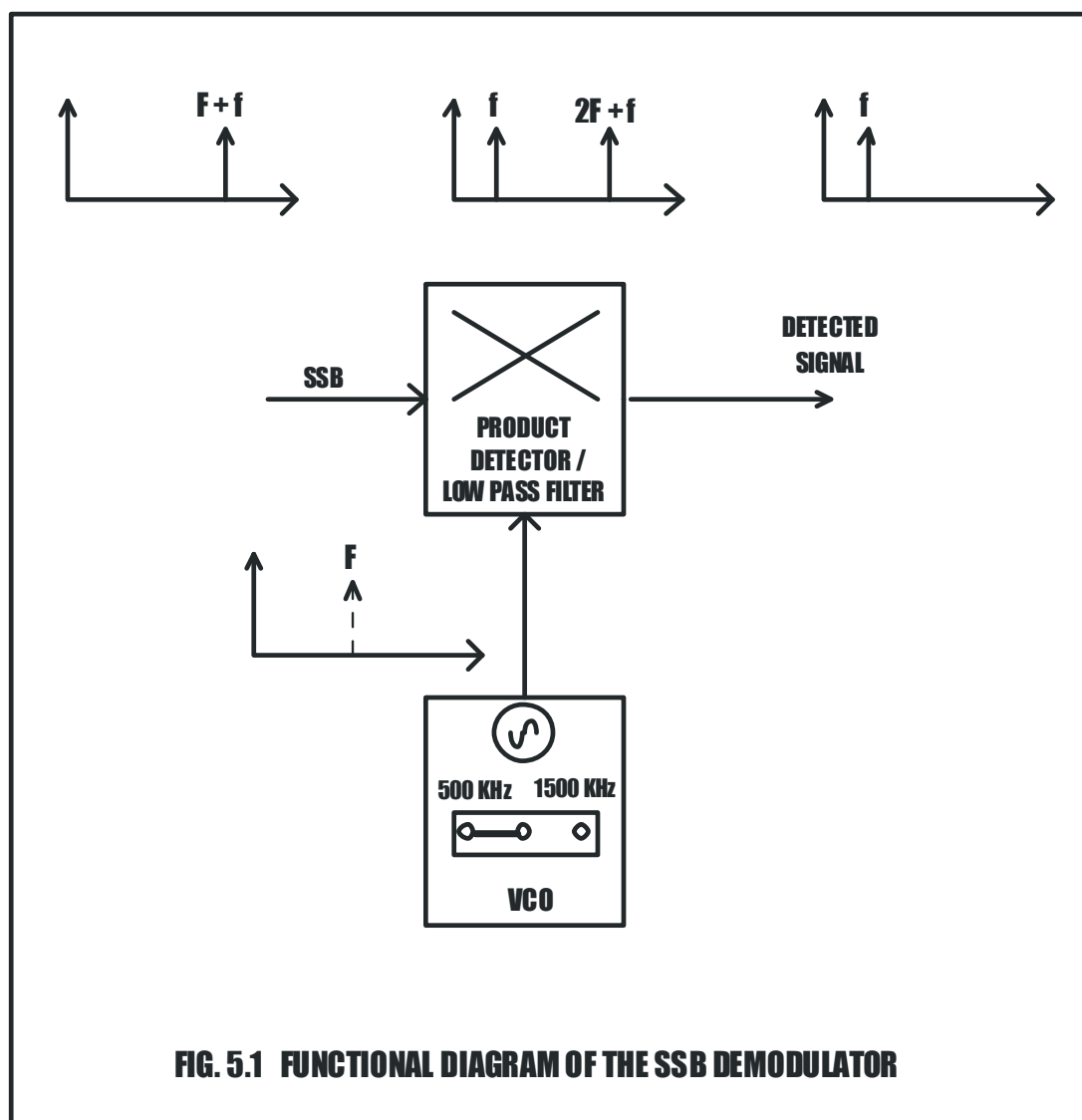
This means that the band extracted by the filter corresponds to the Lower Side Band (FIG. 4.13 B).
11. Increase the frequency of the modulating signal (SINEWAVE) and check that the SSB signal attenuates and tends to a null.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time.

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.

EXPERIMENT NO.5



EXPERIMENT NO: 5

NAME:-

STUDY OF SINGLE SIDEBAND AM RECEPTION

OBJECTIVE:-

- Study of single sideband AM reception using product detector

THEORY:-

5.1 THE RECEIVER

The receiver is of the normal superhet design.

The incoming signal is amplified by the RF Amplifier and passed to the mixer. The other input to the mixer is the local oscillator, which is running at 455KHz above the frequency to which the receiver is tuned. The mixer generates sum and difference signals and the lower of the two is the resulting IF signal occupying a range of frequencies around 455KHz.

The audio information must now be separated from these IF frequencies.

5.2 RECOVERING THE AUDIO SIGNALS

This is achieved by a circuit called an SSB AM decoder. It does the same job as a demodulator or detector in a DSB AM receiver. The SSB AM decoder is slightly more complicated when compared with the DSB equivalent (see FIG. 5.1).

One way of extracting the audio signals is to use a mixer to shift the frequencies just as we have done several times already.

If a mixer combined an input of (audio + 455KHz) with another input of 455KHz the resultant outputs would be the usual 'sum' and 'difference' frequencies.

The product detector and the 455KHz input to the product detector is provided by an oscillator called a VCO.

5.3 SSB SIGNAL DEMODULATION

The SSB signal demodulation requires the presence of the carrier, which must be locally generated in the receiver.

To obtain the starting modulating signal from the modulated signal, multiply the modulating signal and the locally generated carrier, and filter the result to extract the modulating signal. The circuit carrying out the multiplication of the two signals can be the same used to generate the modulation with suppressed carrier in transmission. When used as demodulator the circuit is commonly called product detector.

Let's now examine the demodulator operation. If for example:

$$V_M(t) = K \bullet \cos[(2\pi (F+f) \bullet t]$$

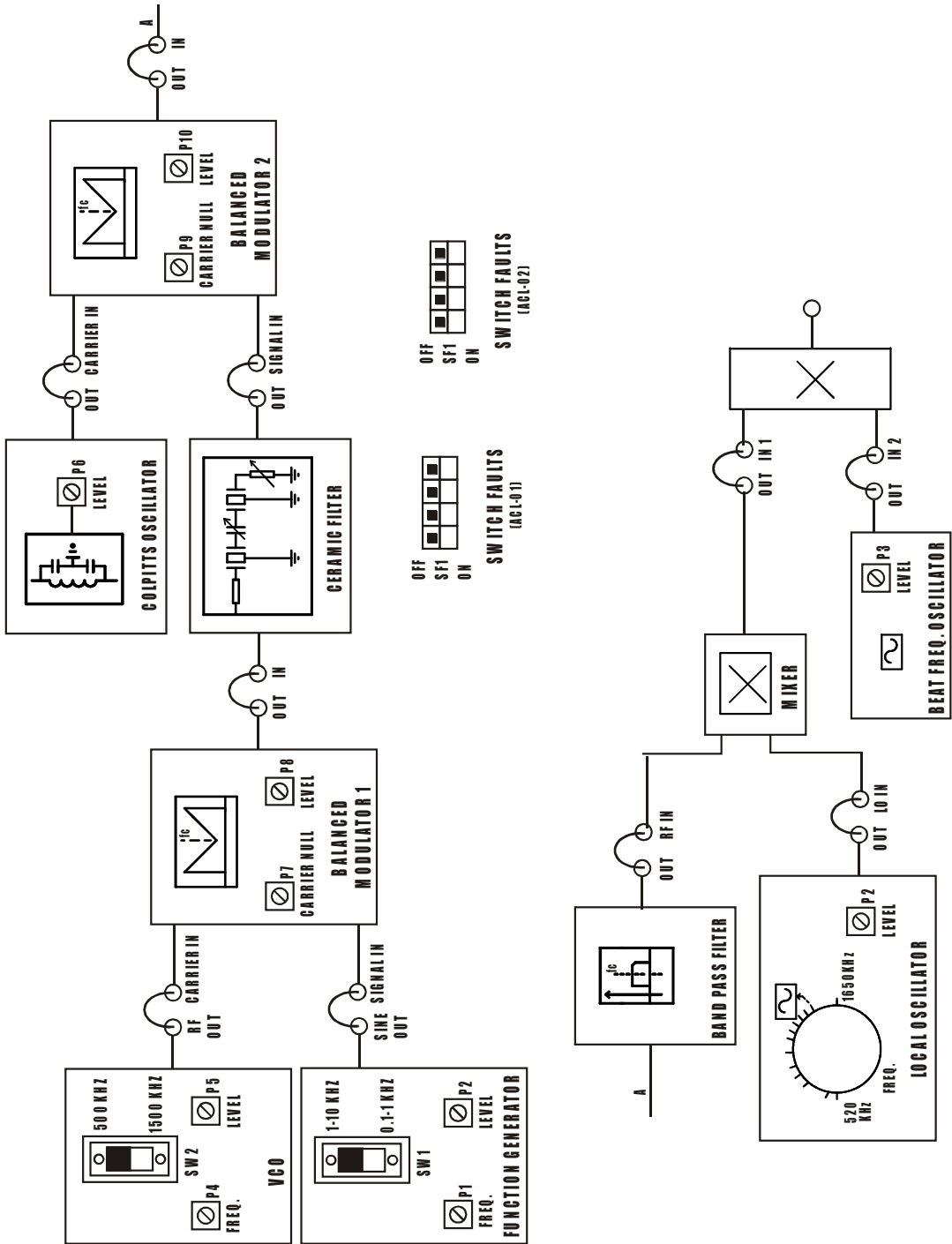


FIG. 5.2 BLOCK DIAGRAM FOR STUDY OF SSB RECEPTION USING PRODUCT DETECTOR VIA CABLE

The SSB signal (upper side band in the example) is:

$$V_c(t) = A \cdot \sin(2\pi F \cdot t)$$

The carrier is locally generated in the receiver. The product $M(t) \cdot v_c(t)$

$$K' \cdot \sin(2\pi f \cdot t)$$

$$K' \cdot \sin(2\pi (2F+f) \cdot t)$$

Then, supposing to have the upper side band and inserting again the carrier, across the output there are the carrier modulator and a really higher frequency, which can be removed with a low pass filter. If the carrier inserted in reception again does not have the same frequency of the carrier suppressed in transmission, the frequency of the demodulated signal is translated by the difference between the two carriers, altering in this way the reception. To reduce these differences to the minimum, the transmission carrier as well as the reception one are usually generated by quartz oscillators. FIG. 5.1 reports the functional diagram of the SSB demodulator.

• STUDY OF SINGLE SIDEBAND AM RECEPTION USING PRODUCT DETECTOR.

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD
- Power supply
- 20MHz Oscilloscope
- Connecting Links
- Frequency meter

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

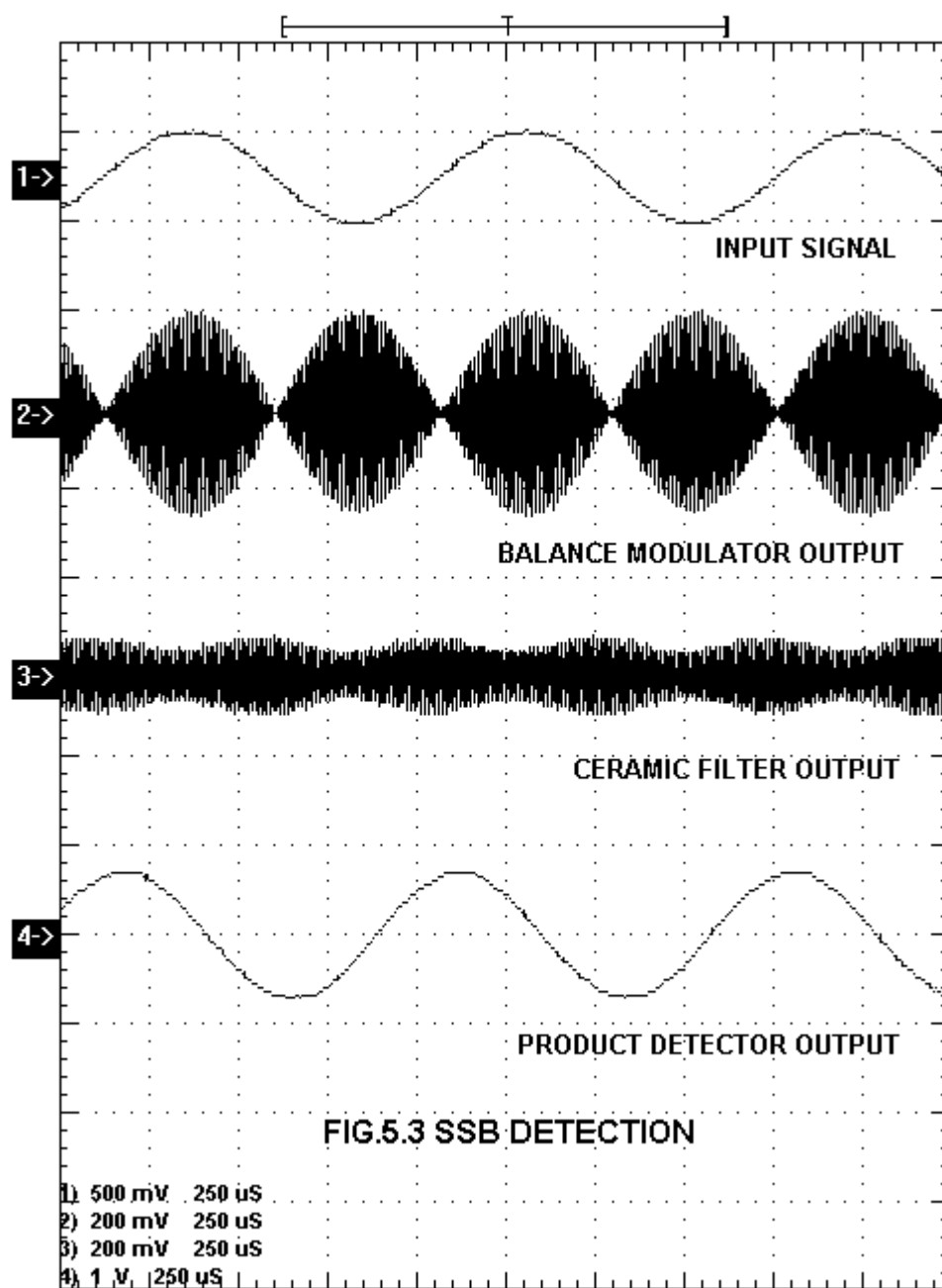
1. Set the connections for SSB Reception as shown in FIG. 5.2.
2. Connect o/p of **FUNCTION GENERATOR** (ACL-AM) **SINE OUT** post to the i/p of Balance Modulator1 (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of **VCO** (ACL-AM) **OUT** post to the input of Balance modulator1 **CARRIER IN** (ACL-AM) post
4. Connect power supply with proper polarity to the kit ACL-AM&ACL-AD, while connecting this; ensure that the power supply is OFF.
 - a. Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR:SINE OUT** POST: LEVEL about 1 Vpp; FREQ. About 3KHz.
 - **VCO**: LEVEL about 2Vpp; FREQ. about 457 KHz.
 - **BALANCED MODULATOR1**: CARRIER NULL pot P7 in central position, so that the modulator is “balanced” and obtain an AM signal across the output with suppressed carrier, pot P8 fully clockwise
 - **BALANCED MODULATOR2**: CARRIER NULL pot P9 in central position, so that the modulator is balanced and obtain an AM signal across the output with suppressed carrier, pot P10 fully clockwise

5. Connect **OUT** post of Balanced Modulator1 to **IN** post of ceramic filter.
6. Observe the SSB signal at the **OUT** post of ceramic filter. You can observe that the filter extracts only one of the two components (sidebands) generated by balance modulator.
7. Connect **OUT** post of CERAMIC FILTER to **SIGNAL IN** post of BALANCED MODULATOR2.
8. Connect **OUTPUT** post of COLPITT'S OSCILLATOR to **CARRIER IN** Post of BALANCED MODULATOR2.
9. Connect **OUT** post of BALANCED MODULATOR2 to **IN** post of BAND PASS FILTER.
10. Connect **OUT** post of BAND PASS FILTER to **IN** post of MIXER.
11. Observe the SSB signal at the **OUT** post of ceramic filter and at the Output of the band pass filter. You can observe that the filter extracts only one of the two components (sidebands) generated by balance modulator.
12. Connect **OUT** post of Band Pass Filter to **RF IN** post of mixer on ACL-AD.
13. Connect **OUT** post of LOCAL OSCILLATOR to **LO IN** post of MIXER; keep pot P2 in fully clockwise position.
14. Connect MIXER **OUT** to **IN1** post of PRODUCT DETECTOR.
15. Connect **OUT** post of BFO to **IN2** post of PRODUCT DETECTOR; keep pot P3 in fully clockwise position.
16. Observe the final output at the OUT post of product detector by connecting this post to channel 2 of CRO. While doing this to observe the output clearly connect the input modulating signal to channel 1 of CRO & tune local oscillator frequency about 1.02MHz or in such a way that output frequency at PRODUCT DETECTOR should follow input frequency. If output frequency of VCO and BFO coincides you will get better results.
17. Increase the frequency of the modulating signal (SINEWAVE) and check that the detected signal attenuates and tends to a null.
Now you can observe that even the frequency of the VCO (ACL-AM) and BFO (ACL-AD) are same, there is a slight drift in the tone generated by the receiver. This is due to the small frequency drifts in the transmitter and receiver oscillator circuits, leading to the changes in the difference frequency produced by the PRODUCT DETECTOR i.e. there would be some phase error in the frequency generated by the VCO and the frequency generated by the BFO. The effect of this phase error is to introduce the distortion in the demodulated signal, where each frequency component of the original message signal undergoes a constant phase shift. This phase distortion is tolerable in voice communication, because the human ear is relatively insensitive to phase distortion. In particular, the presence of phase distortion gives rise to a **Donald duck voice effect**. In the transmission of music and video signals, on the other hand, the presence of this form of phase distortion is totally unacceptable.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time.

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 3 of SF1 in ACL-AD switch fault section to ON position. This will open the resistor (R63) connected to pin 4 of product detector. The output will get disturbed.



EXPERIMENT NO.6

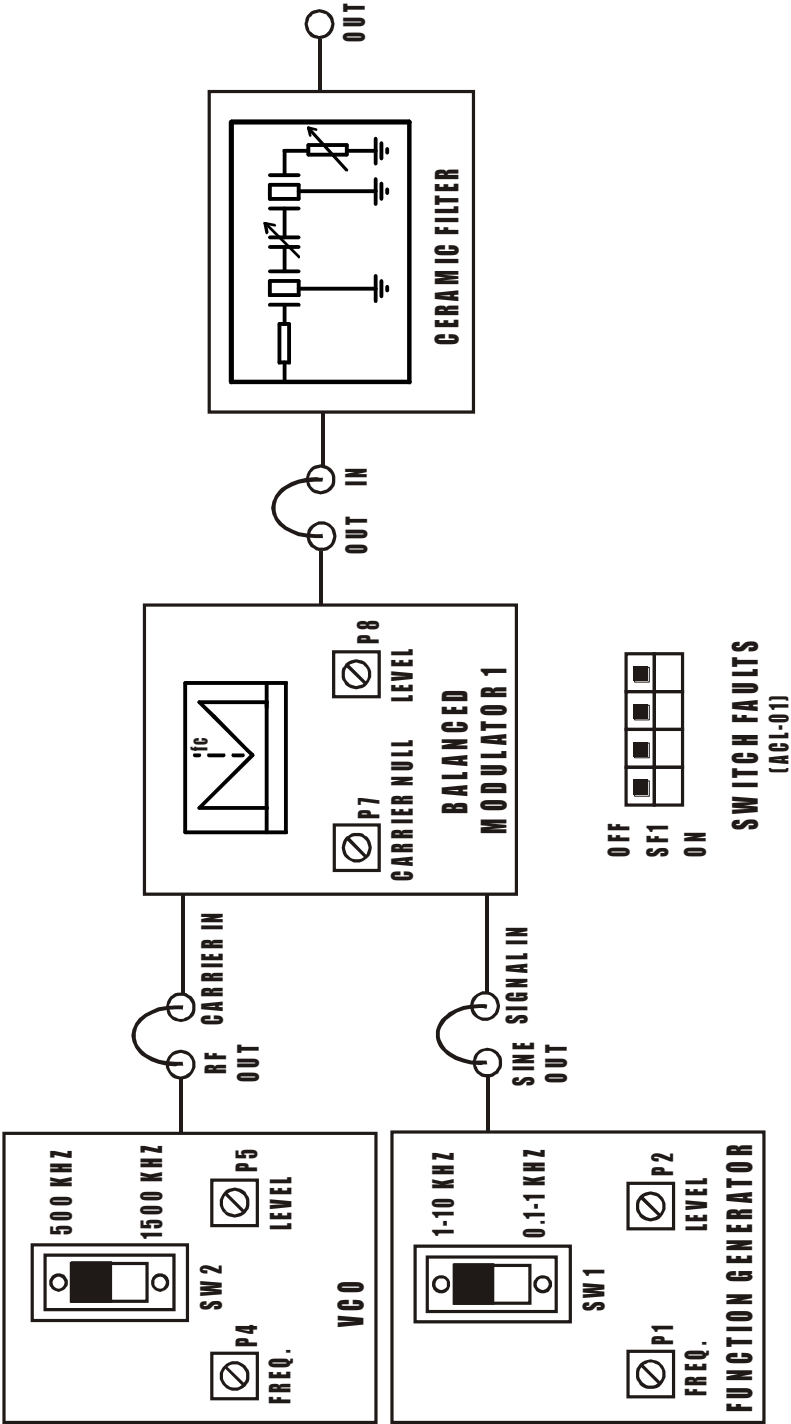


FIG. 6.1 BLOCK DIAGRAM FOR STUDY OF ADJUSTMENT OF TRANSMITTER TUNED CIRCUITS

EXPERIMENT NO: 6

NAME:-

STUDY OF ADJUSTMENT OF TRANSMITTER TUNED CIRCUITS

EQUIPMENT:-

- Module ACL-AM.
- Power supply
- 20MHz Oscilloscope.
- Connecting Links.
- Frequency meter.

NOTE: All adjustments mentioned are for study and reference purpose only.

ADJUSTMENT OF TRANSMITTER TUNED CIRCUITS:

Where signals are to be monitored with an oscilloscope, the scope's input channels should be ac coupled, unless otherwise indicated.

A frequency counter should be used for all frequency measurements.

Use the trimming tool, supplied with the ACL-AM&ACL-AD modules, for trimming inductors. Take care not to turn any inductor's core past its end stop, as this may also result in damage.

VCO

Monitor **OUT** post on the ACL-AM KIT, while using a trimmer tool to adjust transformer **TR1** in the VOLTAGE CONTROL OSCILLATOR block.

By carefully tuning **P4** throughout its range of adjustment, check that the monitored signal is ac at both ends of the adjustment range, frequency range can be adjusted from 350KHz to 500KHz by keeping switch in VCO section on 500KHz and can be adjusted from 350KHz to 1500KHz by keeping switch in VCO section on 1500KHz.

Tune **TR1** so that it is in the center of the 'sine wave' region. Check that the monitored sine wave has amplitude of approximately 2V and a frequency of 1.5MHz. Range of the frequency can be adjusted with the help of transformer **TR1** (± 200 Hz), which is factory set for maximum 1.5MHz frequency.

BALANCED MODULATOR1 TUNED CIRCUIT:

On the ACL-AM KIT, connect the FUNCTION GENERATOR **OUT** post to **SIGNAL IN** post LEVEL about 0.5vpp by **P2** clockwise (MAX) position.

Then turn the CARRIER NULL pot **P7** in the BALANCED MODULATOR1 block to its fully counter-clockwise position.

Monitor post **OUT** and **SIGNAL IN**, by triggering the 'scope, and check that the waveforms appear as shown in FIG. 2E in EXP. NO.2.

Tune POT**P8** until the amplitude of the waveform on post **OUT** is maximum.

While continuing to monitor **OUT** post, turn the carrier null pot **P7** (in the BALANCED MODULATOR1 block) in clockwise direction, until the monitored waveform is as shown in FIG. 4.12 BALANCE MOD OUT in EXP. NO.4.

The amplitude of this waveform is a minimum at points A, B, C etc.

The amplitude of the waveform at these points should be as close to zero as possible to ensure that this is the case, tune **Pot P7** finely until the amplitude of the waveform at these points is as small as possible.

CERAMIC FILTER TUNED CIRCUIT:

For ACL-AM kit,

Switch on the power supply and carry out the following presetting:

- **FUNCTION GENERATOR SECTION:** OUT POST: LEVEL about 1Vpp by POT **P2**; FREQ. About 3KHz by POT **P1**
- **VCO:** LEVEL about 2Vpp by POT **P5**; FREQ. about 457 KHz by POT **P4**
- **BALANCED MODULATOR1:** CARRIER NULL in central position, so that the Modulator is “balanced” and obtain an AM signal across the output with suppressed carrier.

Connect the links as per FIG. 6.1.

Observe the SSB signal at the **OUT** post of ceramic filter. You can observe that the filter extracts only one of the two components (sidebands) generated by balance modulator. To tune this ceramic filter for extracting exact band of frequency, adjust trimmer **CV1** for clear sidebands and maximum amplitude.

OUTPUT AMPLIFIER:

On the ACL-AM kit, Carry out the connections as per fig.2.9 and Carry out the procedure as per experiment **2.B**. Then Observe the AM signal at the output of the amplifier TEST POINT. Output can be controlled by **POT P11**.

We can set transmission signal by means of TR4 for better quality at different frequencies, which is by default set in factory.

EXPERIMENT NO.7

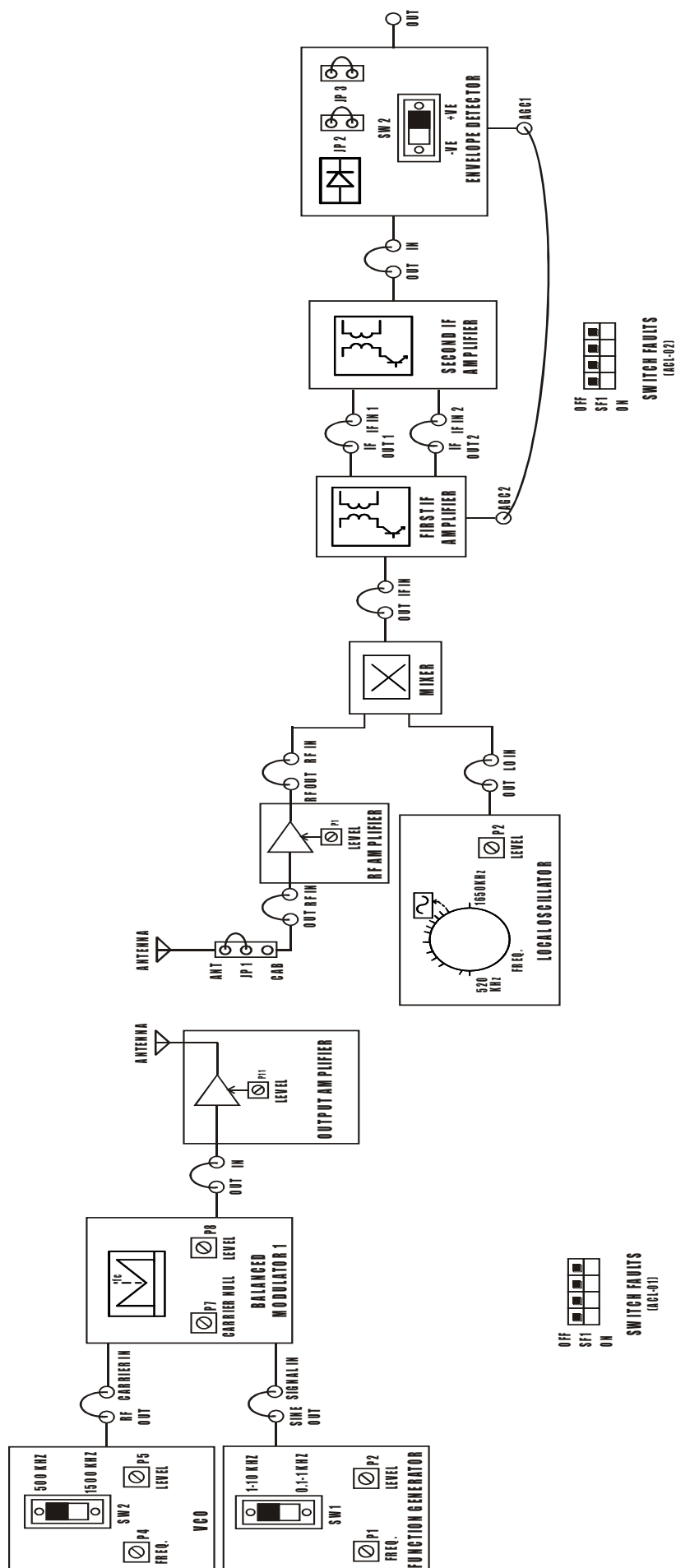


FIG. 7.1 BLOCK DIAGRAM FOR STUDY OF RECEIVER TUNED CIRCUIT FOR DSB

EXPERIMENT NO: 7

NAME:-

STUDY OF ADJUSTMENT OF RECEIVER TUNED CIRCUITS

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD.
- Power supply
- 20MHz Oscilloscope.
- Connecting Links.
- Frequency meter.

NOTE: All adjustments mentioned are for study and reference purpose only.

ADJUSTMENT OF RECEIVER TUNED CIRCUITS:

Where signals are to be monitored with an oscilloscope, the scope's input channels should be ac coupled, unless otherwise indicated. Ensure that X10 oscilloscope probes are used throughout.

A frequency counter should be used for all frequency measurements.

Use the trimming tool, supplied with the ACL-AM and ACL-AD modules, for trimming inductors. Also, take care not to turn any inductor's core past its end stop, as this may also result in damage.

ADJUSTMENT OF R.F. AMPLIFIER TUNED CIRCUIT:

Ensure that the connections as per FIG. 7.1 on the ACL-AD KIT:

- (a) RX INPUT SELECT jumper on **JP1** in **ANT** position
- (b) GAIN preset **P1** (in the R.F. AMPLIFIER block) in its midway position (arrowhead on preset pointing towards top of board)

Set up a connection as per FIG. 7.1 for ACL-AM kit and transmit a sine wave of 0.5V say on 850KHz frequency using DSB modulator. Monitor the output from the R.F. AMPLIFIER block at **OUT** post and follow the steps below:

Turn the Tuning Frequency Dial upto 1300 KHz, until the amplitude of the monitored signal is maximum.

Finally, retune the R.F. amplifier's GAIN preset to its fully clockwise (MAX) position for maximum output of monitor signal.

ADJUSTMENT OF LOCAL OSCILLATOR TUNED CIRCUIT:

Monitor the signal at **OUT** post of LOCAL OSCILLATOR block on the ACL-AD KIT; use a Gang capacitor to adjust frequency in the LOCAL OSCILLATOR block.

By carefully tuning **GANG CAPACITOR** throughout its range of adjustment, check that the monitored signal is ac level at both ends of the adjustment range, frequency range can be adjusted from 900KHz to 2MHz (+/- 200Hz) Level of the signal can be adjusted by using **POTP2**.

Monitor the frequency at **OUT** post of LOCAL OSCILLATOR block.

ADJUSTMENT OF MIXER AND I.F. AMPLIFIER TUNED CIRCUITS:

Carry out the following presetting:

- **LOCAL OSCILLATOR**: LEVEL about 1 Vpp; FREQ. 1000 KHz.
- **VCO** (ACL-AM): LEVEL about 100 mVpp; FREQ. 550 KHz.

Connect the oscilloscope to the output of the mixer **OUT** post and accurately vary the frequency LO until you detect a sine wave signal.

Measure the frequencies of the two input signals and the output signal and check that also the last is the difference of the first two: $F_{IF} = f_{LO} - f_{RF}$.

We can tune trimmer **CV1** until we get clear sine wave, which is equal to F_{IF} .

Also we can do the same procedure to observe signal receiving from antenna.

I.F. AMPLIFIER:

Refer to the connections as done in FIG. 7.1.

- (a) Monitor output of the 1st I.F. AMPLIFIER and tuned transformer TR6 until the amplitude of the monitored signal is maximum 3V.
- (b) Tune transformer TR5 until the amplitude of the monitored signal is maximum.
- (c) Tune transformer TR6 (in the 1st I.F. AMPLIFIER block in component section.) until the amplitude of the monitored signal is once again maximum. So as to recover same DSB AM envelope as seen in transmitter.
- (d) Center frequency of IF AMPLIFIER can be adjusted with the help of trimmer **CV2** and transformer TR6.

All **IFTs**& Trimmers are factory tuned by default.

EXPERIMENT NO.8

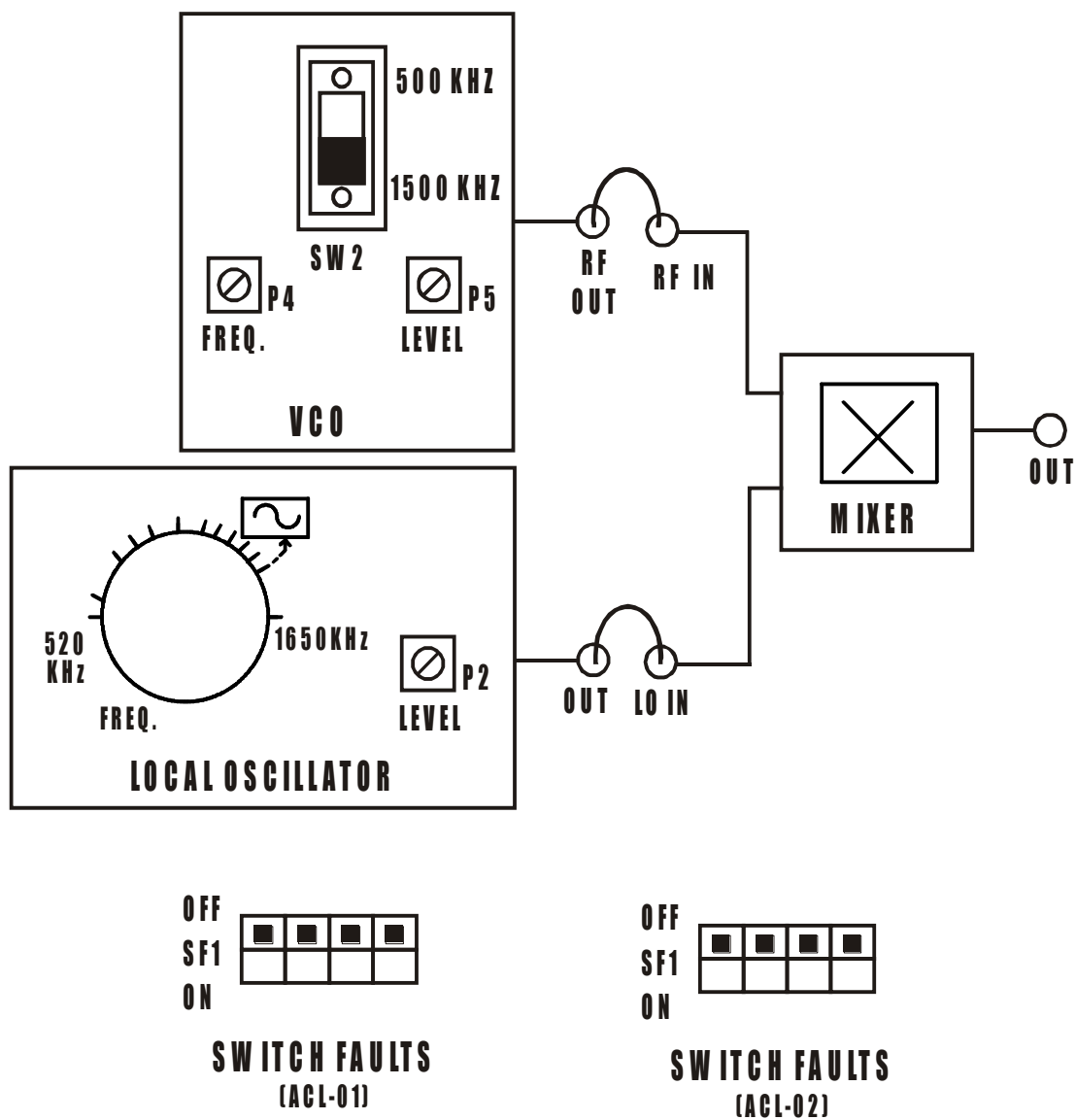


FIG. 8.1 BLOCK DIAGRAM FOR STUDY OF IMAGE FREQUENCY

EXPERIMENT NO: 8

NAME:-

STUDY OF IMAGE FREQUENCIES

OBJECTIVE:-

- A.** To study the frequency relations with the Mixer.
- B.** To study the effect of Image Frequency.

THEORY:

IMAGE FREQUENCIES:

Investigation of Image Frequencies:

This experiment investigates the concept of image frequencies, using the ACL-AD kit. In order to explain what these are, here is a short review of how AM receiver operates.

You will recall that the frequency of the LOCAL OSCILLATOR in the receiver section is set higher than the selected signal frequency by a constant amount, irrespective of the frequency of the selected station. The amount of that constant difference frequency is chosen to be 455KHz, the Intermediate Frequency.

The MIXER in the receiver section mixes the output of the R.F.AMPLIFIER with the output of the LOCAL OSCILLATOR, to extract the 455 KHz difference frequency. This frequency is then passed on to the I.F. Amplifiers, which preferentially amplify signals of around 455KHz.

The output from I.F. AMPLIFIER 2 is then passed on to the detector, in order to recover the original audio signal.

The incoming signal, whose frequency is exactly 455KHz above the frequency of the LOCAL OSCILLATOR, is known as the image frequency.

For every wanted frequency there is a corresponding image frequency. If there is a strong station at or near the image frequency, it will result in irritating whistling noises at the output of the receiver, which will spoil the reception of the wanted station.

In the last section, we saw we could receive a station being broadcast on 550KHz by tuning the local oscillator to a frequency of 1 MHz thus giving the difference (IF) frequency of the required 450KHz.

What would happen if we were to receive another station broadcasting on a Frequency of 1.45MHz?

This would also mix with the local oscillator frequency of 1.45MHz to produce the required IF frequency of 450KHz. This would mean that this station would also be received at the same time as the desired one at 550KHz.

STATION 1:

Frequency 550KHz, Local oscillator 1.0MHz, IF = 450KHz.

STATION 2:

Frequency 1.450MHz, Local oscillator 1.0 MHz, IF = 450KHz.

An 'image frequency' is an unwanted frequency that can also combine with the Local Oscillator output to create the IF frequency.

Notice how the difference in frequencies of the wanted and unwanted stations is twice the IF frequency. In the ACL-AD, it means that the image frequency is always 910KHz above the wanted station.

This is a large frequency difference and even the poor selectivity of the RF amplifier is able to remove the image frequency unless it is very strong indeed. In this case it will pass through the receiver and will be heard at the same time as the wanted station. Frequency interactions between the two stations tend to cause irritating whistles from the loudspeaker.

A. TO STUDY THE FREQUENCY RELATIONS WITH THE MIXER.**EQUIPMENTS:-**

- Modules ACL-AM & ACL-AD.
- Power supply
- Oscilloscope.
- Frequency meter.
- Connecting Links.

NOTE : KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG. 8.1 & Carry out the following connections; make jumper connections as per diagram.
2. Connect the o/p of VCO (ACL-AM) **OUT** post to the i/p of MIXER (ACL-AD) **RF IN** post.
3. Connect local oscillator **OUT** post to **LO IN** of the mixer section of ACL-AD.
4. Connect the power supply with proper polarity to the kit ACL-AM&ACL-AD while connecting this; ensure that the power supply is OFF.
5. Switch on the power supply and Carry out the following presetting:
 - **VCO (ACL-AM):** LEVEL about 100mVpp; FREQ. 550 KHz, Switch on 1500 KHz.
 - **LOCAL OSCILLATOR (ACL-AD):** LEVEL about 1Vpp; FREQ. 1000 KHz.
6. Connect the oscilloscope to the output of the mixer OUT, and accurately vary the frequency of LO until you detect a sine wave signal.
7. Measure the frequencies of the two input signals and the output signal, and check that also the last is the difference of the first two:

$$f_{IF} = f_{LO} - f'_{RF}.$$

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time.

1. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.

B. TO STUDY EFFECT OF IMAGE FREQUENCY

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD.
- Power supply
- Oscilloscope.
- Frequency meter.
- Connecting Links.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG. 8.1 & Carry out the following connections; make jumper connections as per diagram.
2. Repeat the Procedure as explained in A.
3. Switch on the power supply and Carry out the following presetting:
 - **VCO** (ACL-AM): LEVEL about 100mV p-p; FREQ. 1450 KHz, Switch on 1500 KHz.
 - **LOCAL OSCILLATOR**: LEVEL about 1V; FREQ. 1000 KHz.
4. Accurately adjust the above frequency until the output is crossed again by a sine waveform.
5. Assure the frequencies and check that now:

$$f_{IF} = f_{LO} - f'_{RF}$$
6. The last frequency relations indicate that the two signals with different frequency RF are converted to the same frequency IF. If they were contemporarily present, there would be interference between the two and this would make a proper reception of the signal impossible. The unwished frequency is called image frequency. To prevent this inconvenience it is necessary to prevent that the image signal reaches the input of the mixer and this is carried out interposing selective filters between the input of the signal RF and the mixer.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time.

1. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.

EXPERIMENT NO.9

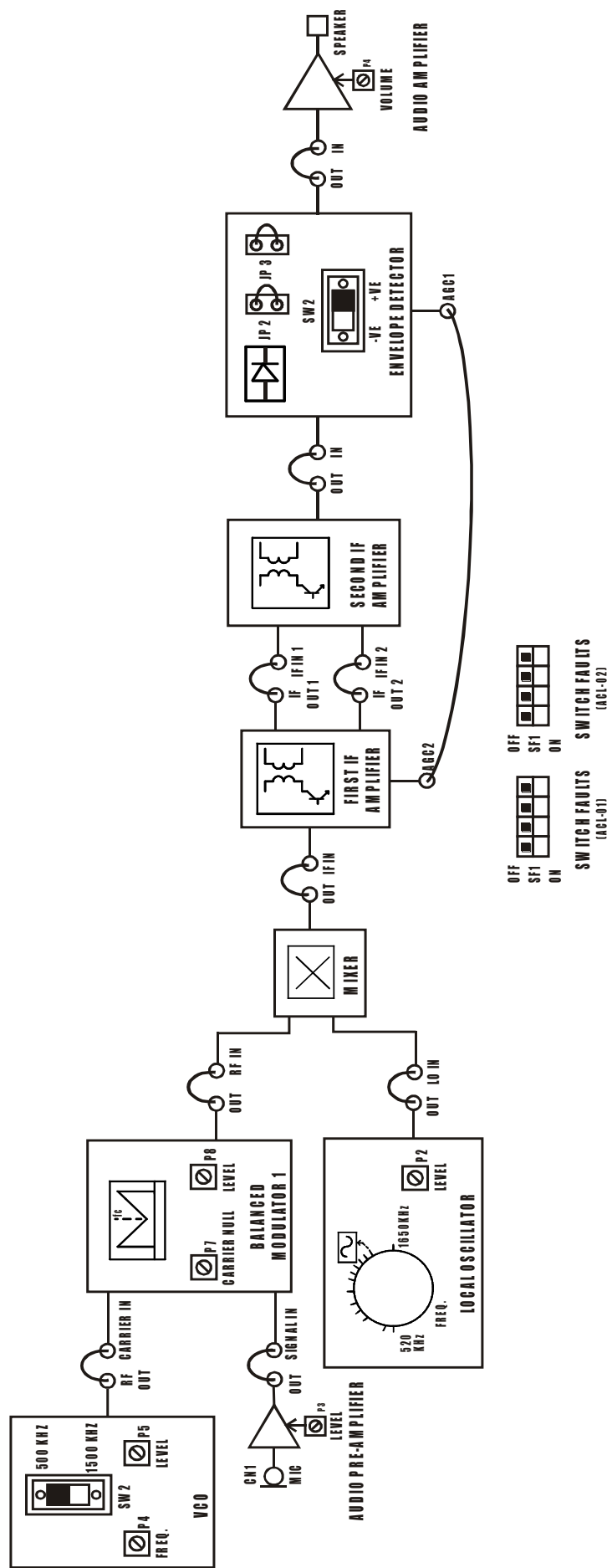


FIG. 9.1 BLOCK DIAGRAM FOR VOICE COMMUNICATION WITH DSB-AM VIA CABLE

EXPERIMENT NO: 9

NAME:-

VOICE TRANSMISSION WITH DSB AM TRANSMISSION/ RECEPTION

OBJECTIVE:-

- A. Voice transmission with DSB AM transmission/ reception via cable.
 - B. Voice transmission with DSB AM transmission/ reception via antenna.
 - C. Voice transmission with SSB AM transmission/ reception.
- A. VOICE TRANSMISSION WITH DSB AM TRANSMISSION/ RECEPTION VIA CABLE.**

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD
- Power supply
- 20MHz Oscilloscope.
- Connecting Links.
- MIC & Speaker.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG. 9.1 & Carry out the following connections.
2. Connect **OUT** post of preamplifier (ACL-AM) to the i/p of Balance Modulator1 (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of VCO (ACL-AM) **OUT** post to the input of Balance modulator1 (ACL-AM) **CARRIER N** post.
4. Connect the power supply with proper polarity to the kit ACL-AM&ACL-AD, while connecting this; ensure that the power supply is OFF.
5. Switch on the power supply and Carry out the following presetting:
 - **VCO:** LEVEL about 2Vpp; **FREQ.** about 850 KHz, Switch on 1500KHz.
 - **BALANCED MODULATOR:** CARRIER NULL completely rotates clockwise or counter clockwise, so that the modulator is “unbalanced” and an AM signal with not suppressed carrier is obtained across the output: Adjust **OUTLEVEL** to obtain an AM signal across the output which amplitude is about 100mVpp.
 - **LOCAL OSCILLATOR (ACL-AD):**1300KHz, 2V.
 - **AUDIO AMPLIFIER:** Volume knob on middle position.
6. Connect local oscillator **OUT** post to **LO IN** of the mixer section.
7. Connect balance modulator out to **RF IN** of mixer section in ACL-AD.
8. Connect mixer **OUT** post to **IF IN** of 1st IF amplifier.
9. Connect **IF OUT1** of 1st IF to **IF IN 1** and **IF OUT2** of 1st IF to **IFIN 2** of 2ND IF AMPLIFIER.
10. Connect **OUT** post of 2nd IF amplifier to **IN** post of envelope detector.
11. Connect post **AGC1** to post **AGC 2**.
12. Connect **OUT** post of Envelope Detector to **IN** post of AUDIO AMPLIFIER and speaker to the speaker jack. If the central frequency of the amplifier and the

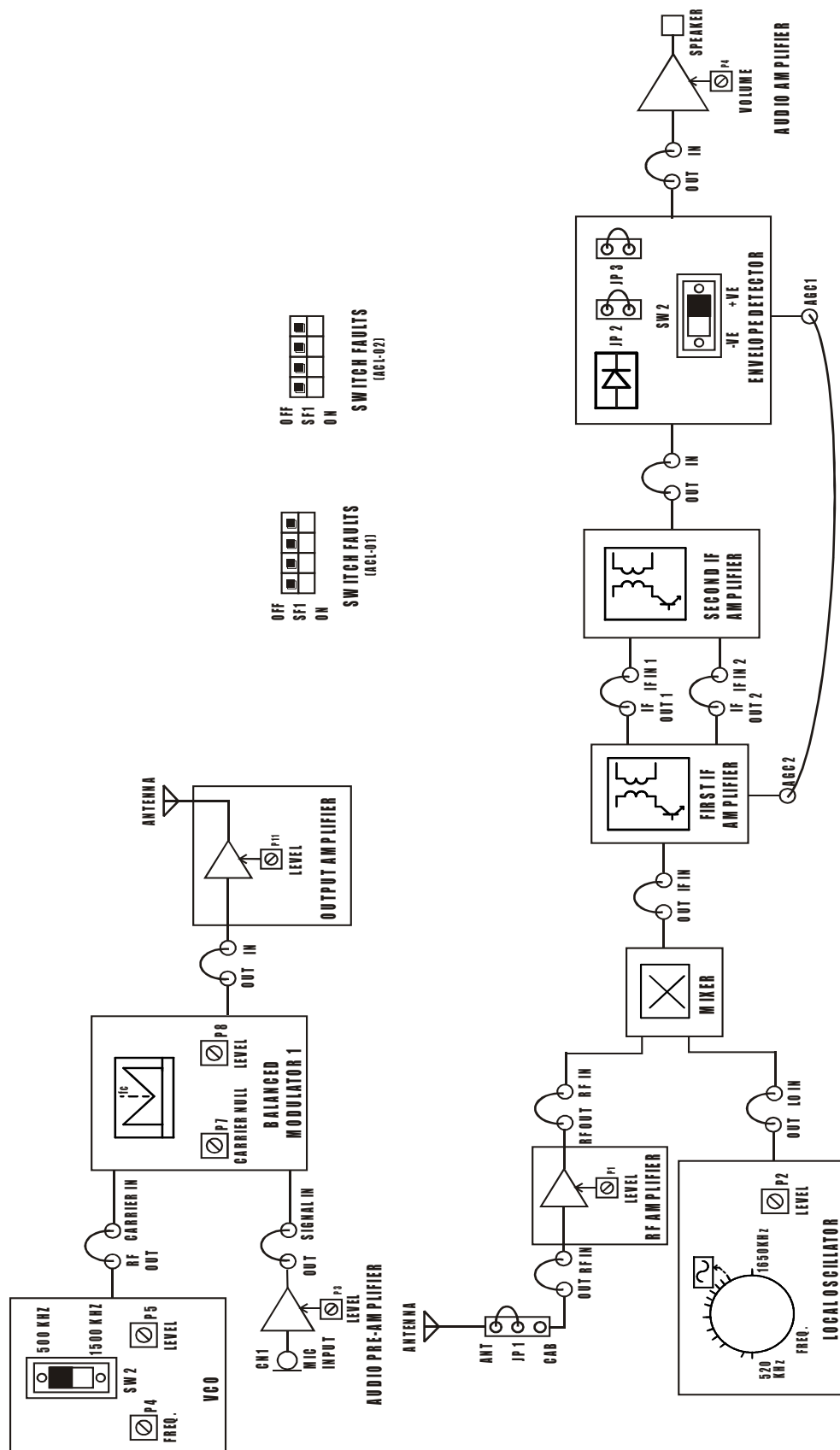


FIG. 9.2 BLOCK DIAGRAM FOR VOICE COMMUNICATION WITH DSB-AM VIA ANTENNA

carrier frequency of the AM signal and local oscillator frequency coincides, you obtain good voice reception on the external speaker. Control local oscillator tuning frequency and volume knob for clear voice.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.
3. Put switch 1 of SF1 in ACL-AD switch fault section to ON position. This will open the capacitor (C35) from the emitter of the first IF amplifier. So the gain will reduce which in turn reduces the output amplitude.
4. Put switch 4 of SF1 in ACL-AD switch fault section to ON position. This will open the diode D1 from the envelope detector, so we will not get the negative envelope output. Also there will not be any effect o AGC signal.

B. VOICE TRANSMISSION WITH DSB AM TRANSMISSION/ RECEPTION VIA ANTENNA.

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD.
- Power supply
- Oscilloscope.
- Connecting Links.
- Headphones.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Refer to the FIG.9.2 and Carry out the following connections.
2. Connect OUT post (ACL-AM) of preamplifier to the i/p of Balance Modulator1 (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of VCO (ACL-AM) **OUT** post to the input of Balance modulator (ACL-AM) **CARRIER IN** post.
4. Connect power supply with proper polarity to the kit ACL-AM&ACL-AD while connecting this; ensure that the power supply is OFF.
5. Switch on the power supply and Carry out the following presetting:
 - **VCO:** LEVEL about 2Vpp; FREQ. About 850 KHz. Switch on 1500KHz.
 - **BALANCED MODULATOR1:** CARRIER NULL completely rotates clockwise or counter clockwise, so that the modulator is “unbalanced” and an AM signal with not suppressed carrier is obtained across the output. Adjust OUTLEVEL to obtain an AM signal across the output which amplitude is about 500mVpp.

- **OUTPUT AMPLIFIER (ACL-AM):** LEVEL fully clock wise.
 - **LOCAL OSCILLATOR (ACL-AD):** 1300KHz, 2V.
 - **RF LEVEL (ACL-AD):** On max. position or adjust as per input signal.
6. Connect the **OUT** post of balance modulator to the **IN** post of output amplifier, in which **OUT** post of amplifier is directly connected to the antenna with JP2.
 7. Connect **RF OUT** post to **RF IN** of MIXER of ACL-AD.
 8. Connect local oscillator **OUT** post to **LO IN** of the mixer section.
 9. Connect mixer **OUT** post to **IF IN** of 1st IF amplifier.
 10. Connect **IF OUT1** of 1st IF to **IF IN 1** and **IF OUT2** of 1st IF to **IF IN 2** OF 2ND IF AMPLIFIER.
 11. Connect **OUT** post of 2nd IF amplifier to **IN** post of envelope detector.
 12. Connect **AGC1** to **AGC 2**.
 13. Connect **OUT** post of Envelope Detector to **IN** post of AUDIO AMPLIFIER and speaker to the speaker jack.
 14. If the central frequency of the amplifier and the carrier frequency of the AM signal and local oscillator frequency coincide, you obtain good voice reception on the external speaker. **Control local oscillator tuning frequency and volume knob for clear voice.**

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time.

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.
3. Put switch 1 of SF1 in ACL-AD switch fault section to ON position. This will open the capacitor (C35) from the emitter of the first IF amplifier. So the gain will reduce which in turn reduces the output amplitude.
4. Put switch 4 of SF1 in ACL-AD switch fault section to ON position. This will open the diode D1 from the envelope detector, so we will not get the negative envelope output. Also there will not be any effect of AGC signal.

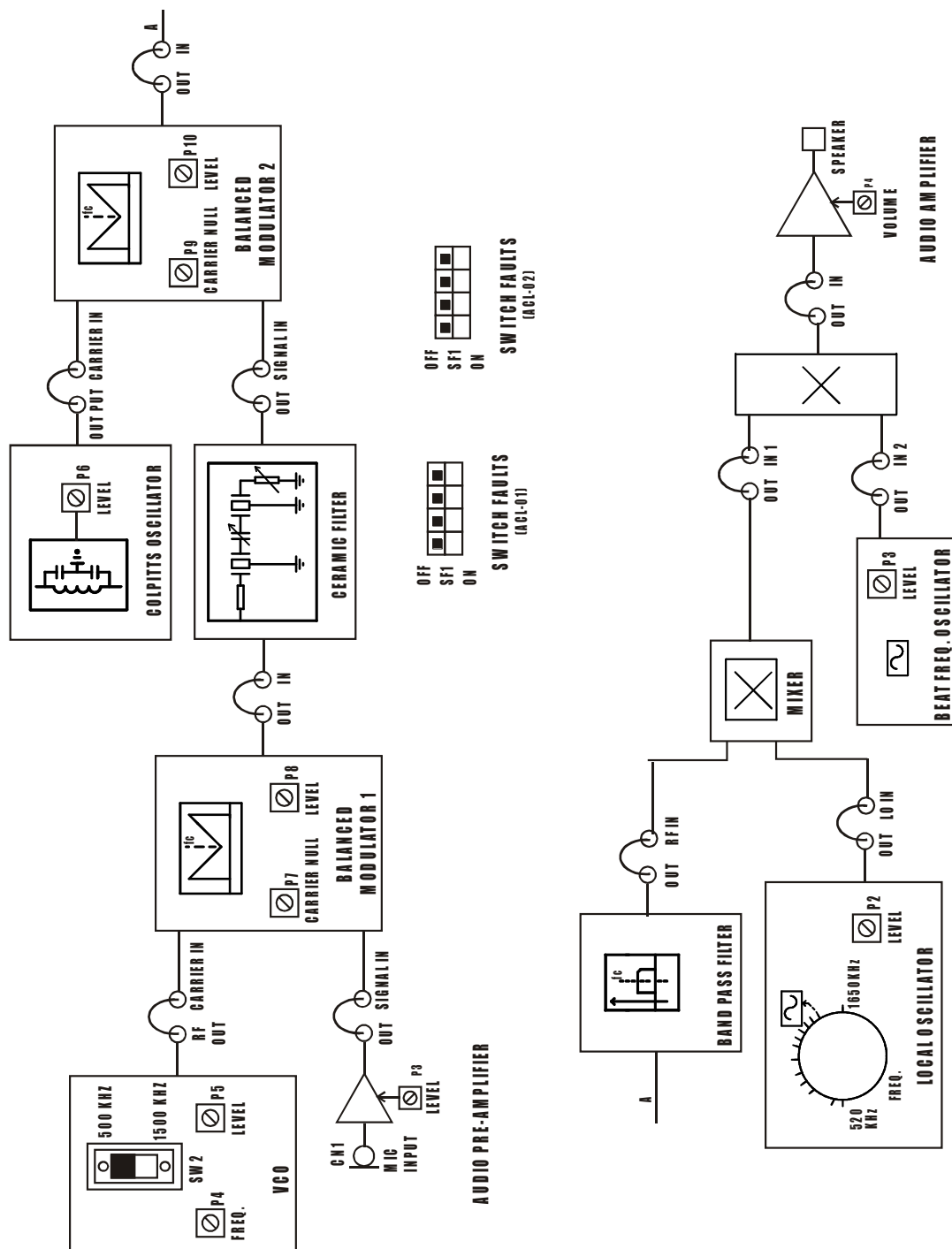


FIG. 9.3 BLOCK DIAGRAM FOR VOICE COMMUNICATION WITH SSB VIA CABLE

C. VOICE TRANSMISSION WITH SSB AM TRANSMISSION/ RECEPTION VIA CABLE

EQUIPMENTS:-

- Modules ACL-AM & ACL-AD
- Power supply
- Oscilloscope.
- Connecting Links.
- MIC & Speaker.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

1. Set the connections for SSB Reception as shown in FIG. 9.3.
2. Connect o/p of FUNCTION GENERATOR (ACL-AM) **SINE OUT** post to the i/p of Balance Modulator1 (ACL-AM) **SIGNAL IN** post.
3. Connect o/p of **VCO** (ACL-AM) **OUT** post to the input of Balance Modulator1 **CARRIER IN** (ACL-AM) post.
4. Connect power supply with proper polarity to the kit ACL-AM & ACL-AD, while connecting this; ensure that the power supply is OFF.
Switch on the power supply and Carry out the following presetting:
 - **FUNCTION GENERATOR:SINE OUT** POST: LEVEL about 1 Vpp; FREQ. About 3KHz.
 - **VCO**: LEVEL about 2Vpp; FREQ. about 457 KHz.
 - **BALANCED MODULATOR1**: CARRIER NULL pot P7 in central position, so that the modulator is “balanced” and obtain an AM signal across the output with suppressed carrier, pot P8 fully clockwise
 - **BALANCED MODULATOR2**: CARRIER NULL pot P9 in central position, so that the modulator is balanced and obtain an AM signal across the output with suppressed carrier, pot P10 fully clockwise
5. Connect **OUT** post of Balance modulator to **IN** post of ceramic filter.
6. Observe the SSB signal at the **OUT** post of ceramic filter. You can observe that the filter extracts only one of the two components (sidebands) generated by balance modulator.
7. Connect **OUT** post of CERAMIC FILTER to **SIGNAL IN** post of BALANCED MODULATOR2.
8. Connect **OUTPUT** post of COLPITT'S OSCILLATOR to **CARRIER IN** Post of BALANCED MODULATOR2.
9. Connect **OUT** post of BALANCED MODULATOR2 to **IN** post of BAND PASS FILTER.
10. Connect **OUT** post of BAND PASS FILTER to **IN** post of MIXER.
11. Observe the SSB signal at the **OUT** post of ceramic filter and at the Output of the band pass filter. You can observe that the filter extracts only one of the two components (sidebands) generated by balance modulator.
12. Connect **OUT** post of BPF to **RF IN** post of mixer on ACL-AD.
13. Connect **OUT** post of LOCAL OSCILLATOR to **LO IN** post of MIXER; keep pot P2 fully clockwise.
14. Connect MIXER **OUT** to **IN1** post of PRODUCT DETECTOR.

15. Connect **OUT** post of BFO to **IN2** post of PRODUCT DETECTOR; keep pot P3 fully clockwise.
16. Observe the final output at the OUT post of product detector by connecting this post to channel 2 of CRO. While doing this to observe the output clearly connect the input modulating signal to channel 1 of CRO & tune local oscillator frequency about 1.02MHz or in such a way that output frequency at PRODUCT DETECTOR should follow input frequency. If output frequency of VCO and BFO coincides you will get better results.
17. Increase the frequency of the modulating signal (SINEWAVE) and check that the detected signal attenuates and tends to a null. Now you can observe that even the frequency of the VCO (ACL-AM) and BFO (ACL-AD) are same, there is a slight drift in the tone generated by the receiver. This is due to the small frequency drifts in the transmitter and receiver oscillator circuits, leading to the changes in the difference frequency produced by the PRODUCT DETECTOR i.e. there would be some phase error in the frequency generated by the VCO and the frequency generated by the BFO. The effect of this phase error is to introduce the distortion in the demodulated signal, where each frequency component of the original message signal undergoes a constant phase shift. This phase distortion is tolerable in voice communication, because the human ear is relatively insensitive to phase distortion. In particular, the presence of phase distortion gives rise to a **Donald duck voice effect**. In the transmission of music and video signals, on the other hand, the presence of this form of phase distortion is totally unacceptable.
18. Then remove **SINE OUT** post and connect it to **OUT** post of AUDIO PRE-AMPLIFIER. Connect the microphone supplied with the kit to **MIC INPUT** jack of ACL-AM.
19. Connect **OUT** post of PRODUCT DETECTOR to **IN** post of AUDIO AMPLIFIER and HEADPHONES to the speaker jack.
20. If the central frequency of the modulator and the carrier frequency of the SSB signal match, you obtain good voice reception. You can use control volume pot P4 for clear voice reception.

SWITCH FAULTS:-

NOTE: Keep the connections as per the procedure. Now put the corresponding switch fault button in ON position and observe the effect on the output. The faults are normally used one at a time.

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R56) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 3 of SF1 in ACL-AD switch fault section to ON position. This will open the resistor (R63) connected to pin 4 of product detector. The output will get disturbed.

SWITCH FAULTS:-

Switch fault section provides 4 switches each on ACL-AM and ACL-AD.

These switches can be used to simulate fault conditions in various parts of the circuit. The faults are normally used one at a time. To ensure that the ACL kits are fully operational all switches should be set to OFF position before use.

1. Put switch 1 of SF1 in ACL-AM switch fault section to ON position. This will short the resistor (R57) in parallel with the bias resistor. The balance modulator 1 output gets distorted.
2. Put switch 2 of SF1 in ACL-AM switch fault section to ON position. This will short the 1K resistor (R32) in VCO. So when the frequency selector switch is in 1500 KHz position, we will get the constant frequency.
3. Put switch 3 of SF1 in ACL-AM switch fault section to ON position. This will short the coupling capacitor (C13) in colpitt's oscillator. The frequency and the shape of the output get changed.
4. Put switch 4 of SF1 in ACL-AM switch fault section to ON position. This will short the feedback resistor (R87) in output amplifier. The output amplitude gets reduced.
5. Put switch 1 of SF1 in ACL-AD switch fault section to ON position. This will open the capacitor (C35) from the emitter of the first IF amplifier. So the gain will reduce which in turn reduces the output amplitude.
6. Put switch 2 of SF1 in ACL-AD switch fault section to ON position. This will open the capacitor (C49) in AUDIO AMPLIFIER, which will result in disturbed audio output.
7. Put switch 3 of SF1 in ACL-AD switch fault section to ON position. This will open the resistor (R63) connected to pin 4 of product detector. The output will get disturbed.
8. Put switch 4 of SF1 in ACL-AD switch fault section to ON position. This will open the diode D1 from the envelope detector; O/P will get disturbed in both the envelope. Also there will not be any effect of AGC signal.

EXPERIMENT NO.10

EXPERIMENT NO: 10

NAME:-

STUDY OPERATION OF DIODE DETECTOR

OBJECTIVE:-

To study operation of diode detector:-

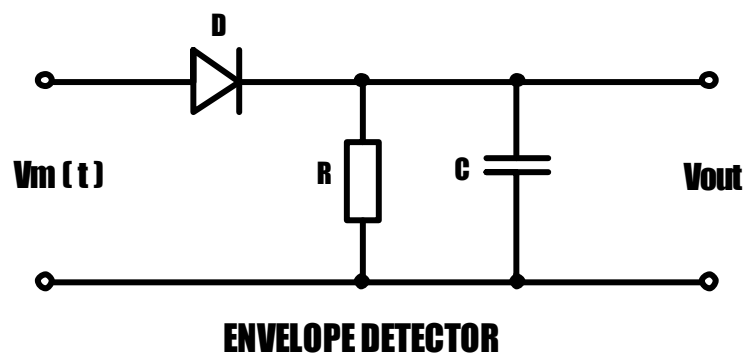
EQUIPMENTS:-

- Modules ACL-AM & ACL-AD
- Power supply
- 20MHz Oscilloscope, voltmeter.
- Connecting Links.

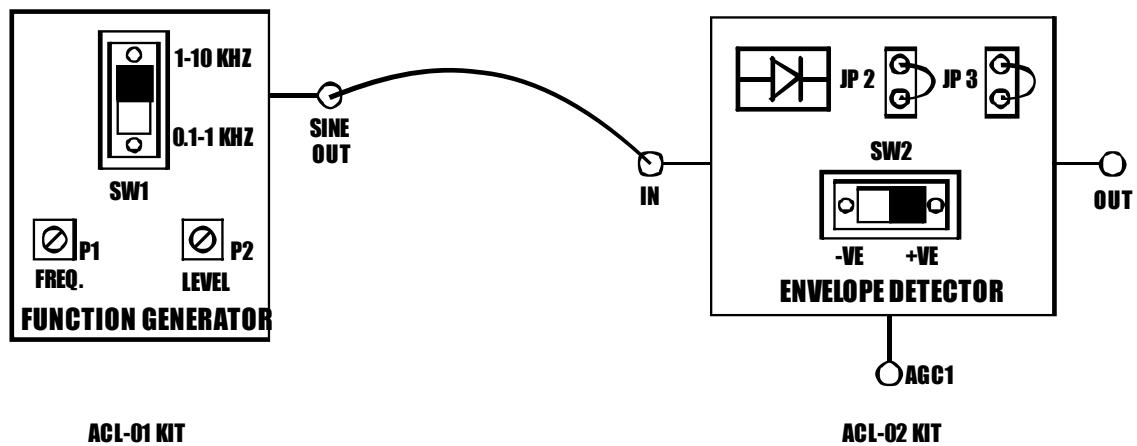
THEORY:-

DIODE DETECTOR (ENVELOPE DETECTOR):

The extraction of the modulating signal from an AM signal can be carried out using an envelope detector. Consider, in fact, the AM signal shown in FIG. 2A, and note that the modulating signal constitutes the envelope of the waveform reported.



The most common envelope detector consists of a diode followed by a RC filter. Its operation is analogous to the one of a half-wave rectifier, as the output voltage follows the maximum values of the carrier. As the amplitude of the carrier is variable, by properly choosing R and C, the output of the detector can faithfully reproduce these variations.

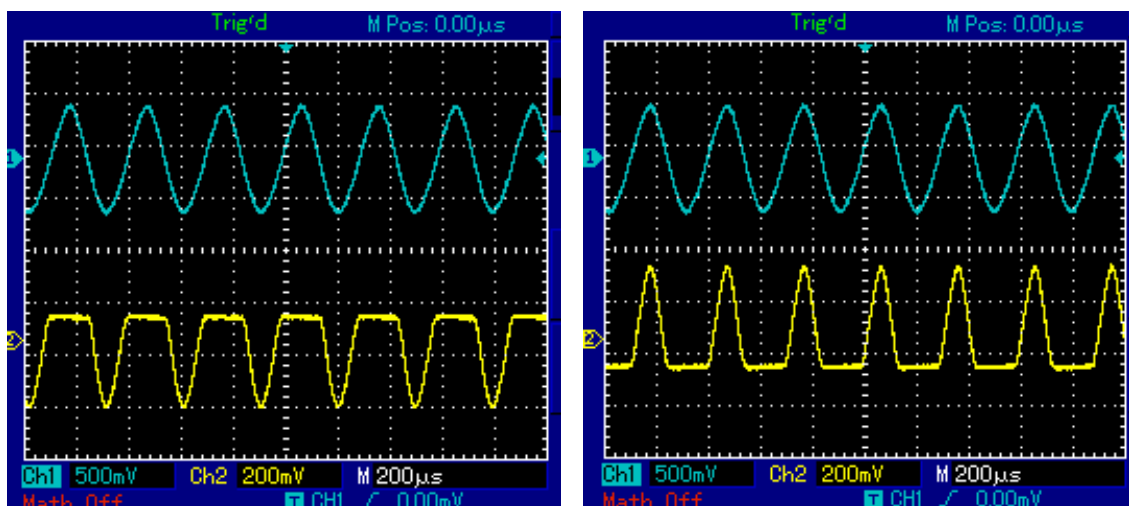


BLOCK DIAGRAM FOR STUDY OF ENVELOPE DETECTOR

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

- 1) Refer to the block diagram for DIODE DETECTOR (ENVELOPE DETECTOR).
- 2) Connect SINE OUT post in function GENERATOR from ACL-AM to IN post of the ENVELOPE DETECTOR in ACL-AD.
- 3) Vary the frequency of SINE OUT using pot P1 in ACL-AM kit.
- 4) Keep the switch SW2 on –VE and +VE position and accordingly observe the detectors response.



CONCLUSION:-

The most common envelop detector consists of a diode followed by a RC filter. Its operation is analogous to the one of a half-wave rectifier, as the output voltage follows the maximum value of the carrier

EXPERIMENT NO.11

EXPERIMENT NO: 11

NAME:-

STUDY OPERATION OF MIXER

OBJECTIVE:-

To Study Operation of Mixer:-

EQUIPMENTS:-

- Modules ACL-AM&ACL-AD
- Power supply
- 20MHz Oscilloscope, voltmeter.
- Connecting Links.

THEORY:-

This experiment investigates the concept of image frequencies, using the ACL-AD kit. In order to explain what these are, here is a short review of how AM receiver operates.

You will recall that the frequency of the LOCAL OSCILLATOR in the receiver section is set higher than the selected signal frequency by a constant amount, irrespective of the frequency of the selected station. The amount of that constant difference frequency is chosen to be 455 KHz, the Intermediate Frequency.

The MIXER in the receiver section mixes the output of the R.F.AMPLIFIER with the output of the LOCAL OSCILLATOR, to extract the 455 KHz difference frequency.

The incoming signal, whose frequency is exactly 455 KHz above the frequency of the LOCAL OSCILLATOR, is known as the image frequency.

For every wanted frequency there is a corresponding image frequency. If there is a strong station at or near the image frequency, it will result in irritating whistling noises at the output of the receiver, which will spoil the reception of the wanted station.

In the last section, we saw we could receive a station being broadcast on 550KHz by tuning the local oscillator to a frequency of 1 MHz thus giving the difference (IF) frequency of the required 450KHz.

What would happen if we were to receive station broadcasting on a Frequency of 1.45MHz?

This would also mix with the local oscillator frequency of 1.45MHz to produce the required IF frequency of 450 KHz. This would mean that this station would also be received at the same time as the desired one at 550 KHz.

STATION 1:

Frequency 550 KHz, Local oscillator 1.0MHz, IF = 450 KHz.

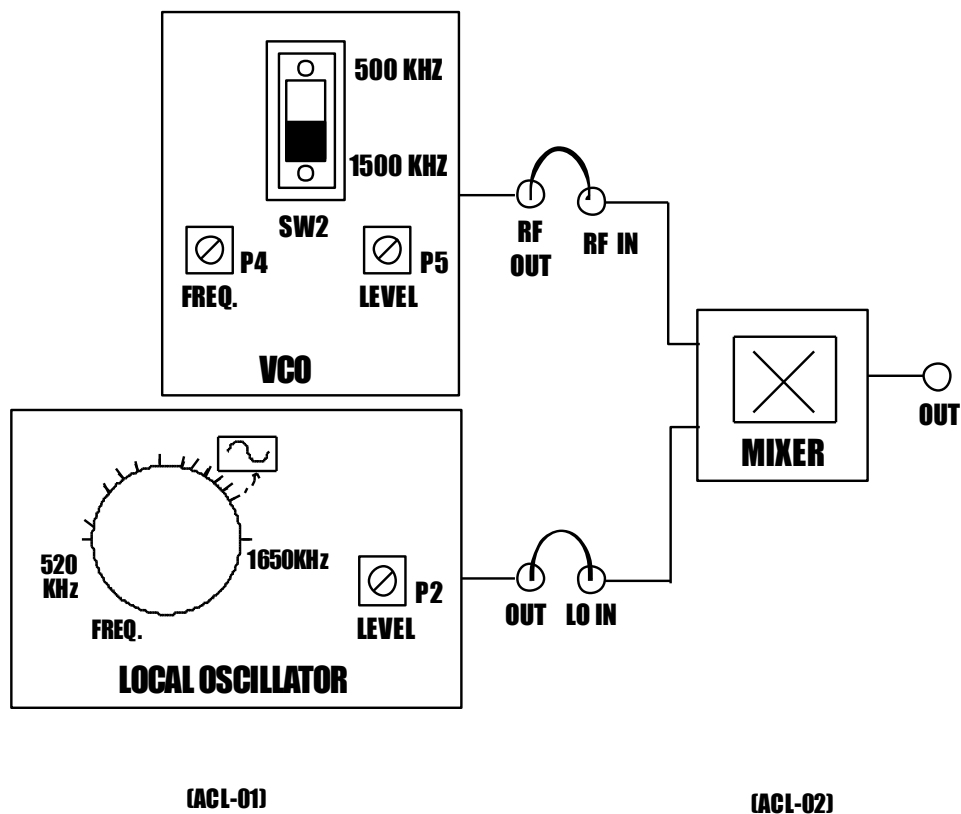
STATION 2:

Frequency 1.450MHz, Local oscillator 1.0 MHz, IF = 450 KHz.

An 'image frequency' is an unwanted frequency that can also combine with the Local Oscillator output to create the IF frequency.

Notice how the difference in frequencies of the wanted and unwanted stations is twice the IF frequency. In the ACL-AD, it means that the image frequency is always 910 KHz above the wanted station.

This is a large frequency difference and even the poor selectivity of the RF amplifier is able to remove the image frequency unless it is very strong indeed. In this case it will pass through the receiver and will be heard at the same time as the wanted station. Frequency interactions between the two stations tend to cause irritating whistles from the loudspeaker.



BLOCK DIAGRAM FOR STUDY OF MIXER

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.

PROCEDURE:-

Refer to block diagram for MIXER in expt.8.

- 1) From ACL-AM kit Connect RFOUT post of VCO to RFIN post of MIXER in ACL-AD kit.
- 2) Connect OUT post of LOCAL OSCILLATOR in ACL-AM kit to LO IN post of MIXER in ACL-AD kit.
- 3) From ACL-AM kit, Keep switches SW2 on 1500 KHz position. Adjust FREQ of VCO for 1.5MHz.

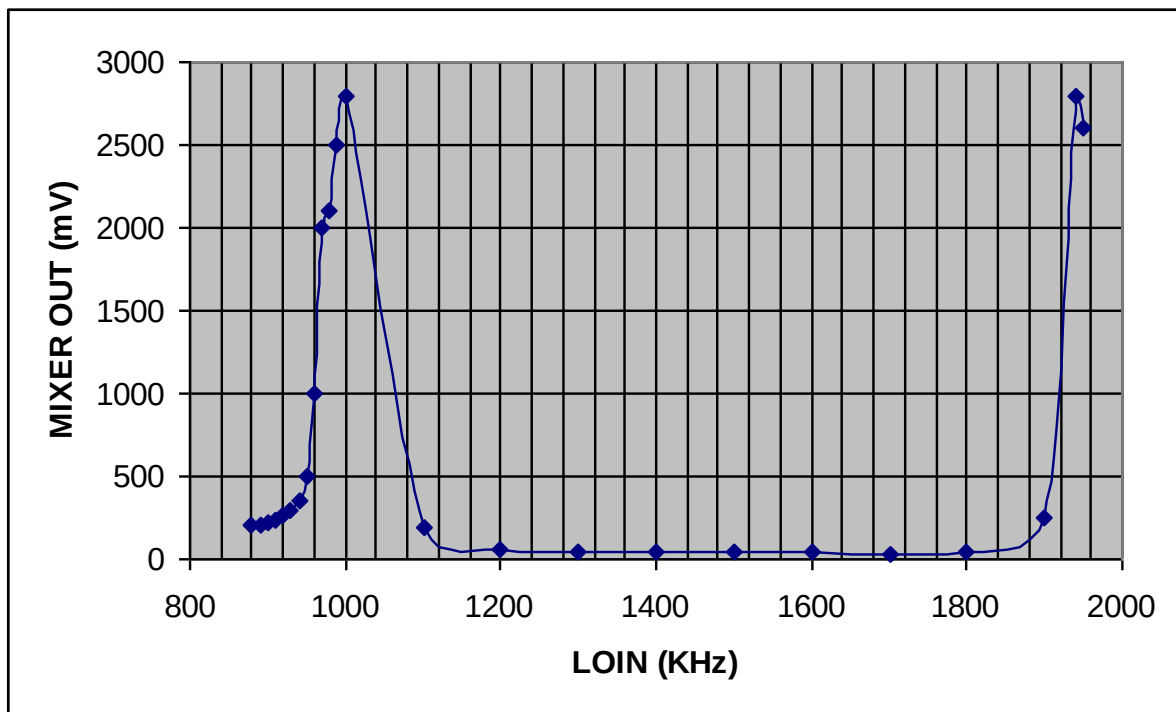
- 4) From ACL-AD kit, Rotate LEVEL pot P2 fully clockwise.
- 5) Start Adjusting LOCAL OSCILLATOR frequency, starting from its minimum position say for 880KHz in interval of 10KHz and note down the MIXER OUT voltage. Also notice that where you get maximum peak-peak amplitude of a sine wave on OUT post of MIXER (at around 1MHz). Keep on adjusting further and note down the readings till you get second higher frequency where you get maximum peak-peak amplitude of a sine wave on OUT post of MIXER.
- 6) Plot the graph of LOIN frequency versus MIXER OUT voltage.

Sample Table:-

For VCO = 1.5MHz, 500mV.

LOIN (KHz)	MIXER OUT (mV)
880	200
890	200
900	215
910	230
920	260
930	290
940	360
950	500
960	1000
970	2000
980	2100
990	2500
1000	2800
1100	190
1200	60
1300	50
1400	50
1500	50
1600	40
1700	36
1800	40
1900	250
1940	2800
1950	2600

Note: - Readings shown above in the table are only a sample. It may vary from kit to kit and the user.

**CONCLUSION:-**

From the graph plotted find out the image frequencies for the mixer for 1.5MHz of VCO and the formula $f_{IF} = f_{LO} - f_{RF}$ is verified

EXPERIMENT NO.12

EXPERIMENT NO: 12**NAME:-****STUDY OPERATION OF AGC****OBJECTIVE:-**

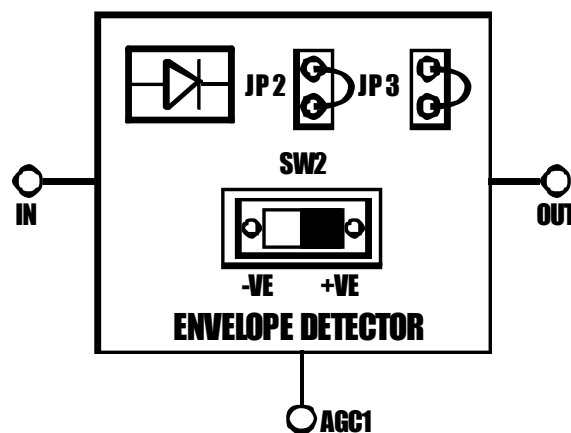
To study operation of AGC:-

EQUIPMENTS:-

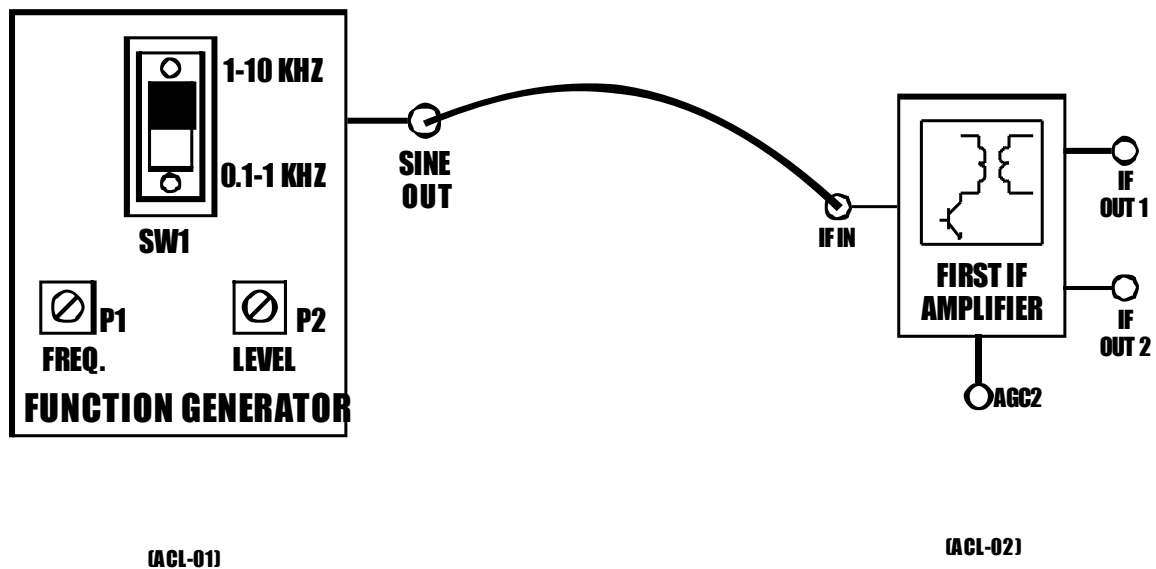
- Modules ACL-AM & ACL-AD
- Power supply
- 20MHz Oscilloscope
- Voltmeter.
- Connecting Links.

NOTE: KEEP ALL THE SWITCH FAULTS IN OFF POSITION.**PROCEDURE:-**

- 1) Apply DC voltage from -2.5V to +2.5V at IN post of the ENVELOPE DETECTOR in ACL-AD kit.



- 2) Note down the voltage at AGC1 post of EVELOP DETECTOR IN ACL-AD kit with respect to DC voltage applied to ENVELOPE DETECTOR.
- 3) It will be observed from the above readings that as the input voltage to envelop detector increases in positive direction, the AGC voltage at AGC1 post will generate more negative voltage to compensate.
Similarly as the input voltage to envelop detector decreases in negative direction, the AGC voltage at AGC1 post will generate more positive voltage to compensate.
- 4) Apply sine wave from SINE OUT post of FUNCTION GENERATOR in ACL-AM to IF IN post of FIRST IF AMPLIFIER in ACL-AD kit.



- 5) Apply DC voltage from -2.5V to +2.5V at AGC2 post of FIRST IF AMPLIFIER in ACL-AD kit and note down the readings at IF OUT1 post.
- 6) It will be observed from the above readings that as the input voltage to AGC2 post increases from -2.5V to 2.5V, the IF OUT1 voltage will also goes on increasing.

Sample Table:-

IN (mVdc) to ENVELOP DETECTOR	AGC1 (mV)
200	-5.2
400	-23.3
600	-46
800	-65.6
1000	-99.6
1200	-118.8
1400	-158
1800	-211
2000	-247
2200	-266
2300	-288

Note: - Readings shown above in the table are only a sample. It may vary from kit to kit and the user.

CONCLUSION:-

Thus it been observed and verified that how positive or negative feedback voltage is generated for AGC, and then how the AGC is achieved.